

WELCOME TO SUGARCUBE™



Thank You for purchasing the SugarCube **SC2-Plus**.

Your experience of high-end audio is about to get a lot sweeter when all your vinyl will sound like mint.

These pages provide an overview of the set-up and control of the **SC2-Plus**.

For any usage questions,
please contact:
support@sweetvinyl.com

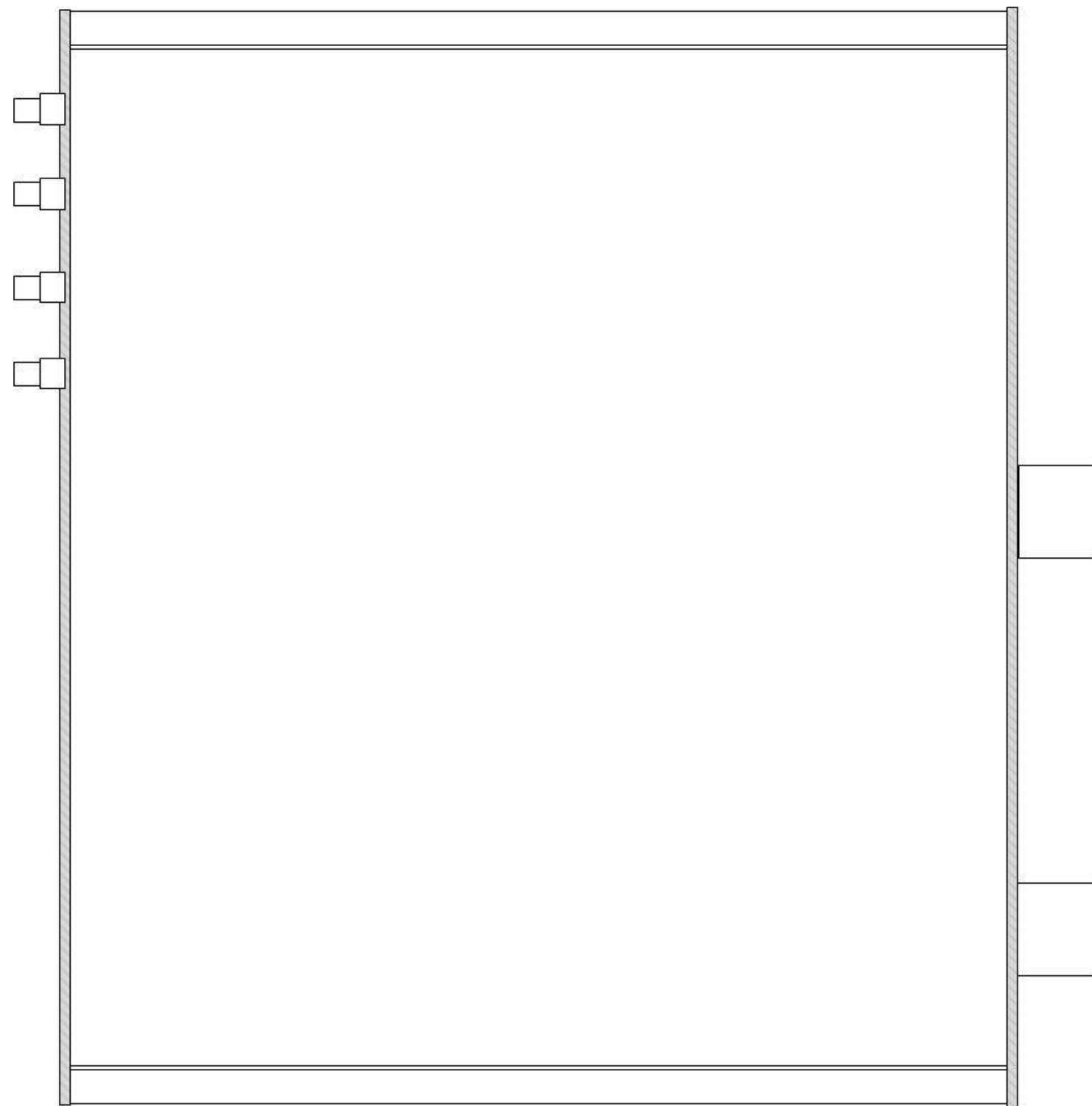


Section 1

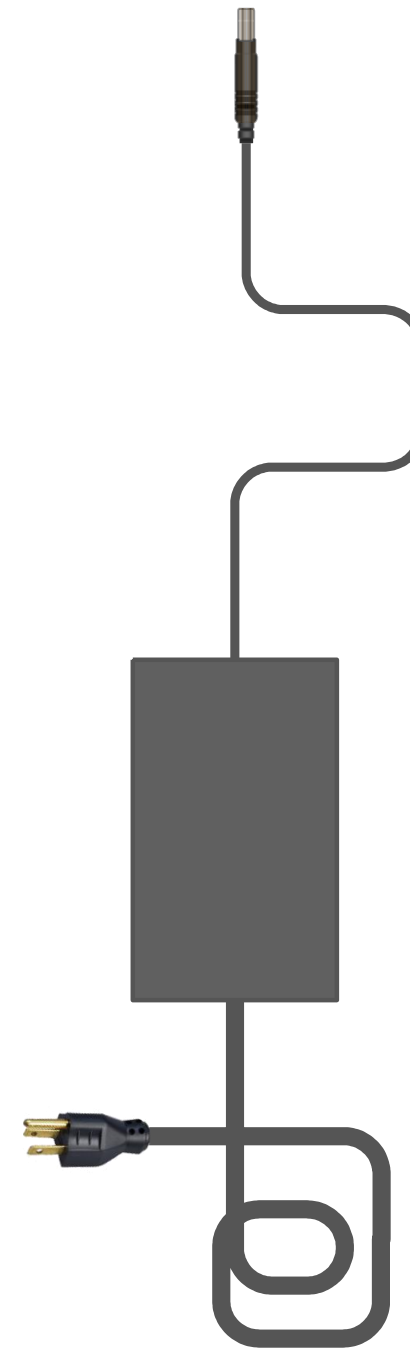
Unpacking Your SugarCube



1.1 Unpacking Your SugarCube



SugarCube SC2-Plus Unit



**AC Power Adapter,
5V, 4A DC Output**



Wifi USB Adapter



**USB
Stick**

Please download the **SugarCube App** from your mobile device's app store or use a **Browser** to control your **SugarCube** remotely.

This adapter connects the SugarCube to your WiFi network or you can use a wired Internet connection if available.

USB-stick contains the full SugarCube Setup and User Guide.

1.2 SC2-Plus Front-Panel Controls



HEADPHONE JACK, VOLUME, MUTE

Headphone amp operates independently and does not affect the level of the line in / out audio.

BYPASS

When pressed and the LED is lit, an internal relay has shunted the **LINE-IN** directly to **LINE-OUT**, bypassing all internal circuitry.

STRENGTH

Controls the strength of the click and pop detection and removal process. Almost all LPs will sound great at a setting of 5 or 6.

Note: we recommend that you do NOT adjust **STRENGTH** all the way to 10, you will rarely need this much. Less is more.

CLICK REMOVE 'IN'

Push on, push off: when the LED is lit, SugarCube is processing audio to remove clicks and pops (the **BYPASS** LED will go out.)

CLICK MONITOR

When pressed and the LED is lit, the SugarCube suppresses the music so that you can better hear the clicks and pops. This helps set the **STRENGTH** level for particularly noisy LPs.

STATUS

Green when storage plugged in and ready. Blinks red while recording.

USB3

Plug in USB stick or drive
To record & playback
Storage.

RECORD

Press to start recording
Press to stop.

1.3 SC2-Plus Back-Panel

DIGITAL OUTPUT

Toslink and SPDIF outputs at 192K/24bit.

DC POWER IN

Connect the supplied adapter to this port.

POWER ON/OFF

Note: when power is off, an internal relay bypasses **LINE IN** to **LINE OUT**.



LINE OUTPUT

Outputs to your line stage (pre-amp). When the front panel **BYPASS** button is pressed, the audio from **LINE IN** passes directly to this **LINE OUT** through a mechanical audiophile-grade relay.

LINE INPUT

Accepts the output of your phono pre-amp. Please do NOT directly plug your turntable into this port unless it has a built-in phono pre-amp with a line-level output.

USB COMPUTER PORT

Use this port to link your computer or music server to for enhanced ADC and DAC functionality. Detailed instructions later in guide.

USB PORT FOR WIFI ADAPTER

Do either of the above when controlling the SC2-Plus on the SugarCube iOS or Android app or from your desktop.

WIRED LAN

If an Ethernet LAN cable is available, connect to this port instead of plugging in the WiFi adapter.

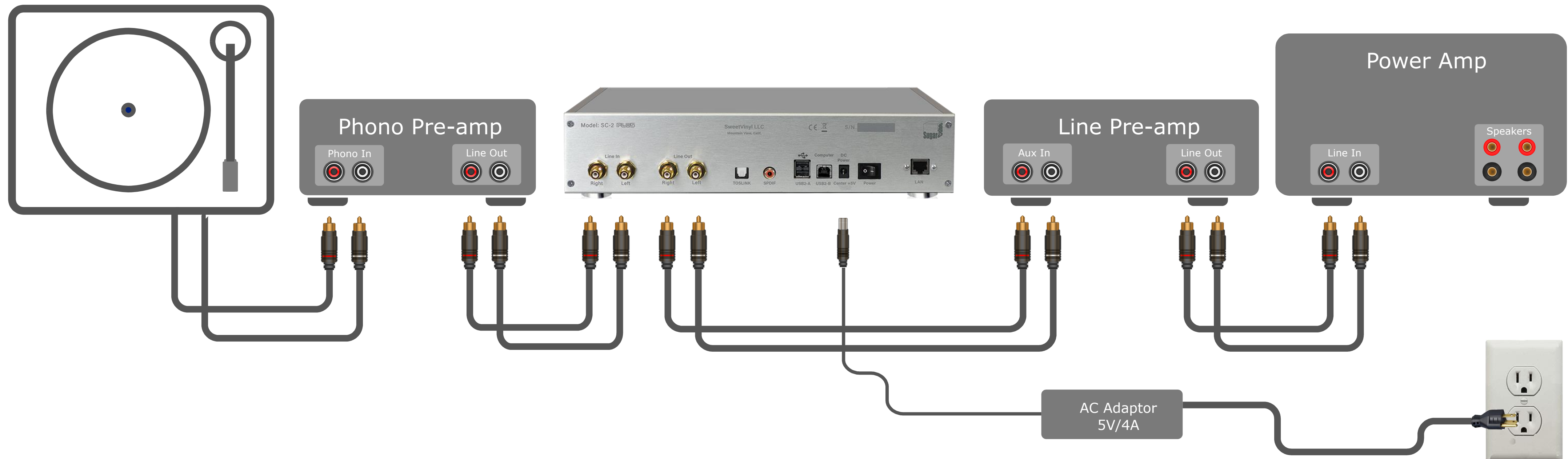
Section 2

Connecting SugarCube to Your Sound System

PLEASE BE SURE ALL
EQUIPMENT IS
TURNED OFF
BEFORE ATTEMPTING
INSTALLATION



2.1 Connecting Your SugarCube



Connecting in series between a phono pre-amp and line-stage (pre-amp).

1. Plug your SugarCube power adapter into a power source.
2. Connect the output of your phono pre-amp to the **LINE IN** of the SugarCube.
3. Connect the **LINE OUT** of the SugarCube to an appropriate line-level input of your line-stage/pre-amp.

NOTE: **DO NOT CONNECT** THE **LINE OUT** OF THE **SUGARCUBE** TO THE **PHONO INPUT** ON YOUR EQUIPMENT.

2.2 Connecting Your SugarCube

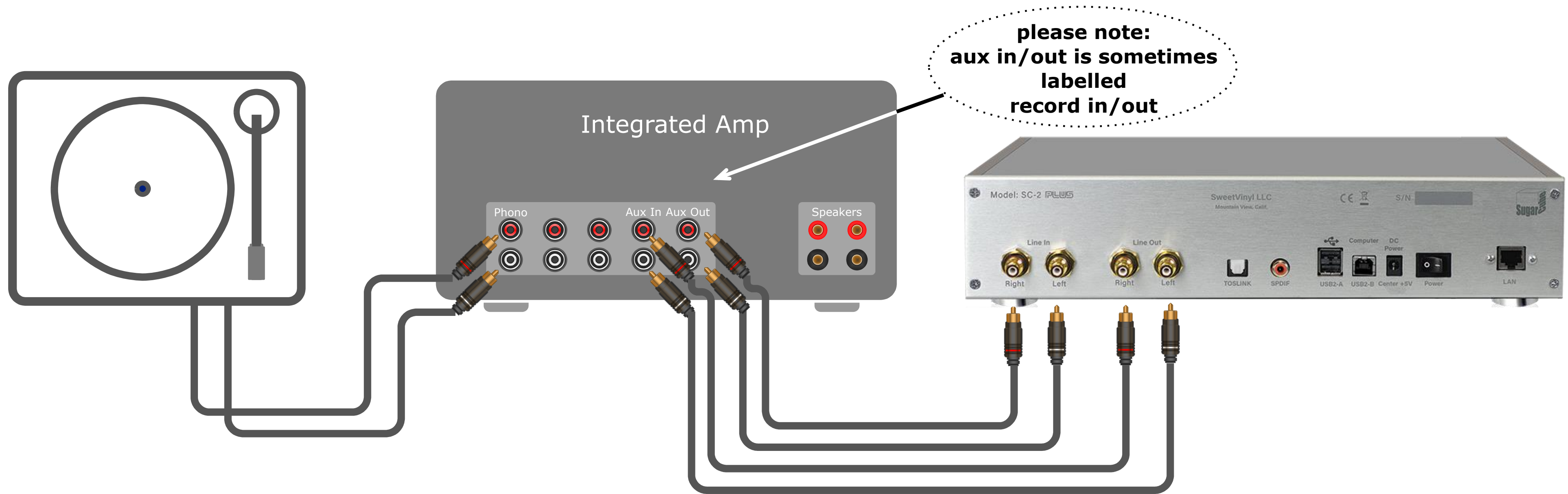


Connecting in series between a pre-amp, with phono input, and power amp

1. Connect the output of your pre-amp to the **LINE IN** of the SC1-Plus.
2. Connect the **LINE OUT** of the SC1-Plus to the input of your power amp.

2.3 Connecting Your SugarCube

Integrated Amp: using the 'tape loop'



Connecting to an integrated amplifier via Aux In/Out (or Rec In/Out).

1. Connect **AUX OUT** (or REC OUT) of your integrated amp (or processor) to the **LINE IN** of the SC1-Plus.
2. Connect **LINE OUT** of the SC1-Plus to **AUX IN** (or **REC IN**) of your integrated amp or processor.

NOTE: DO NOT CONNECT THE **LINE OUT** OF THE SUGARCUBE TO THE **PHONO INPUT** ON YOUR EQUIPMENT.

2.4 Connecting Your SugarCube

Integrated Amp: connecting between the preamp and power amp

Step 1: TURN OFF THE INTEGRATED AMP

Step 2: Wait a minute or two

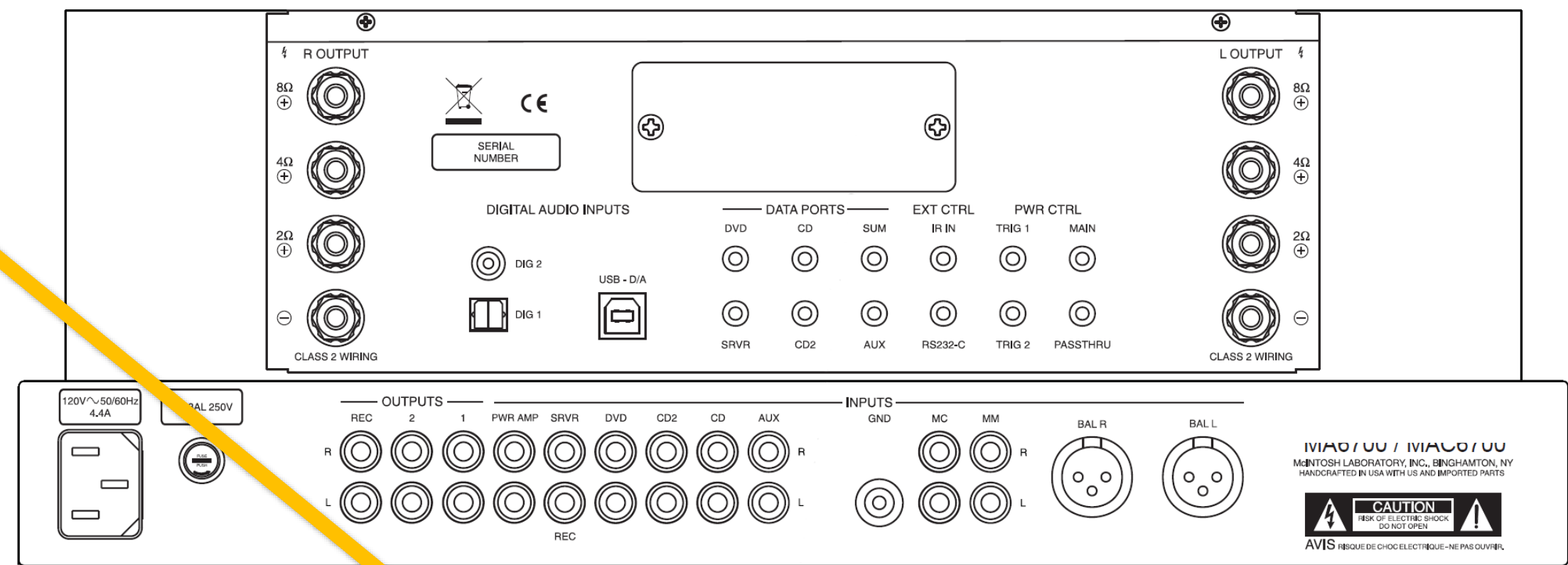
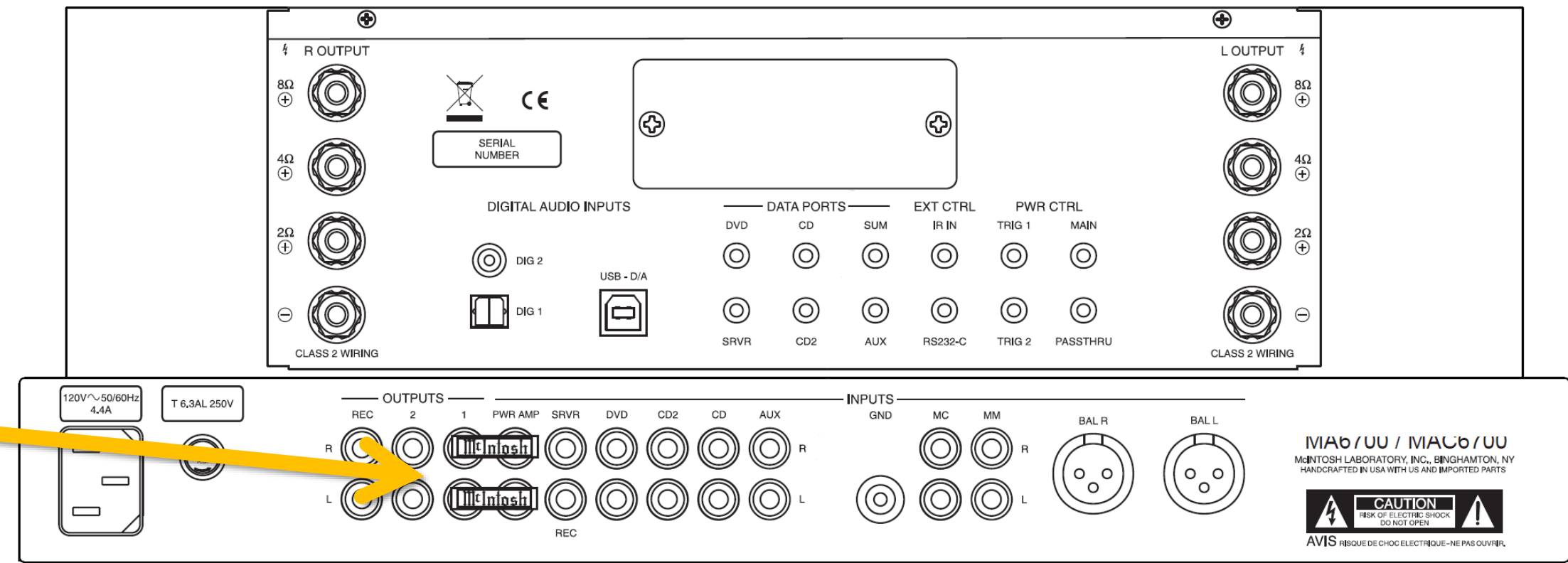
Step 3: locate the preamp-power amp jumpers

Step 4: DOUBLE-CHECK THAT THE AMP IS OFF

Step 5: Remove the jumpers

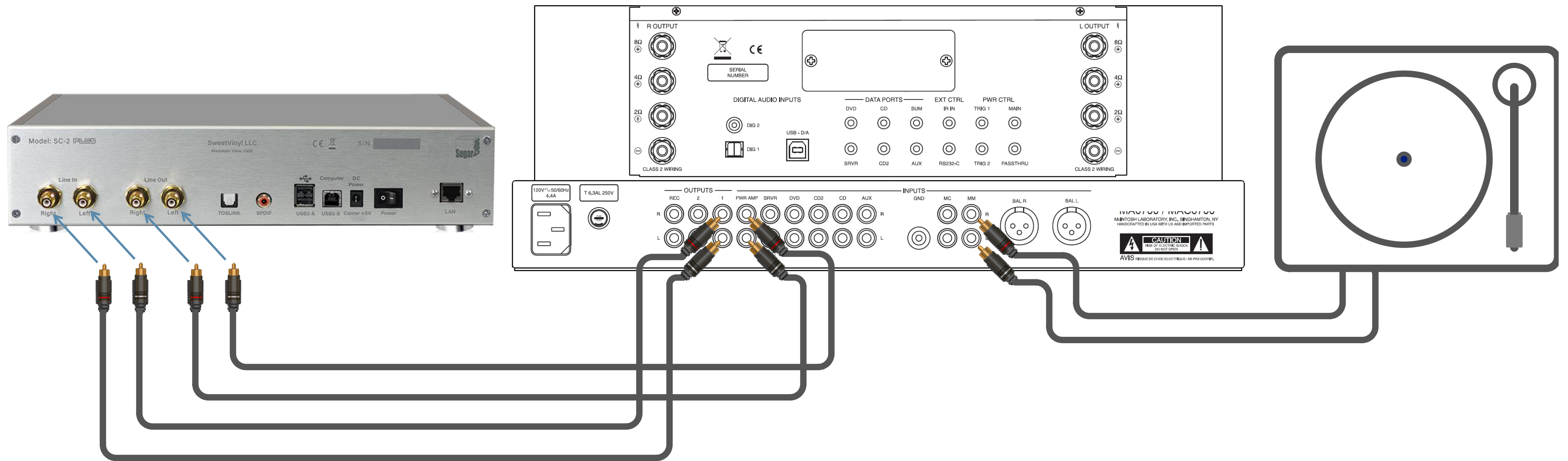
Step 6: Put the jumpers away in a safe place

Step 7: NEXT PAGE...



2.5 Connecting Your SugarCube

Integrated Amp: connecting between the preamp and power amp, cont.



Connecting to an integrated amplifier between the pre-amp and power-amp.

1. Connect **OUTPUT** (this is the pre-amp) of your integrated amp to the **LINE IN** of the SugarCube.
2. Connect **LINE OUT** of the SugarCube to **PWR AMP** (power amp input) of your integrated amp.
3. **ALWAYS TURN ON THE SUGARCUBE FIRST THEN THE AMP AND TURN OFF THE AMP FIRST THEN THE SUGARCUBE.**
4. If the integrated amp has a headphone output, it will likely have its own amplifier and may not be in the loop with the SugarCube. If that is the case, you will not hear the SugarCube in the headphones.
5. The SugarCube draws only about 11 watts, you can leave it on full time if you wish.
6. Not all integrated amps have a line out / power amp in or an Aux Out / In. If this is the case, you may need an external phono pre-amp.

2.6 Connecting Your SugarCube

Integrated Amp **WITHOUT** an AUX OUT/IN (or REC OUT/IN, etc.)



Many contemporary integrated amplifiers no longer have what was known as a 'tape loop' (back in the day of the cassette and reel-to-reel tape). Also, most modern home theatre processors that do have what are called 'effects loops' will **not** send audio from its phono input through the loop. In these cases, there is not a way to connect to SugarCube directly to these devices while using its built-in phono stage.

The solution is to add a phono stage (pre-amp) between your turntable and the SugarCube. Then connect the Line Out of the SugarCube to any available input on the integrated amp (except for the phono input, of course.)

2.7 Connecting Your SugarCube

Connecting turntables with built-in phono pre-amps



For turntables with a built-in phono pre-amp, be sure to set the **PHONO/LINE** switch to **LINE** before plugging into the SugarCube.

Directly connecting the line-level output from your turntable.

1. Connect the turntable with a **built-in** phono pre-amp to **LINE IN**.
2. Be sure the turntable is set to **LINE** not **PHONO**.
3. Connect the **LINE OUT** of the SugarCube to any line-level input of your line stage, pre-amp, or integrated amp.

NOTE: SORRY TO BE A BROKEN RECORD, SO TO SPEAK, BUT DO NOT CONNECT THE **LINE OUT** OF THE SUGARCUBE TO THE **PHONO INPUT** ON YOUR EQUIPMENT.

2.8 Starting Up Your SugarCube



(1) SugarCube just powered on



(2) SugarCube is ready to go
(please expect about **three minutes to ready.**)



(3) Hold the 'Bypass' button for a few seconds to see LAN address
(and WiFi SSID if using wireless.)

Please note: When you power up your SugarCube, it will take about **three minutes** to complete its startup procedures. During this time, the display will light up but remain blank. We apologize for the wait but we have some seriously sophisticated software loading and it simply takes additional time. When you hear the results of our efforts, you will agree that it was worth the wait.

Section 3

Networking the SugarCube

- Part 1: Ethernet and WiFi
- Part 2: Chrome Browsers
- Part 3: Apple iOS
- Part 4: Android





3.1 Using Ethernet or WiFi with SugarCube

3.1.1 Pairing your SugarCube



While setting up the connection between the SugarCube and the mobile or web browser SugarCube app, you will be prompted to enter a **Pair Code**, please just use the code **1111**. (**Note:** you can push in and release the '**Strength**' knob to obtain a unique pairing code but there is no need for this extra step, it remains to support legacy systems.)

3.1.2 Connecting Your SugarCube to a LAN or WiFi



If you would like to use the iOS or Android app or web browser to control your SugarCube SC2-Plus, please connect either the WiFi adapter supplied with your SugarCube or plug into a wired connection.

NOTES:

- THE SUGARCUBE SC2-PLUS BASIC OPERATIONS ARE FULLY FUNCTIONAL BY MEANS OF THE FRONT PANEL CONTROLS WITHOUT THE NEED TO CONNECT TO A NETWORK.
- SOFTWARE UPDATES ARE AUTOMATIC AND PROVIDED WHEN THE SUGARCUBE IS CONNECTED TO A WIFI OR ETHERNET NETWORK.

3.1.3 Connecting Your SugarCube via Wired Ethernet



- When connecting via wired Ethernet you need only open the iOS or Android App or follow a few steps to establish a browser connection.

NOTES:

- THE SUGARCUBE SC2-PLUS BASIC OPERATIONS ARE FULLY FUNCTIONAL BY MEANS OF THE FRONT PANEL CONTROLS WITHOUT THE NEED TO CONNECT TO A NETWORK.
- SOFTWARE UPDATES ARE AUTOMATIC AND PROVIDED WHEN THE SUGARCUBE IS CONNECTED TO A WIFI OR ETHERNET NETWORK.

3.1.4 Connecting Your SugarCube via WiFi



- If using the WiFi adapter you will either use the iOS or Android App to setup WiFi, or use Ethernet and a browser to configure the WiFi parameters.

NOTES:

- THE SUGARCUBE SC2-PLUS BASIC OPERATIONS ARE FULLY FUNCTIONAL BY MEANS OF THE FRONT PANEL CONTROLS WITHOUT THE NEED TO CONNECT TO A NETWORK.
- SOFTWARE UPDATES ARE AUTOMATIC AND PROVIDED WHEN THE SUGARCUBE IS CONNECTED TO A WIFI OR ETHERNET NETWORK.



3.2 Using a **Browser** with SugarCube

3.2.1 Controlling the SugarCube from a Web Browser

Finding the IP address of your SugarCube



2

Press in and hold 'BYPASS' for 3 to 5 seconds

Your SugarCube will then enter a display mode where it will show you its IP address, among other information.

You will use an address that looks like this example. It usually will start with 192 or with 10.

3



Plug in an Ethernet Cable

1

3.2.2 Controlling the SugarCube from a Web Browser

Entering the IP address with addition information into the browser



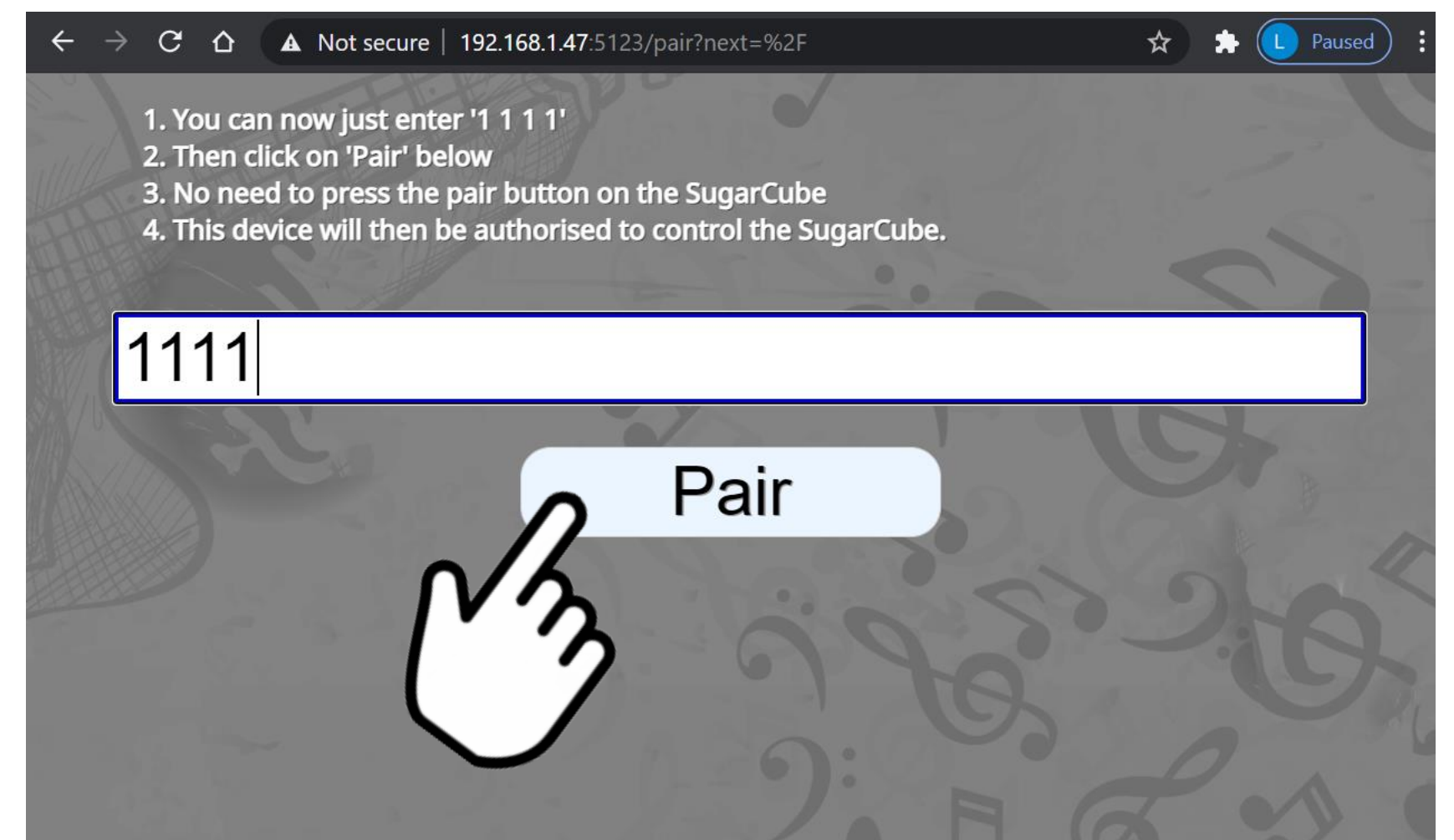
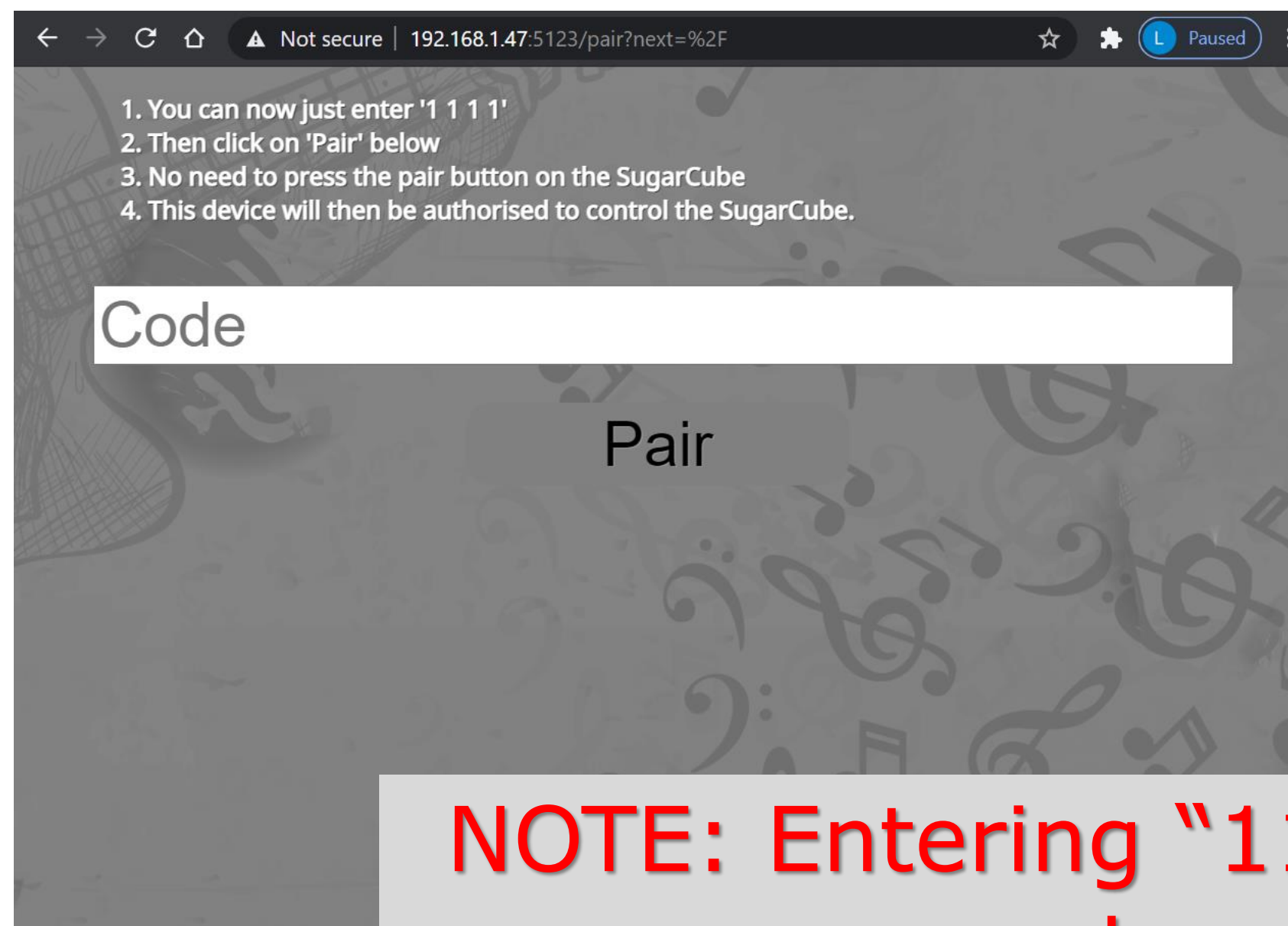
Important Note: the SugarCube web interface is compatible with **Google Chrome, Microsoft Edge, & Apple Safari** web browsers. We cannot guarantee compatibility with any other brand.

(1) Type the SugarCube's IP address into the **address bar** then add a colon ':' and '5123'.

(2) Press 'Enter' and the SugarCube webpage will appear in the browser.

PLEASE NOTE: the '5123' is a fixed port number and NOT the serial number of your SugarCube.

3.2.3 Controlling the SugarCube from a Web Browser



NOTE: Entering "1111" as a pairing code will be accepted on recent versions of the SugarCube software

1

(1) You will see a screen appearing in your browser to accept a pairing code.

2

(2) Then, enter the four-digit pairing code displayed on the SugarCube display and click on 'Pair'.

3.2.4 Controlling the SugarCube from a Web Browser



Important Note: the SugarCube web interface is compatible with **Google Chrome, Microsoft Edge, & Apple Safari** web browsers. We cannot guarantee compatibility with any other brand.

The SugarCube user interface will appear in the browser.

3.2.5 Controlling the SugarCube from a Web Browser

Setting up WiFi (in place of wired Ethernet), 1 of 4

Settings

State: normal
SSID: <Not Configured>
Info: No wireless adapter present

SSID
Password

Site survey

No wifi seen.

	SSID	Signal	Channel	Security	Access Points
Use	1qa2ws3ed	87%	6 40	WPA2	2
Use	219Guest	20%	11	WPA1 WPA2	1
Use	219SecureIoT	34%	6	WPA1 WPA2	1
Use	219SecureWireless	30%	6	WPA1 WPA2	1
Use	BlueCart Wifi	30%	6	WPA2	1
Use	BlueCart Wifi - Guest	50%	6	WPA2	1
Use	SmugMug	84%	40	WPA2	1
Use	SmugMugs	84%	40	WPA2	1
Use	sweetvinyl	100%	9 48	WPA2	4
Use	Tap Guest	34%	11	WPA1 WPA2	1

S/N: Model: SC-2

USB DC In Power LAN

Pair 12v / 2 amp center (+)

Made in U.S.A.

(1) If wired Ethernet is not available in your sound room, then take the SugarCube to your router just for these WiFi setup steps.

(2) You need both Ethernet and the WiFi stick plugged into the SugarCube.

(3) Find your WiFi name on the list of SSIDs.

3.2.6 Controlling the SugarCube from a Web Browser

Setting up WiFi (in place of wired Ethernet), 2 of 4

State: normal

Ssid: <Not Configured>

Info: disconnected

SSID

Password

Site survey

	SSID	Signal	Channel	Security	Access Points
Use	1qa2ws3ed	87%	6 40	WPA2	2
Use	219Guest	20%	11	WPA1 WPA2	1
Use	219SecureIOT	34%	6	WPA1 WPA2	1
Use	219SecureWireless	30%	6	WPA1 WPA2	1
Use	Wifi	30%	6	WPA2	1
Use	Wifi - Guest	50%	6	WPA2	1
Use		84%	40	WPA2	1
Use	SmugMugs	84%	40	WPA2	1
Use	sweetvinyl	100%	9 48	WPA2	4
Use	Tap Guest	34%	11	WPA1 WPA2	1

(4) Please select your network by clicking on the 'Use' button next to your SSID.

Inf disconnected

SSID

Password

Site survey

	SSID	Signal	Channel	Security	Access
Use	1qa2ws3ed	80%	6 40	WPA2	
Use	219Guest	44%	11	WPA1 WPA2	
Use	219SecureIOT	30%	6	WPA1 WPA2	
Use	219SecureWireless	34%	6	WPA1 WPA2	
Use	BlueCart Wifi	34%	6	WPA2	
Use	SmugMug	87%	6 40	WPA2	

(5) Next, enter the password for your WiFi then click on 'Connect.'

3.2.7 Controlling the SugarCube from a Web Browser

Setting up WiFi (in place of wired Ethernet), 3 of 4

Press the pair button on your SugarCube for at least 5 seconds to start the setup wifi network.

State: connect

Ssid: s\ **<your wifi>**

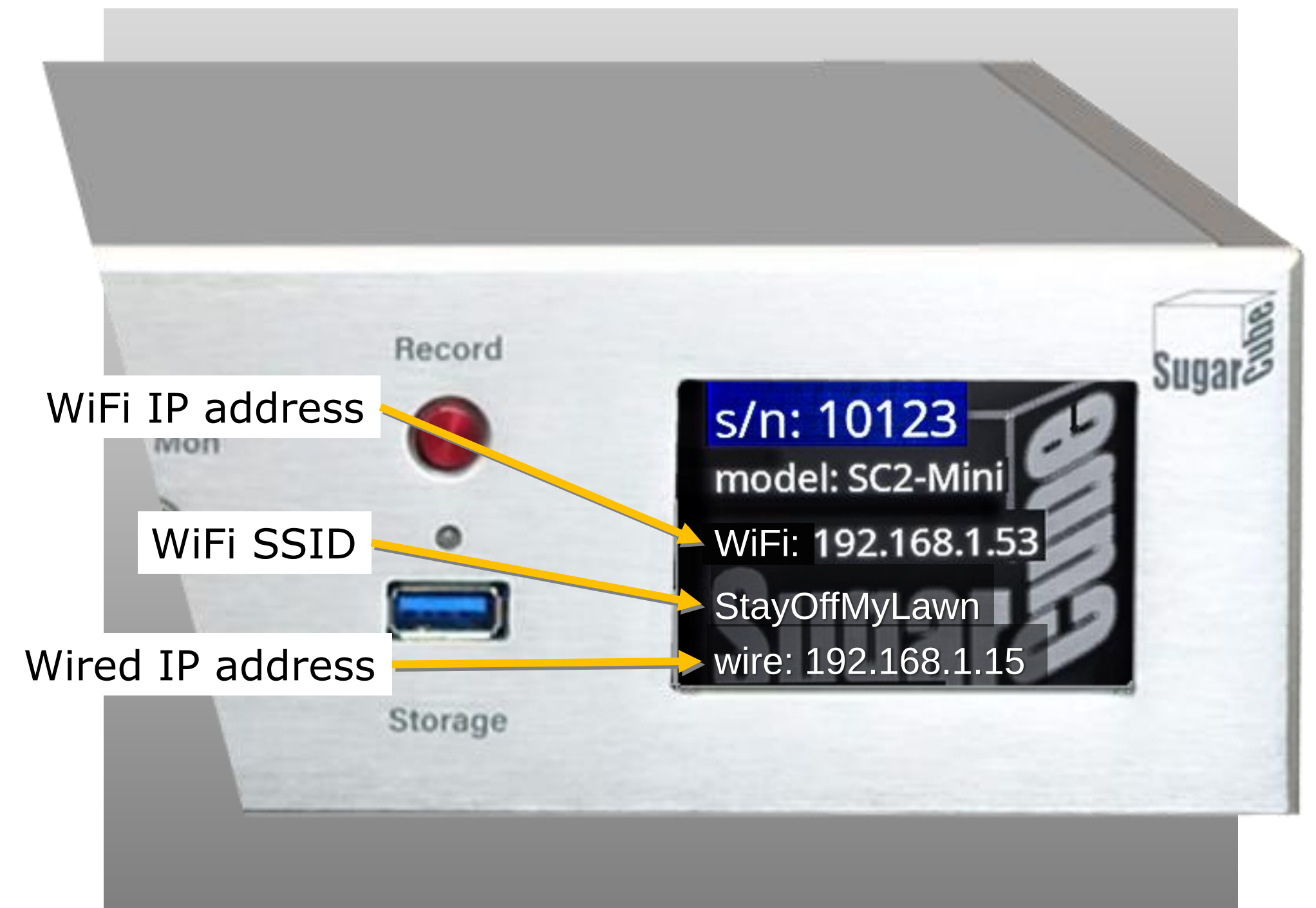
Info: connecting (getting IP configuration)

SSID sw **<your wifi>**

Password **<password>**

Site survey

	SSID	Signal	Channel	Security	Access Points
Use	1qa2ws3ed	87%	6 40	WPA2	2
Use	219Guest	20%	11	WPA1 WPA2	1
Use	219SecureIoT	34%	6	WPA1 WPA2	1
Use	219SecureWireless	30%	6	WPA1 WPA2	1
Use	BlueCart Wifi	30%	6	WPA2	1
Use	BlueCart Wifi - Guest	50%	6	WPA2	1
Use	SmugMug	84%	40	WPA2	1
Use	SmugMugs	84%	40	WPA2	1

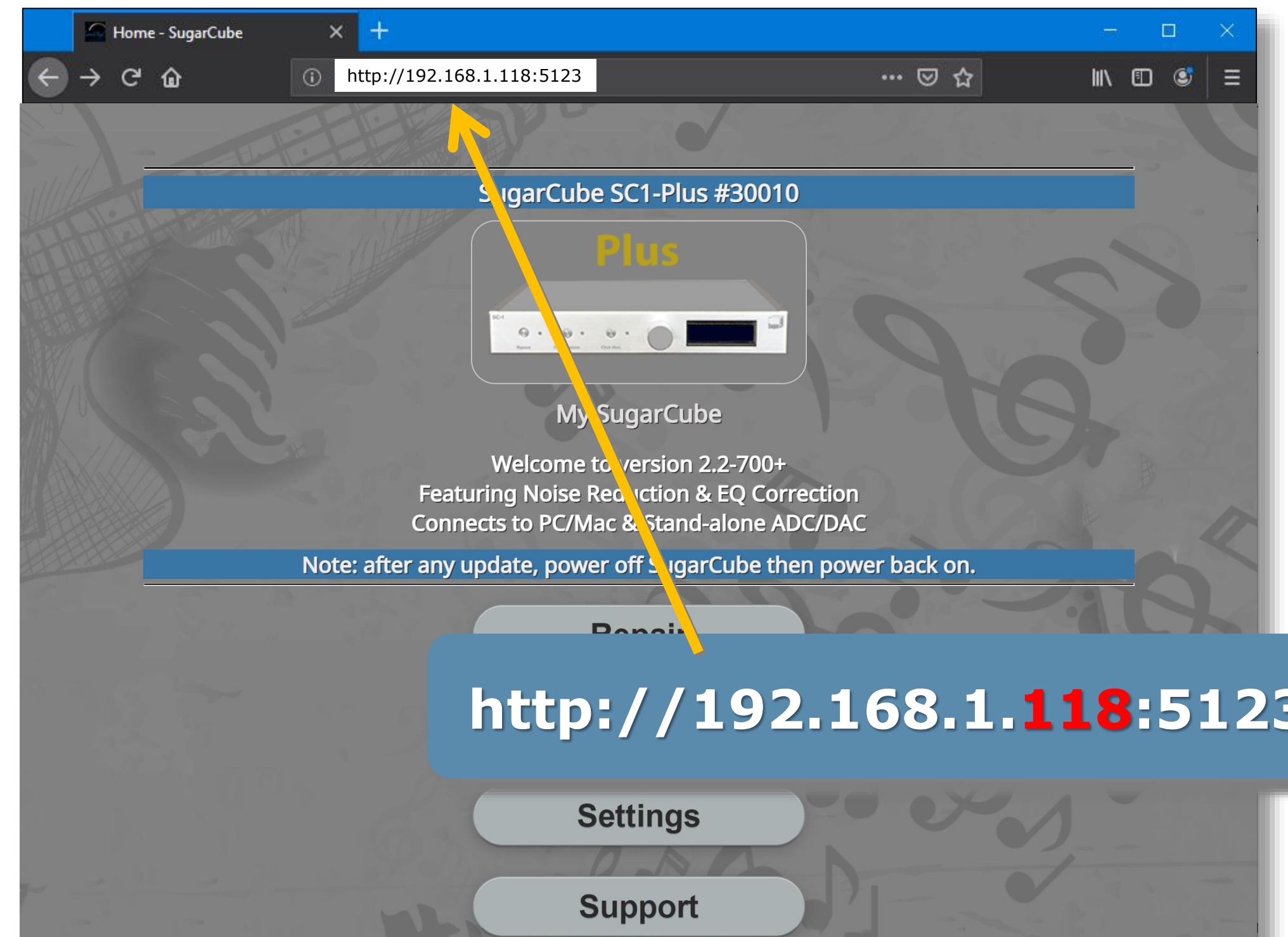


(6) The system will indicate that it is connecting to your WiFi.

(7) Once connected, you will now see both your new WiFi address (w) and your currently wired Ethernet IP address (e). Remember, you are currently communicating with the SugarCube via the wired IP address.

3.2.8 Controlling the SugarCube from a Web Browser

Setting up WiFi (in place of wired Ethernet), 4 of 4



(9) So, when you disconnect the wire, your browser will no longer communicate. You will have to change the IP address in your browser, in this example, to the WiFi address ending in '.118'. (Once the wire is disconnected, the front panel display will show only the WiFi IP address.)

(10) Usually, you need only to change the last group of numbers and press 'Enter', and you should reconnect (sometimes you may have to re-enter the entire address).

(11) Please note, if your SugarCube cannot connect to your WiFi, contact support@sweetvinyl.com with the details of your system (WiFi make and model and the serial number of your SugarCube.)

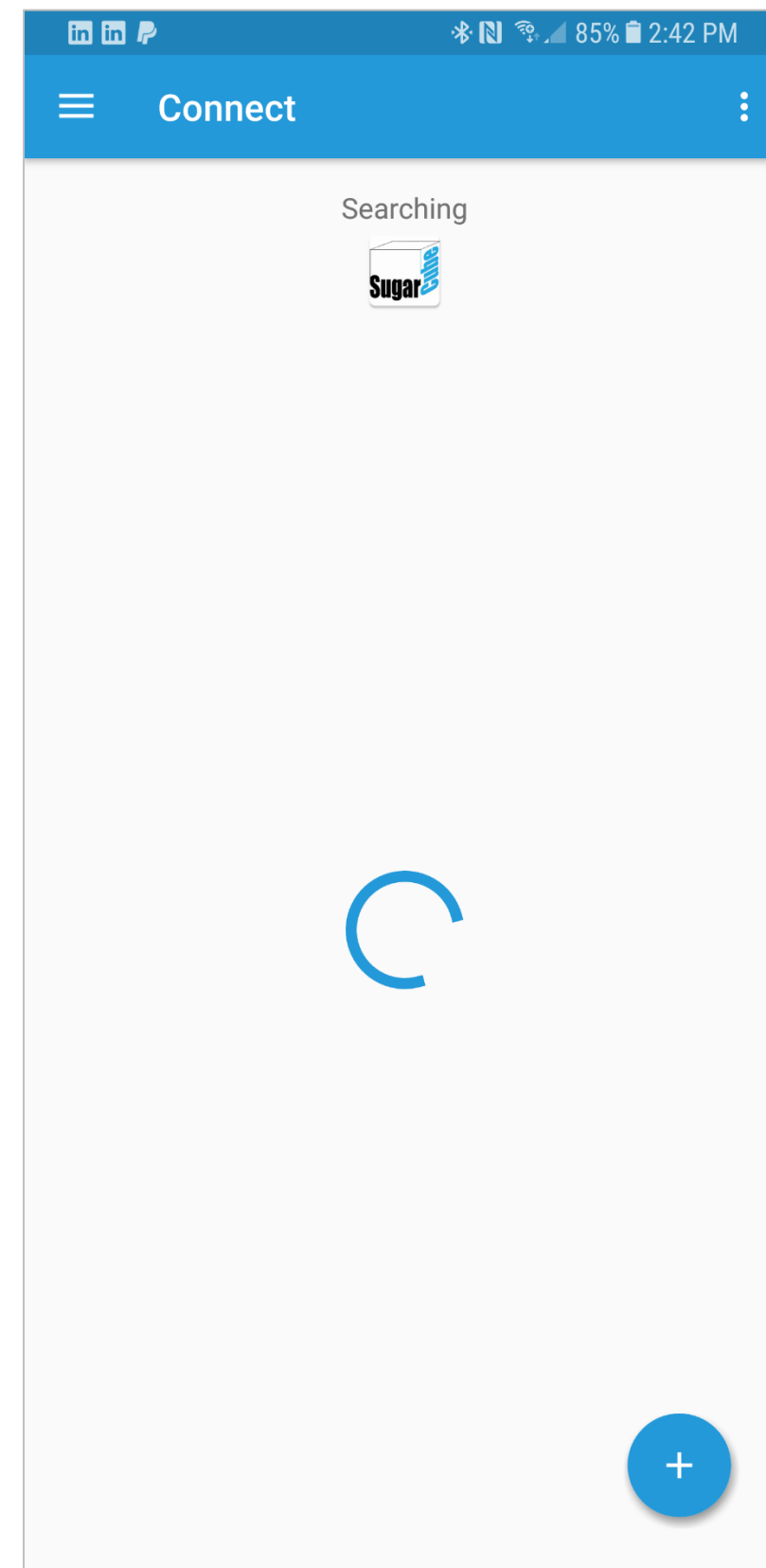


3.3 Using **iOS App** with SugarCube

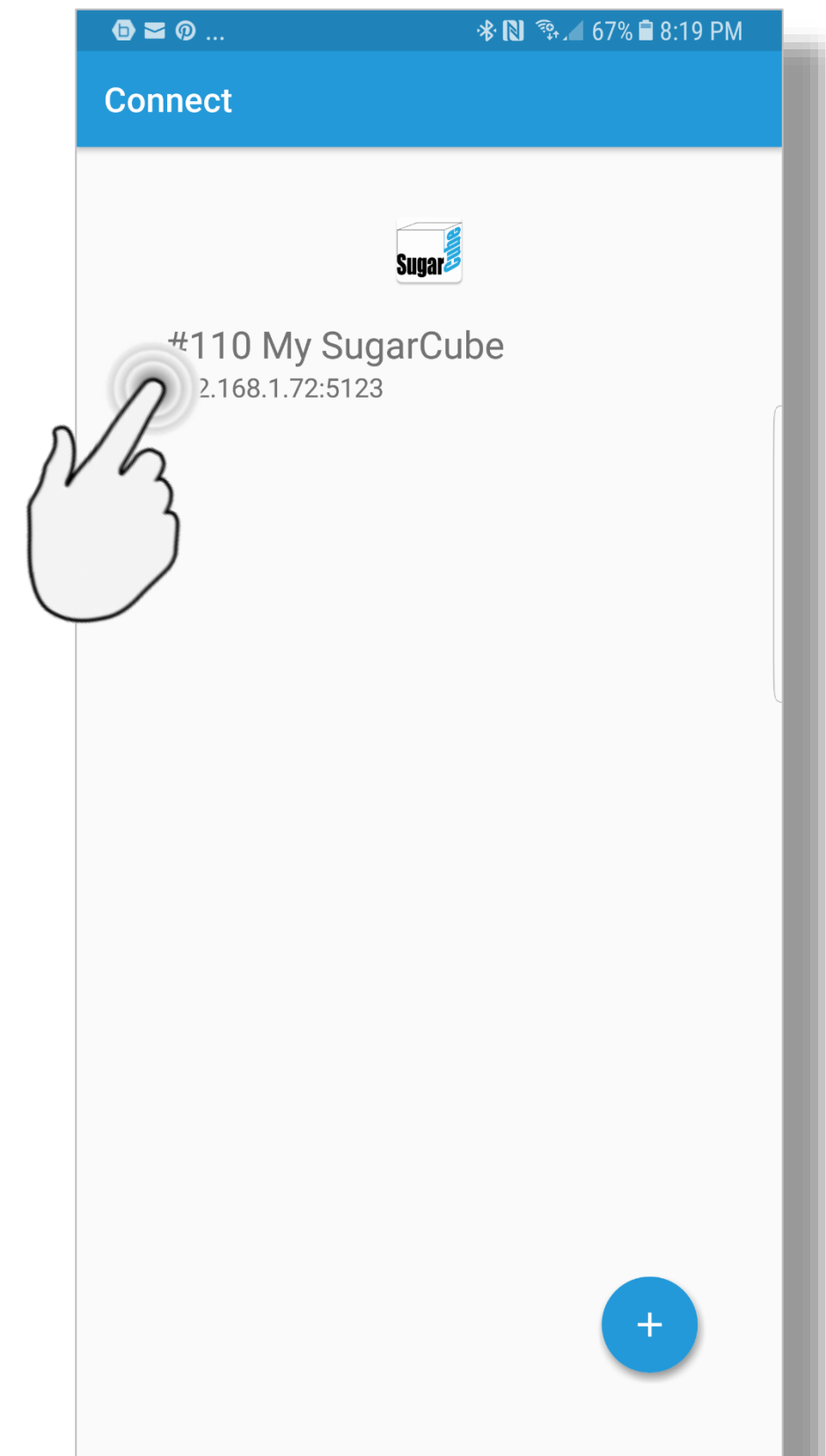
3.3.1 Connecting App with SugarCube on **Wired Ethernet**



(1) App is loading

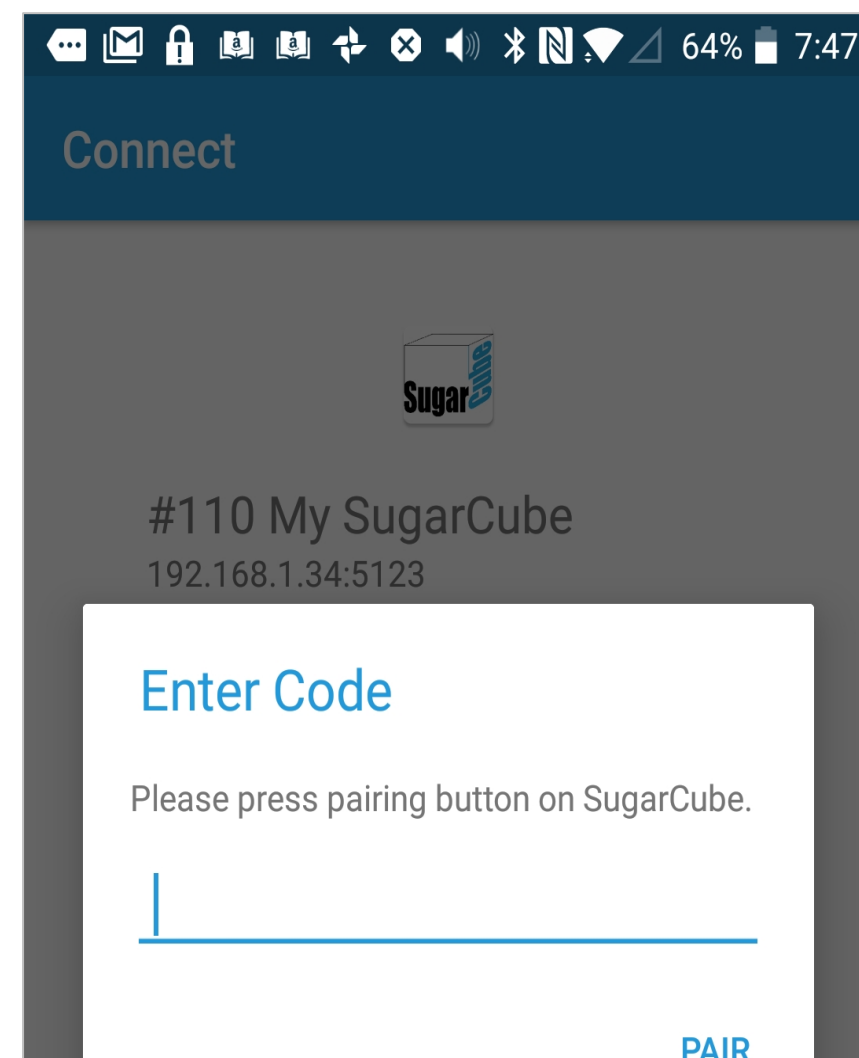


(2) App is searching for SugarCubes on the LAN



(3) App has found a SugarCube and will now initiate a connection.

3.3.2 Connecting App with SugarCube on Wired Ethernet

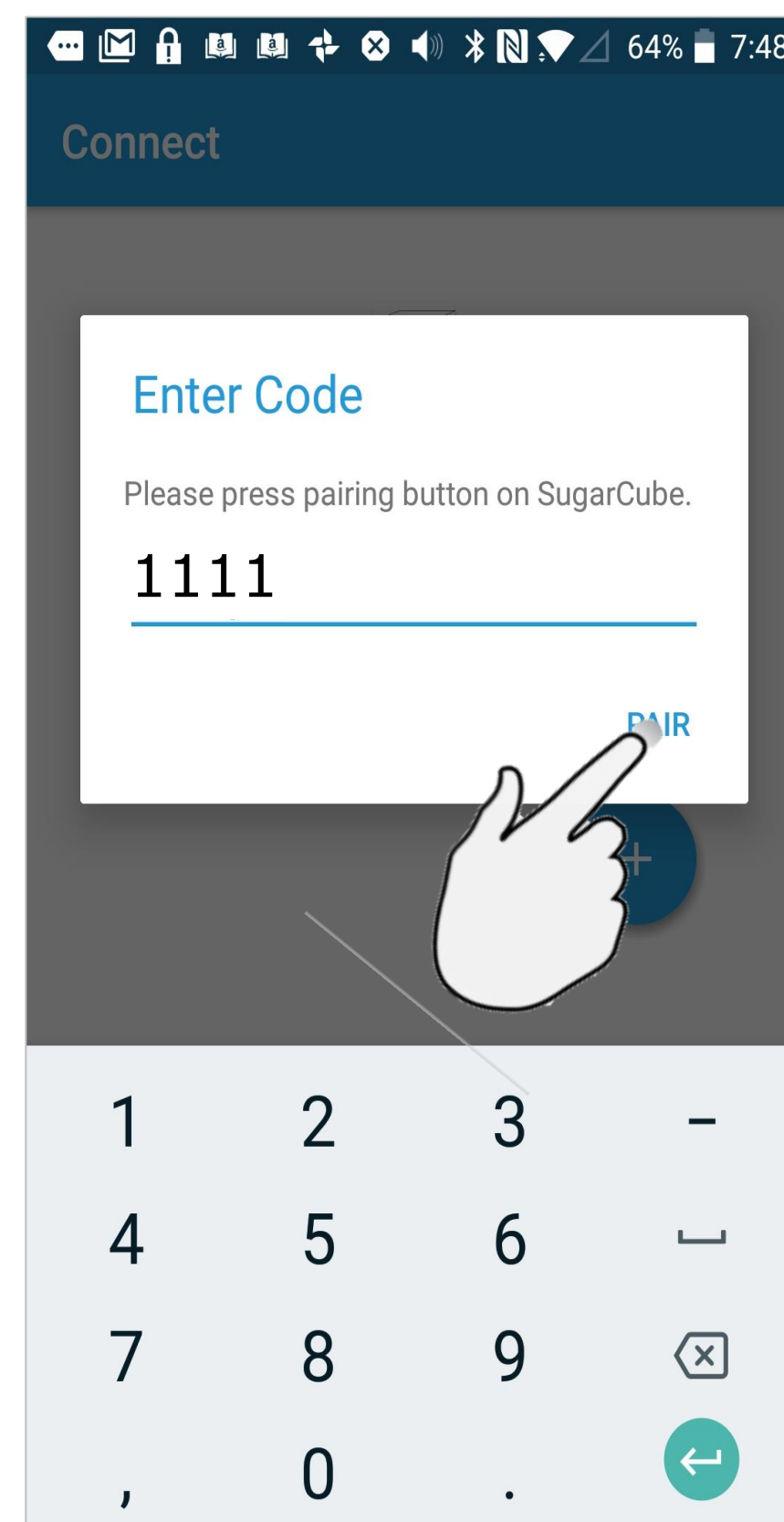


For pairing, enter:

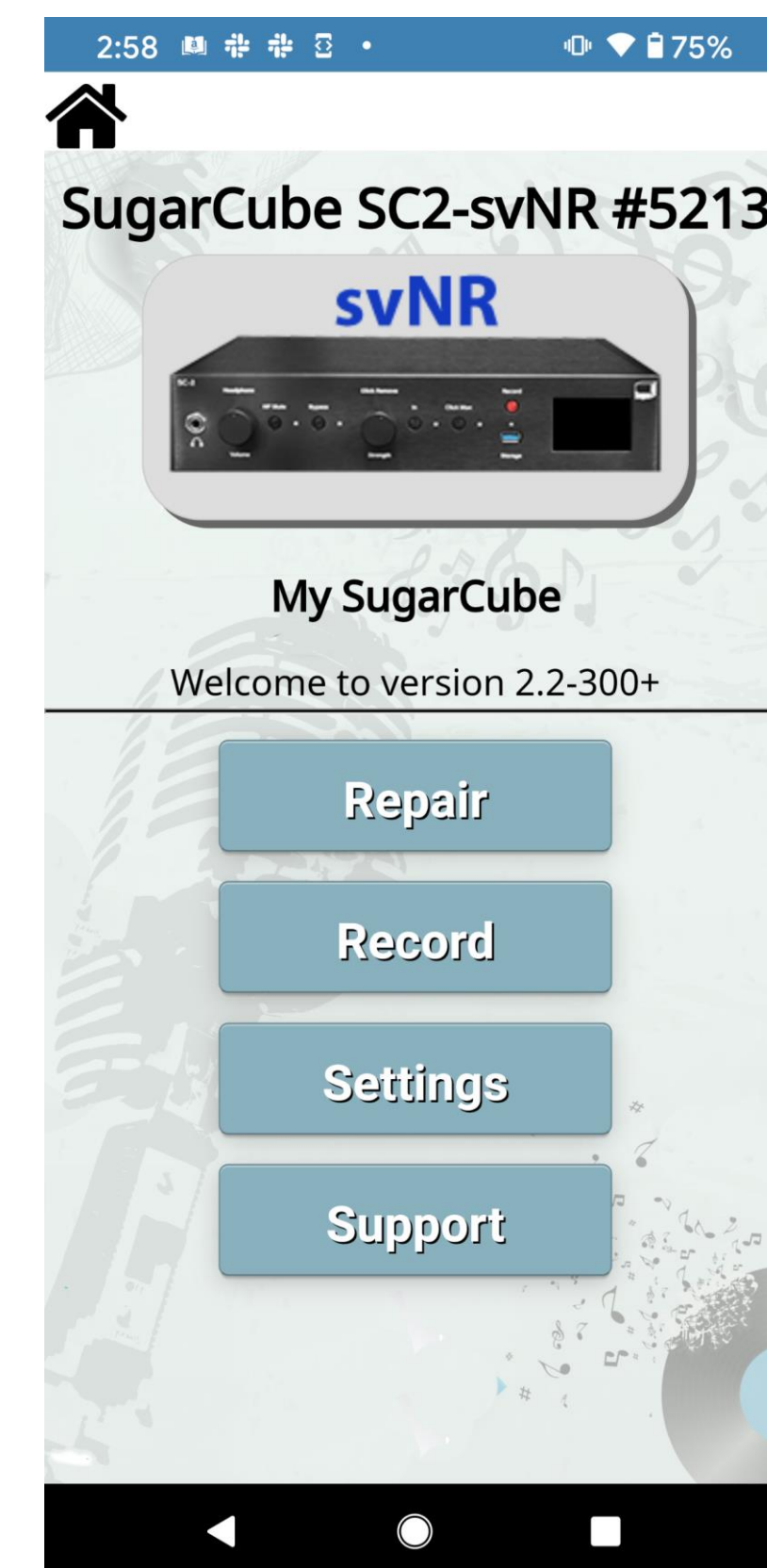
1 1 1 1

as a pairing code
(you may still get a code
from your SugarCube,
if you wish).

(4) The app needs you to enter a pairing code, press the knob on the front panel.

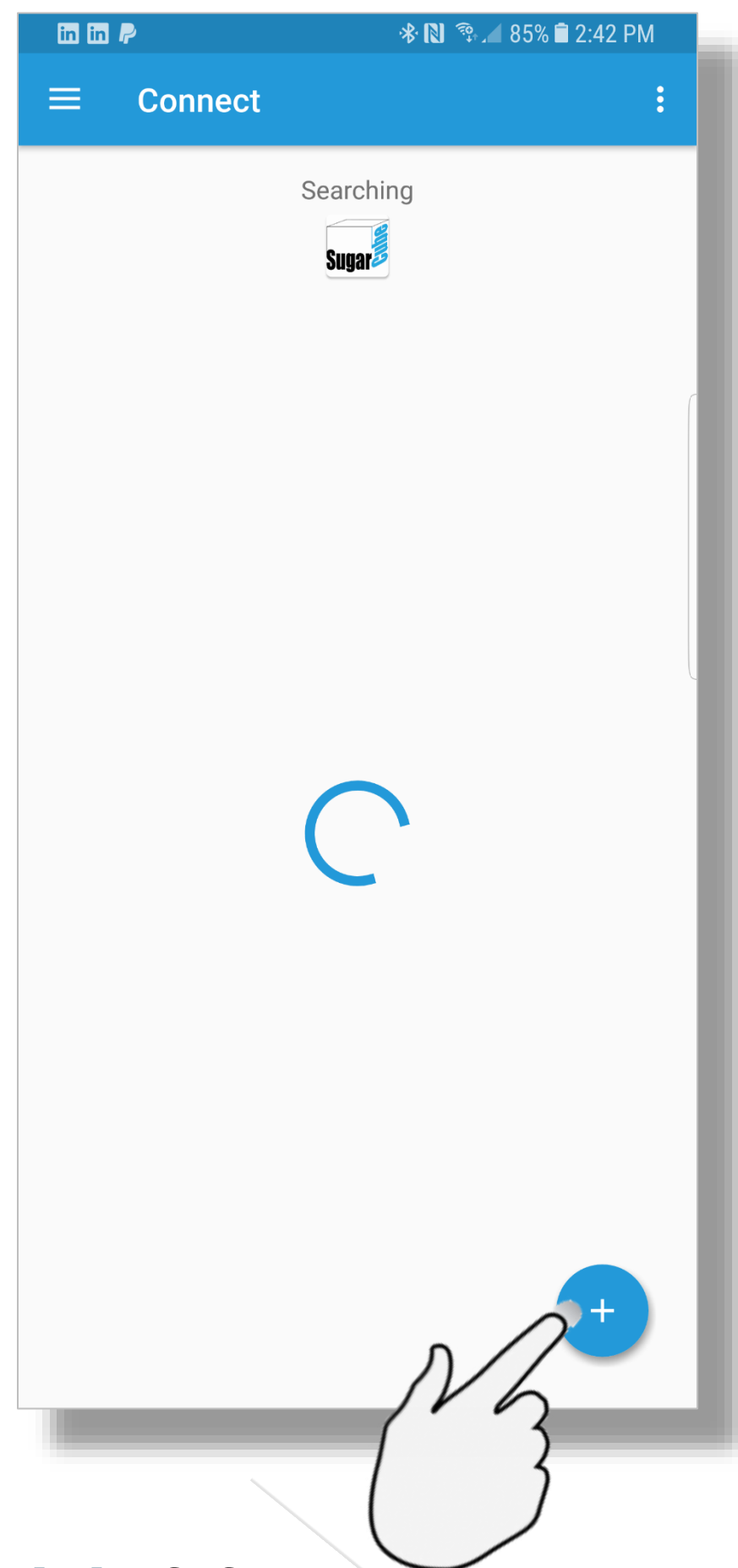


(5) App needs you to enter the pair code '**1111**' or what is displayed on the front panel display then touch '**PAIR**'.

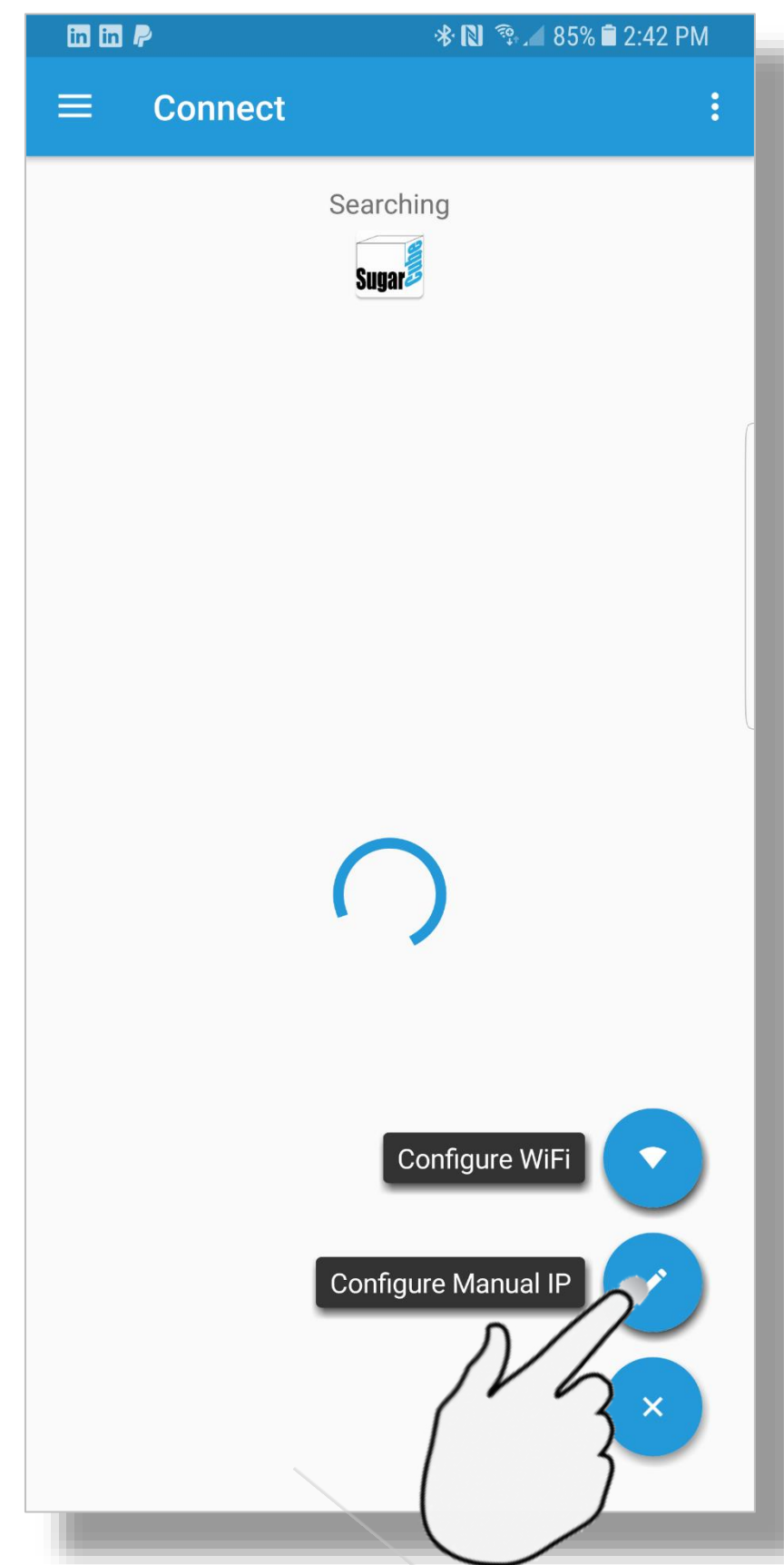


(6) App is linked to SugarCube and ready for use.

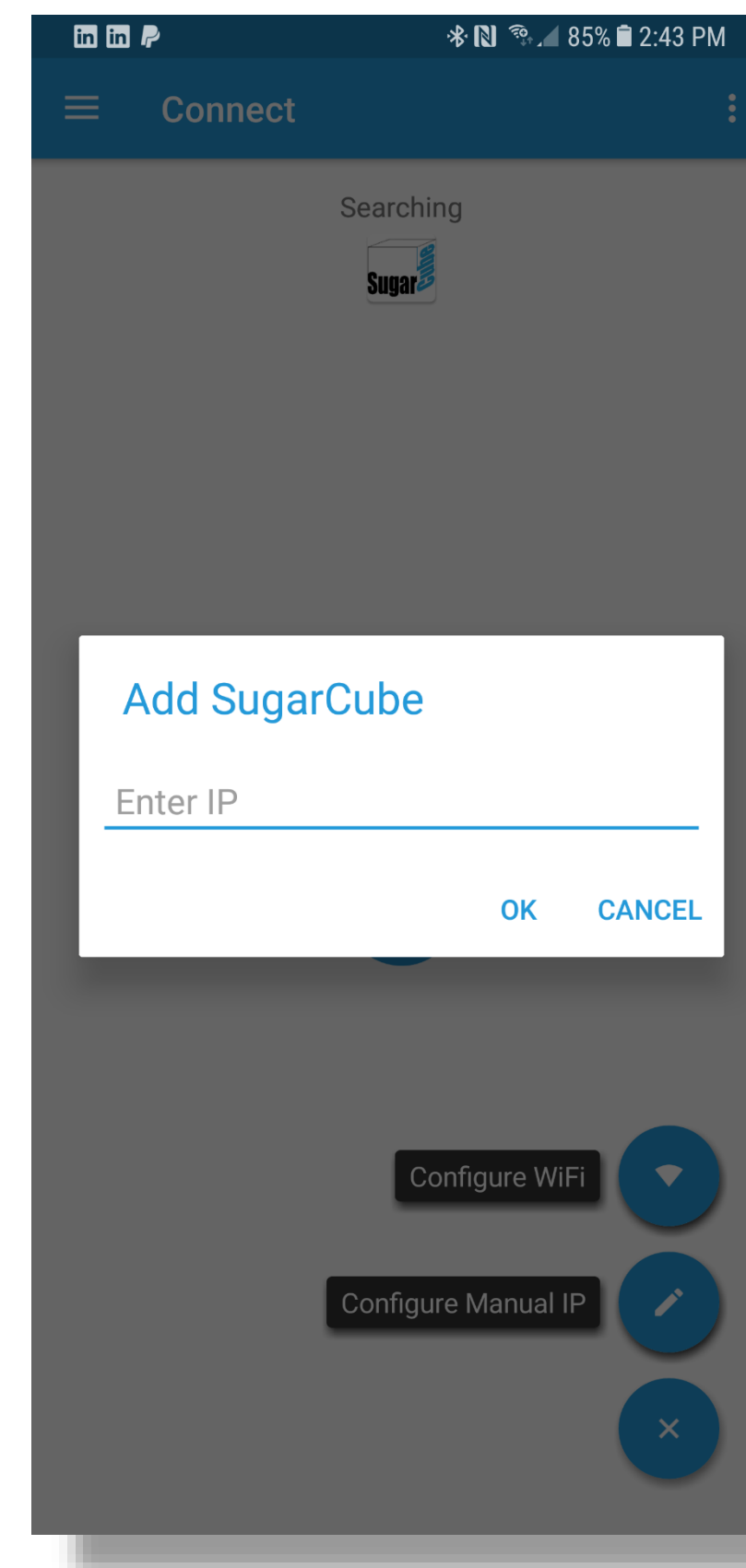
3.3.3 Connecting App with SugarCube Using IP Address



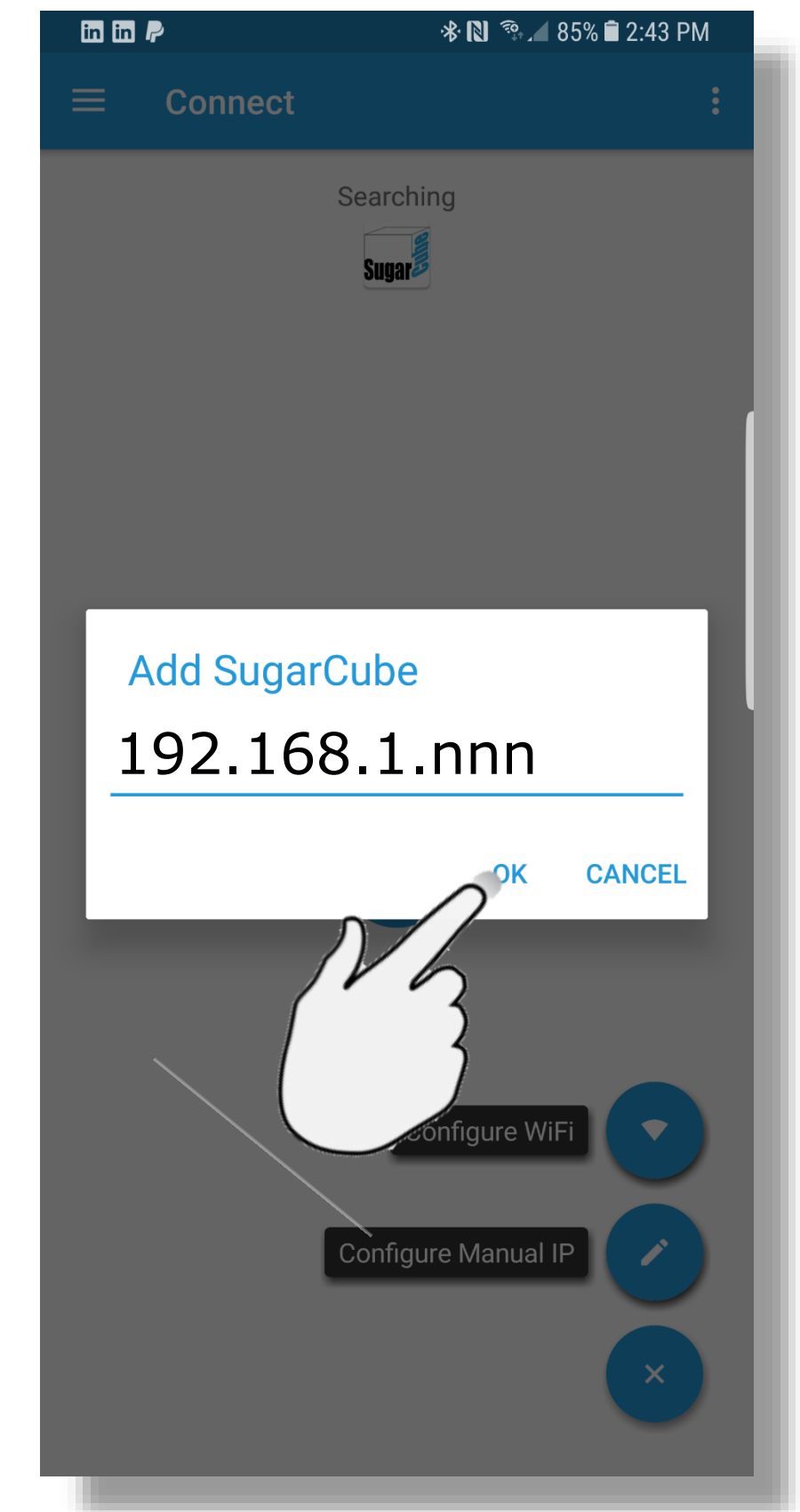
(1) If, for any reason, you wish to manually connect to your SugarCube touch the menu button in the lower right of the screen.



(2) Select 'Configure Manual IP'.

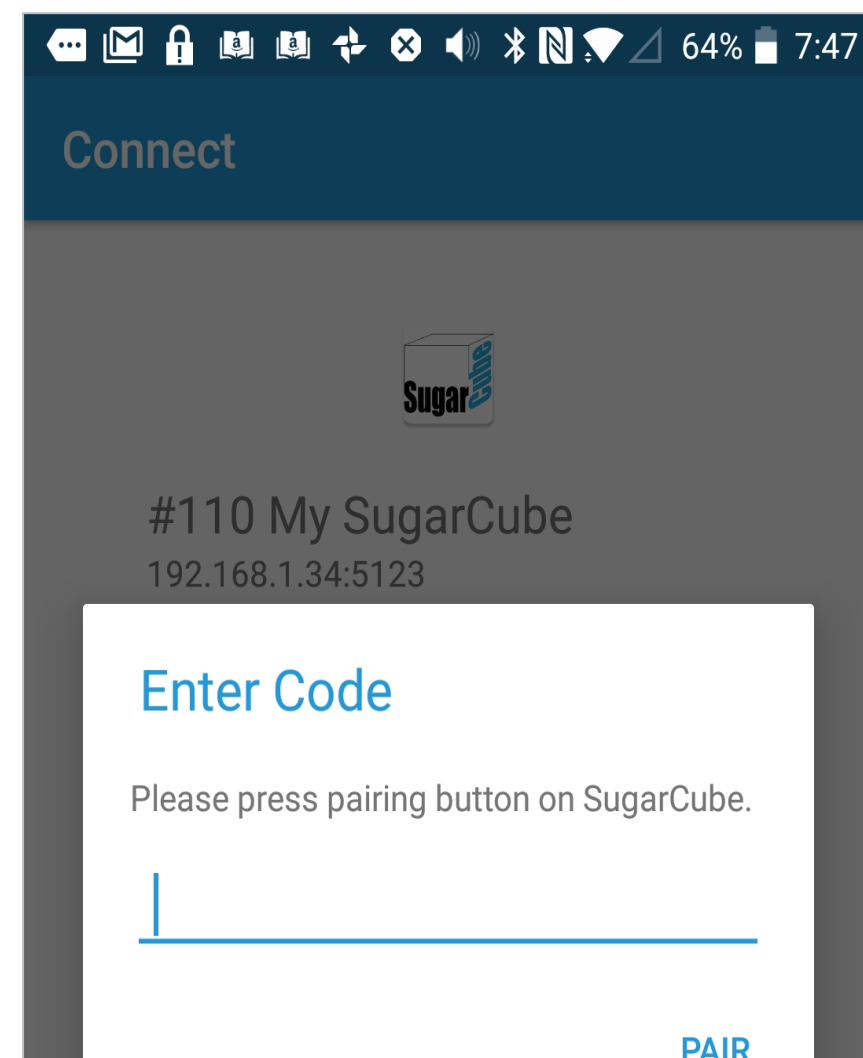


(3) You will then see a window to enter the IP address of the SugarCube. This address can be seen by pressing and holding 'Bypass' for a few seconds.



(4) Enter the IP address as displayed on the front panel display.

3.3.4 Connecting App with SugarCube Using IP Address

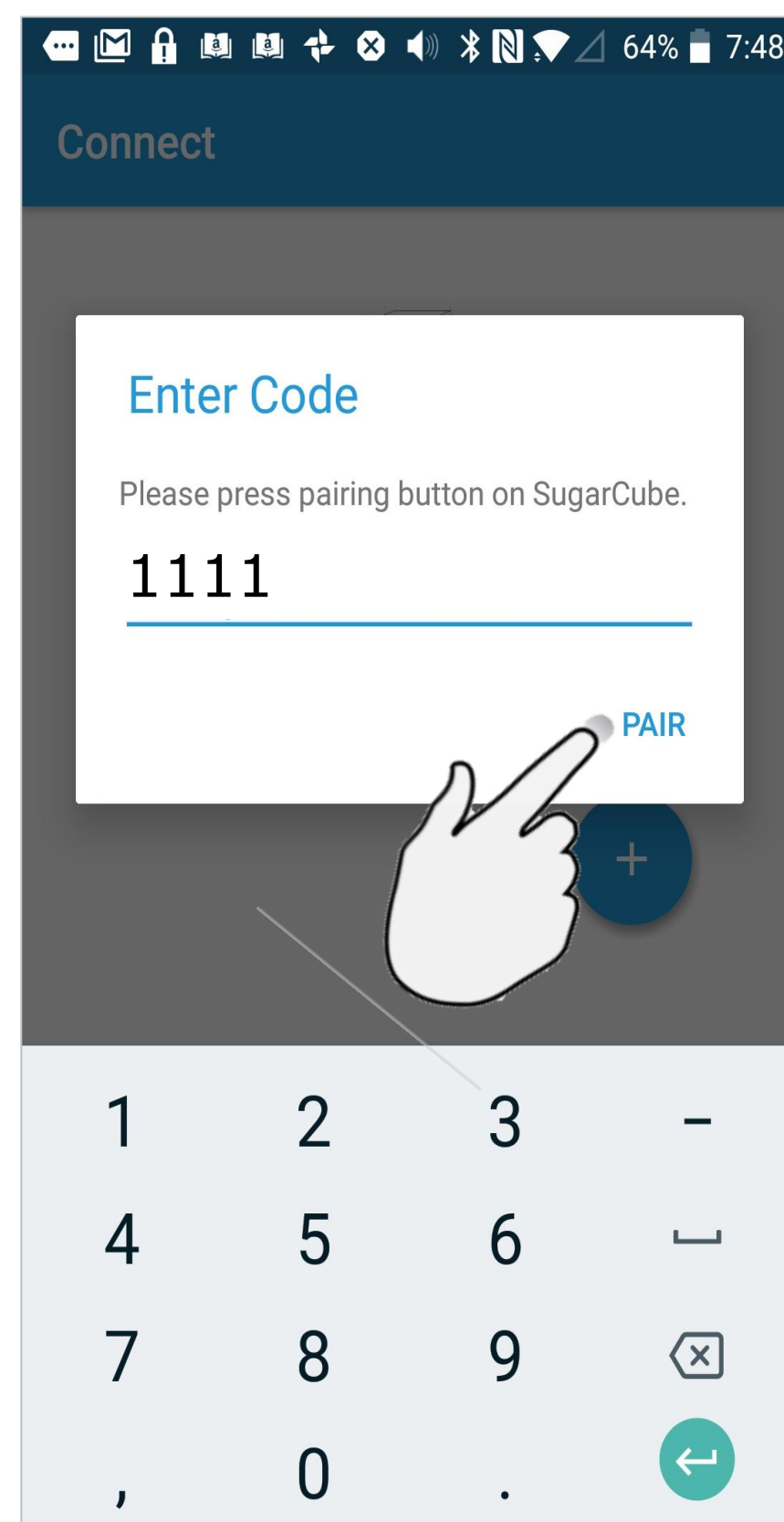


For pairing, enter:

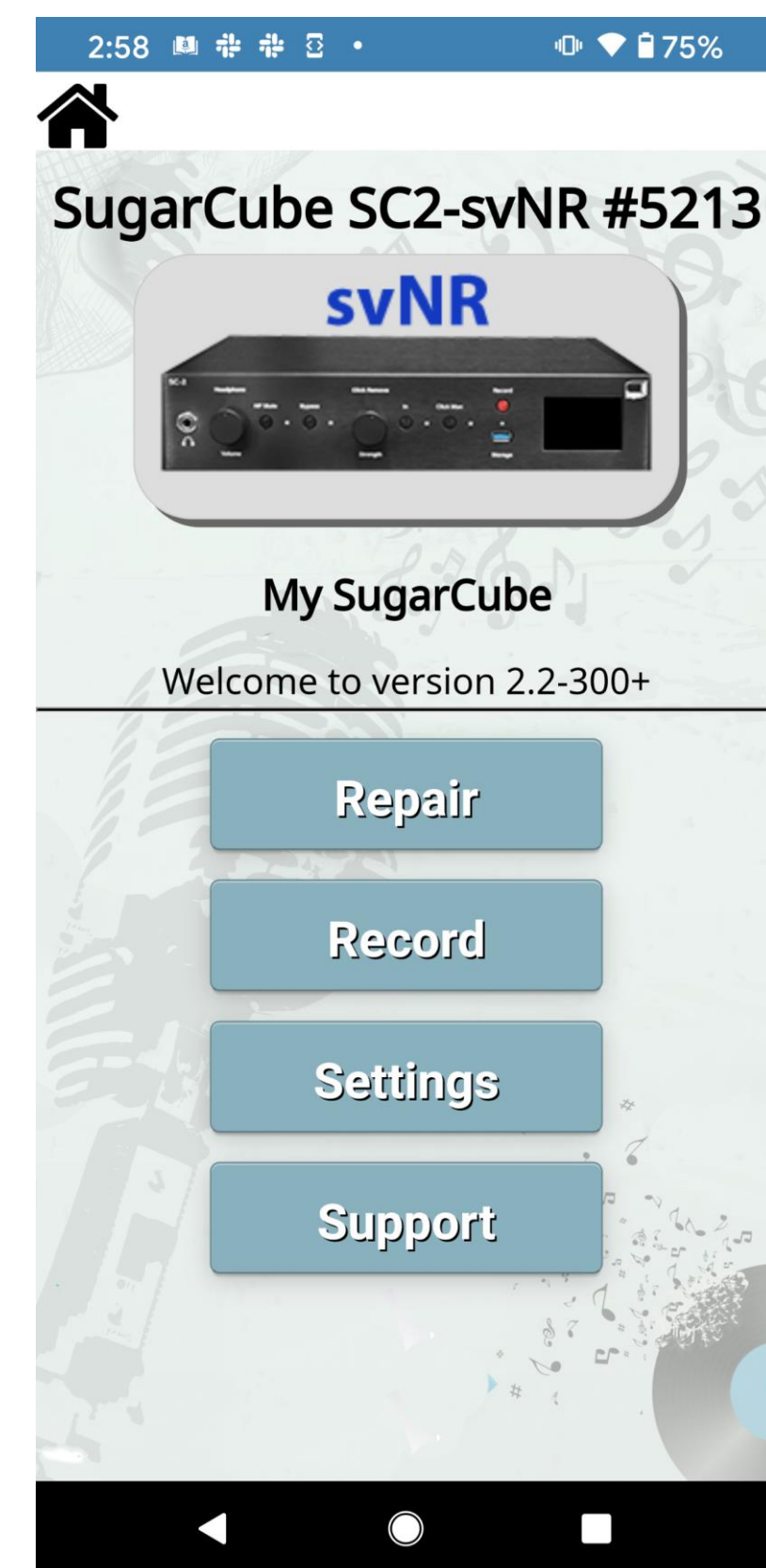
1 1 1 1

**as a pairing code
(you may still get a code
from your SugarCube,
if you wish).**

(5) The app need you to press the knob on the front panel.



(6) App needs you to enter the pair code displayed on the front panel display then touch 'PAIR'.

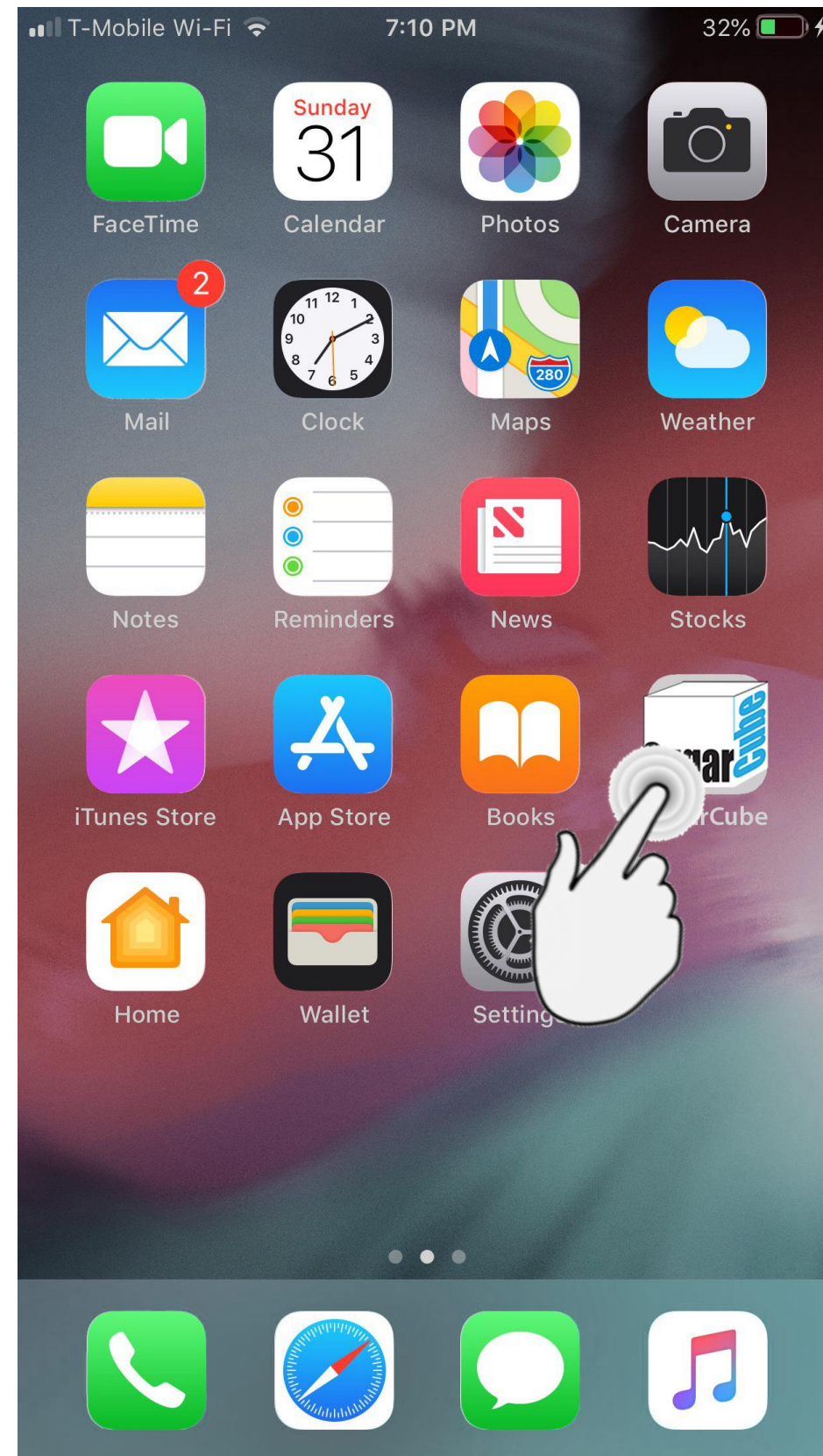


(7) App is linked to SugarCube and ready for use.

3.3.5 Setting up iOS device to control SugarCube with **WiFi**

1. iPhone, iPad, Touch.
2. Network with WiFi your Apple device can communicate with and that can be connected to your SugarCube with WiFi or wired Ethernet.
3. From the Apple App Store, you will be able to download the 'SugarCube' app to your Apple mobile device. (Look for: SugarCube by SweetVinyl.)
4. The WiFi dongle supplied with your SugarCube.

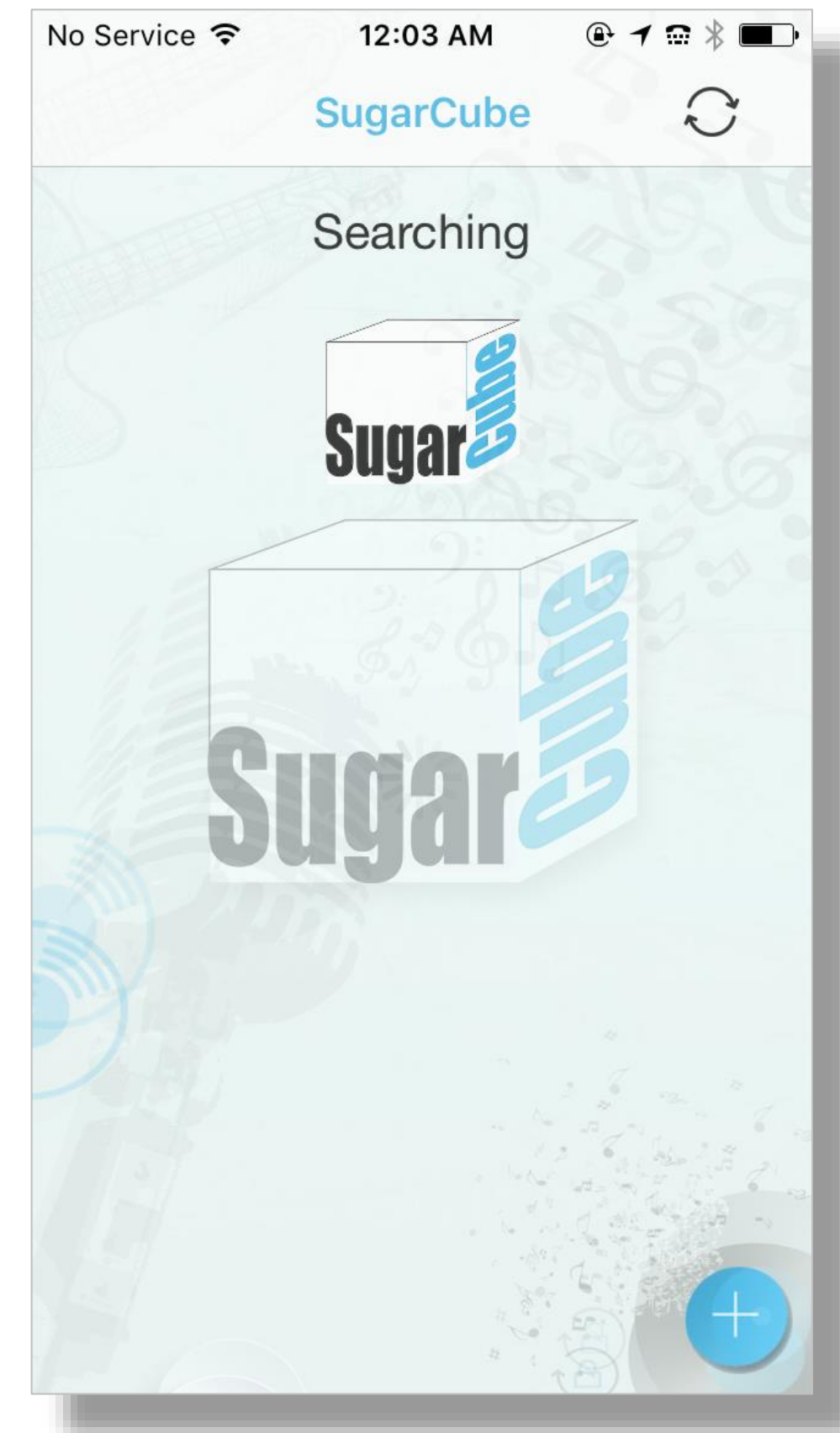
3.3.6 Connecting App with SugarCube on WiFi



(1) Launch the SugarCube app

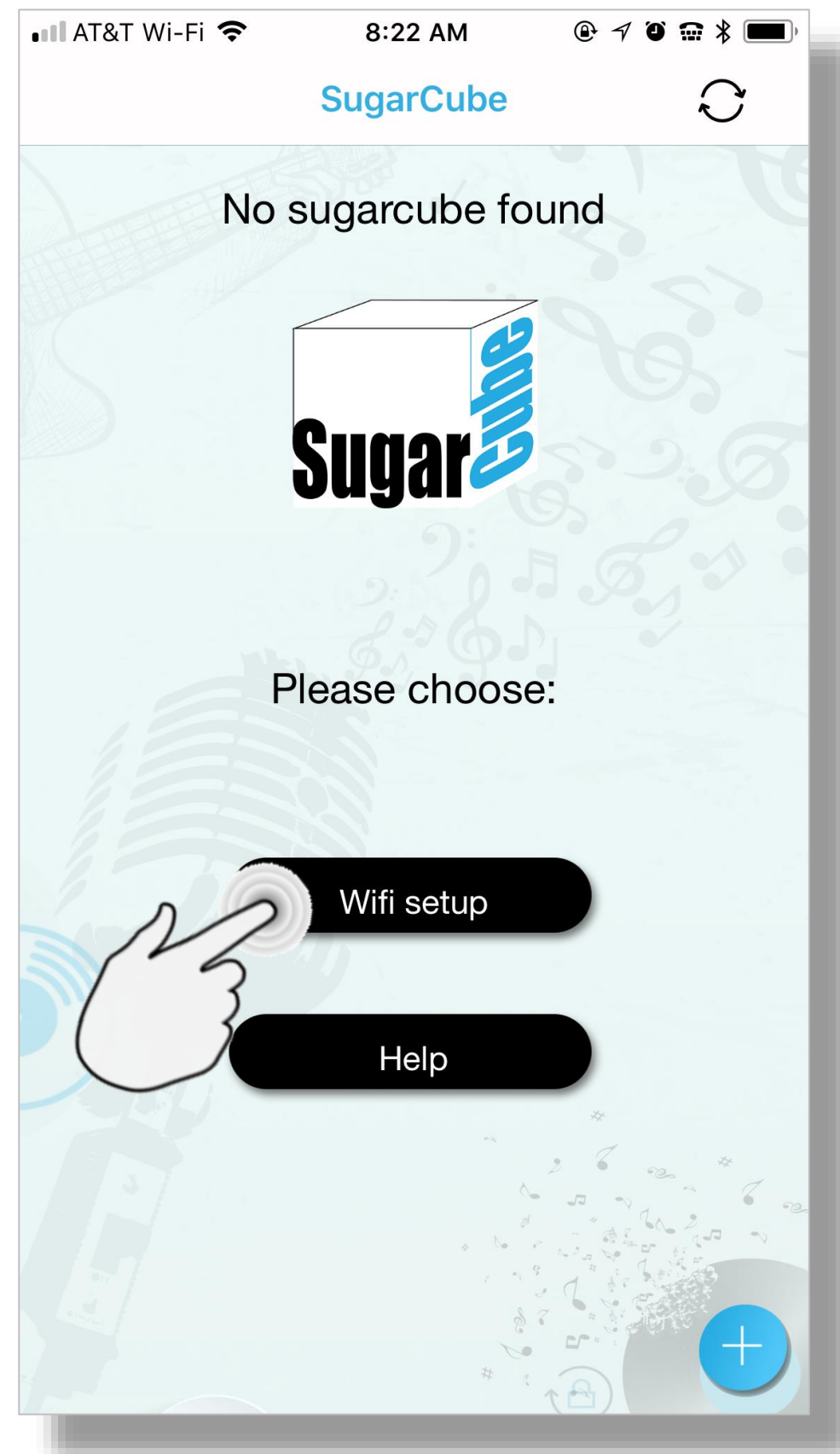


(2) App is loading

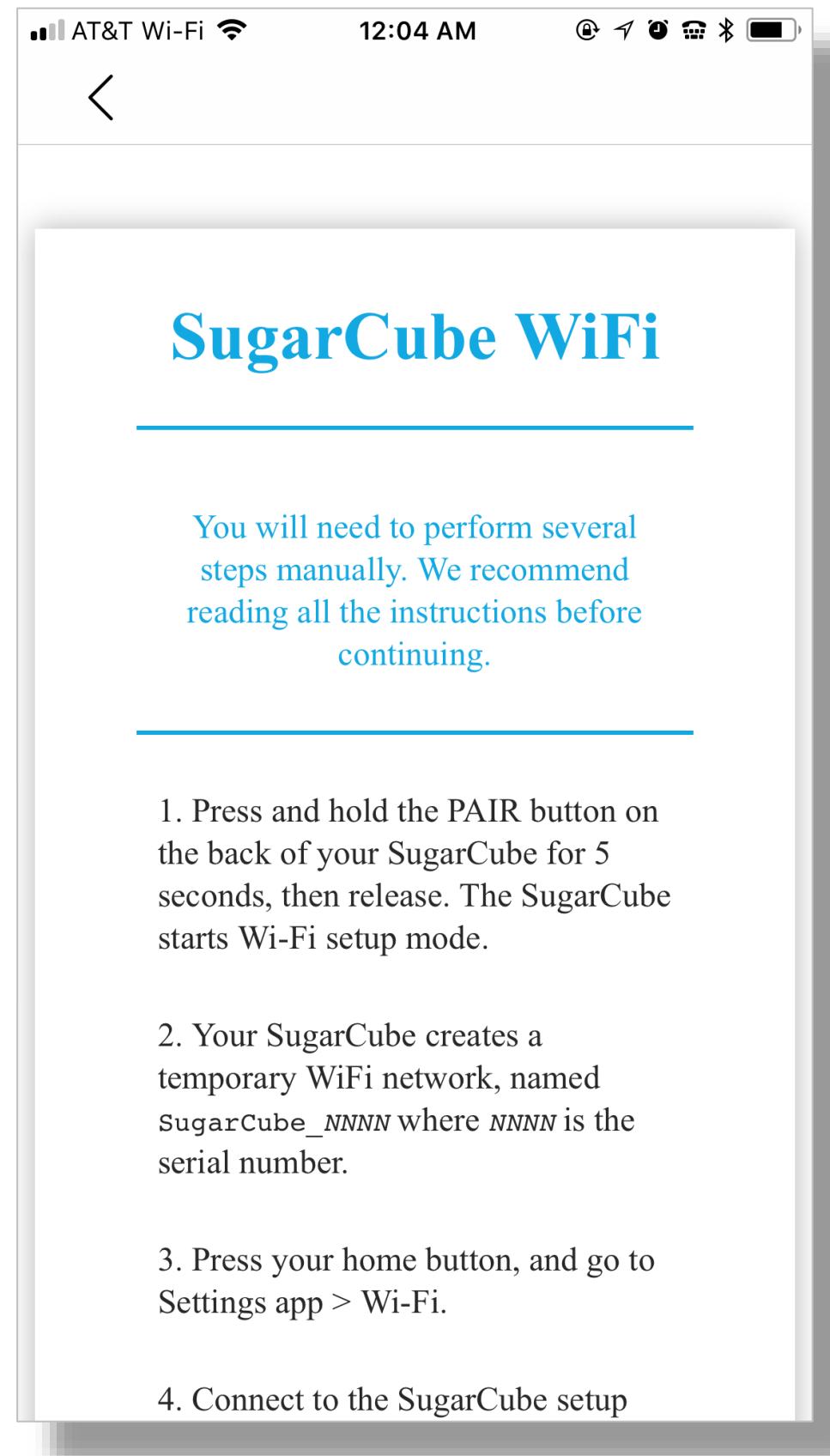


(3) App is searching for SugarCubes already setup on the LAN if any. For the first time user, nothing will be found, of course.

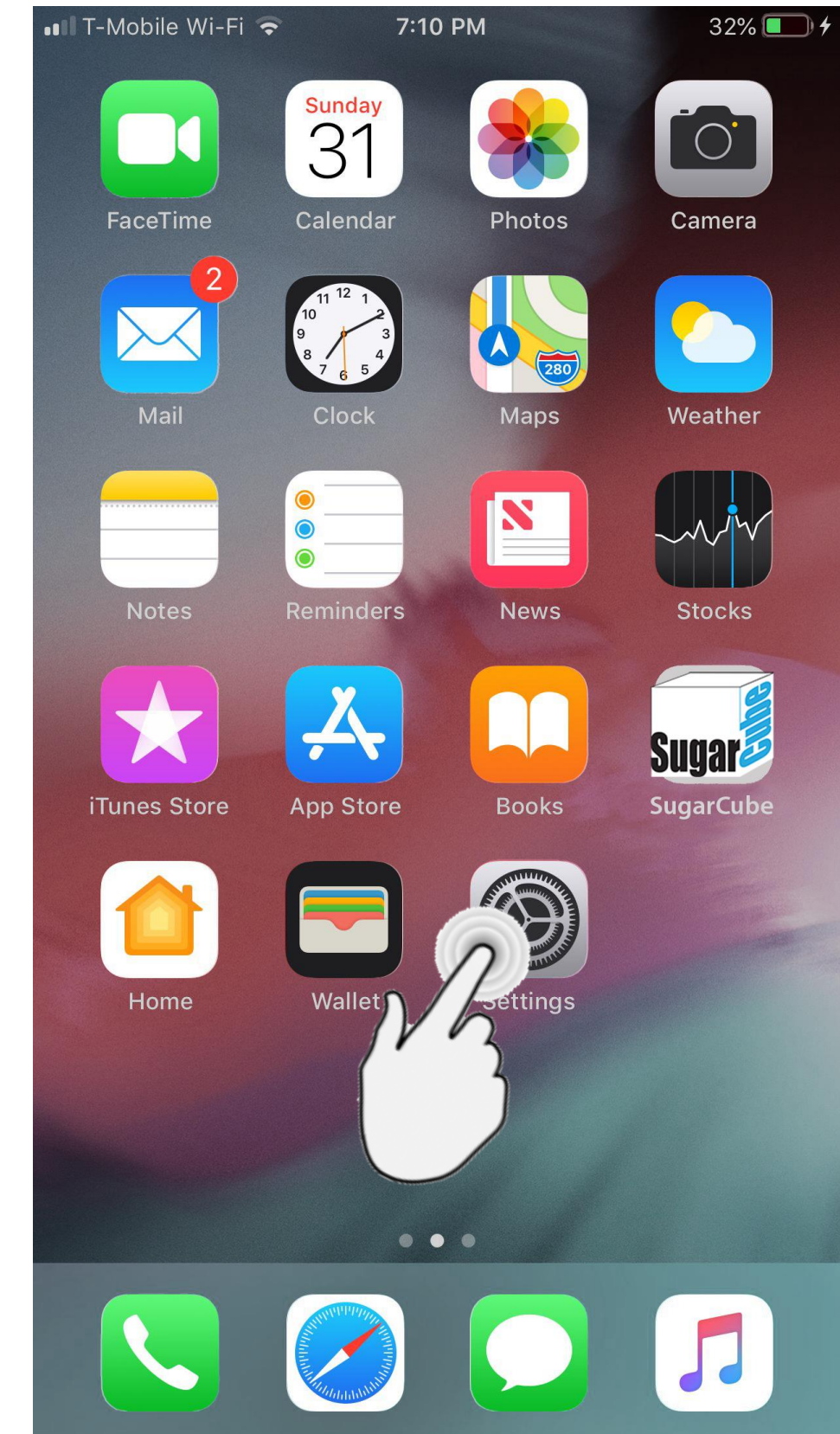
3.3.7 Connecting App with SugarCube on WiFi



(4) No SugarCube is found so the app offers the option to run WiFi setup.

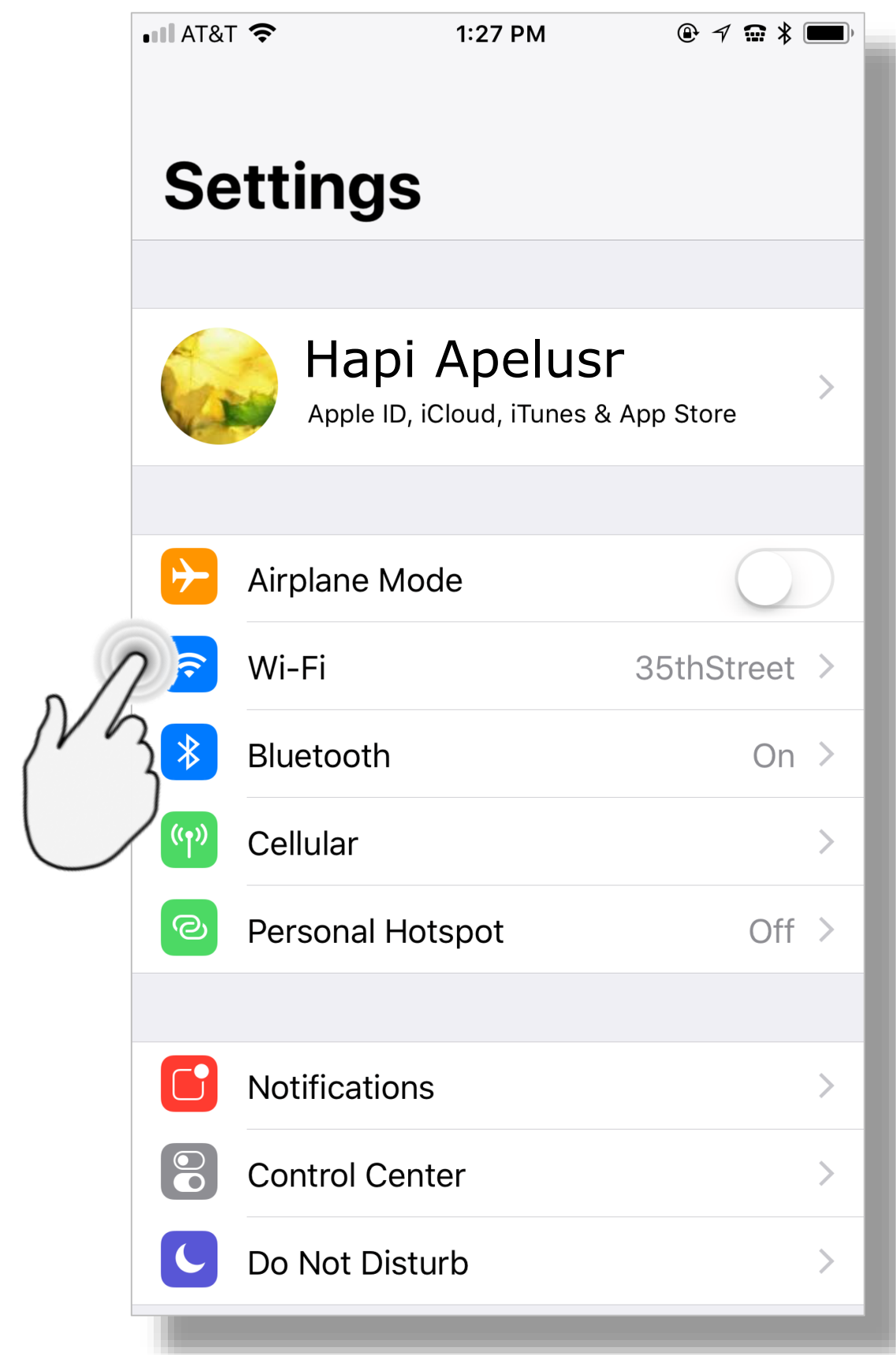


(5) The user is instructed to press and hold the front panel knob for 5 seconds to put the SugarCube into 'WiFi Setup Mode' which will be displayed on the front panel.

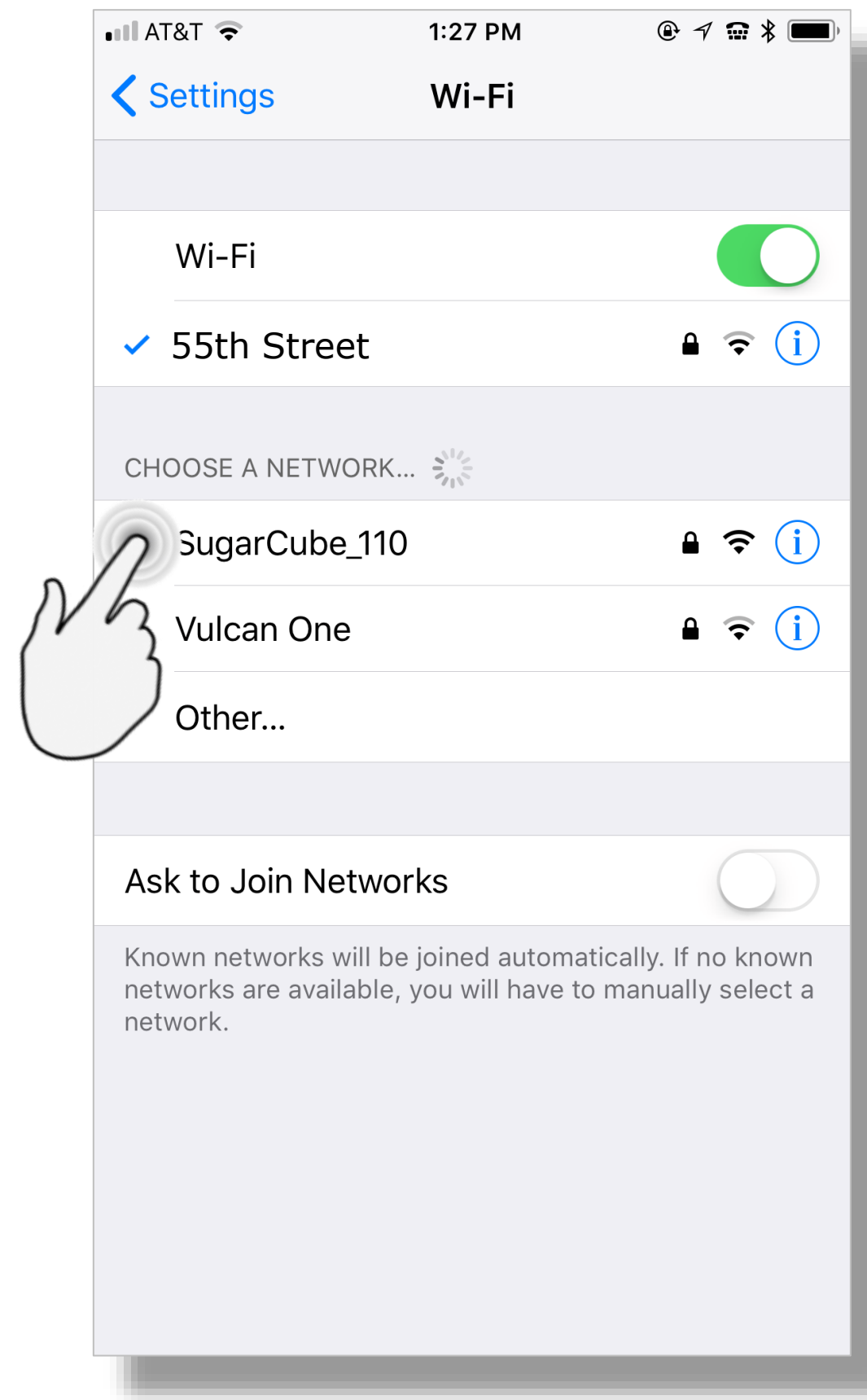


(6) The instructions direct the user to press the 'Home' key to get back to the 'Home' screen and then launch 'Setup'.

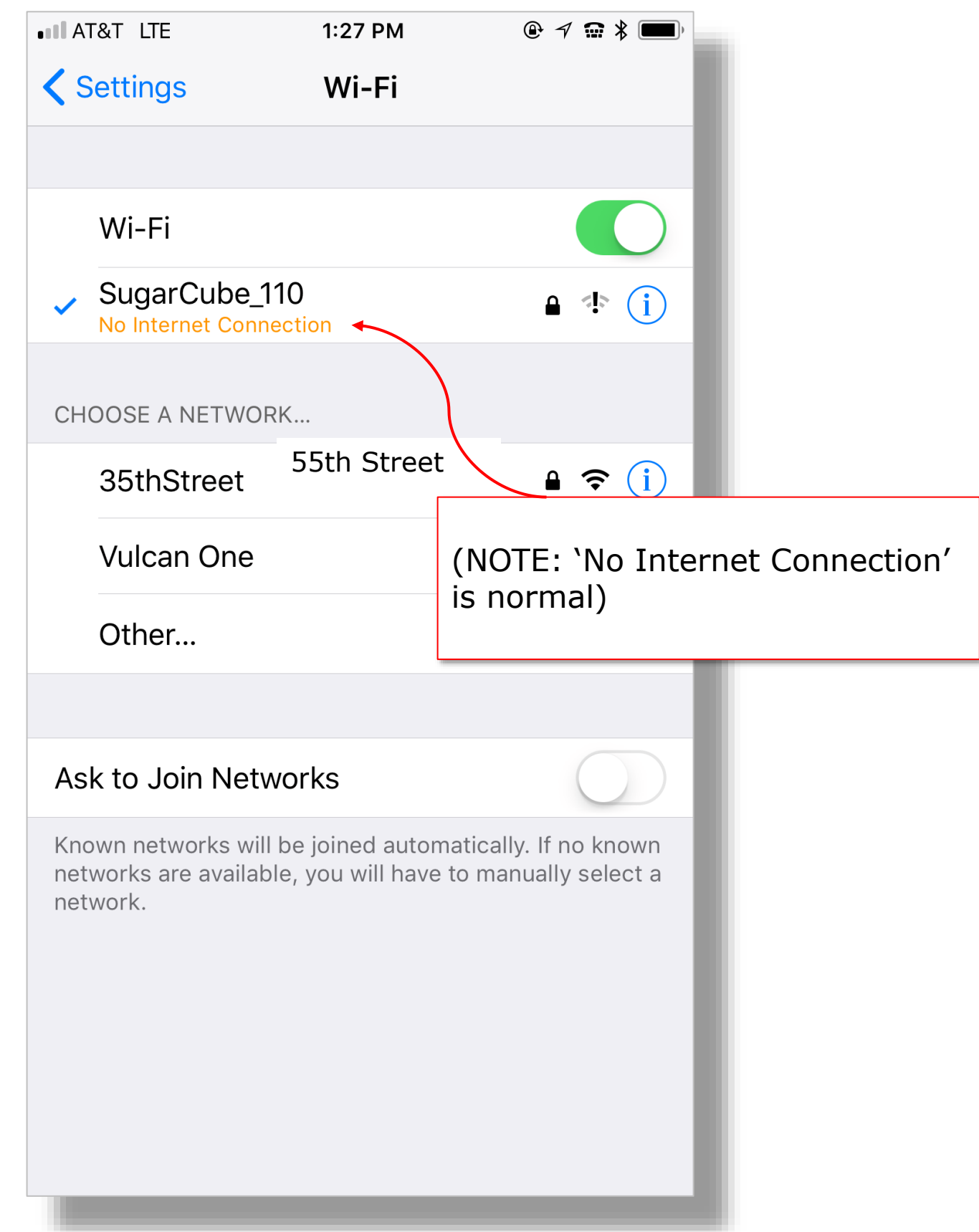
3.3.8 Connecting App with SugarCube on WiFi



(7) When in 'Settings', select 'Wi-Fi'

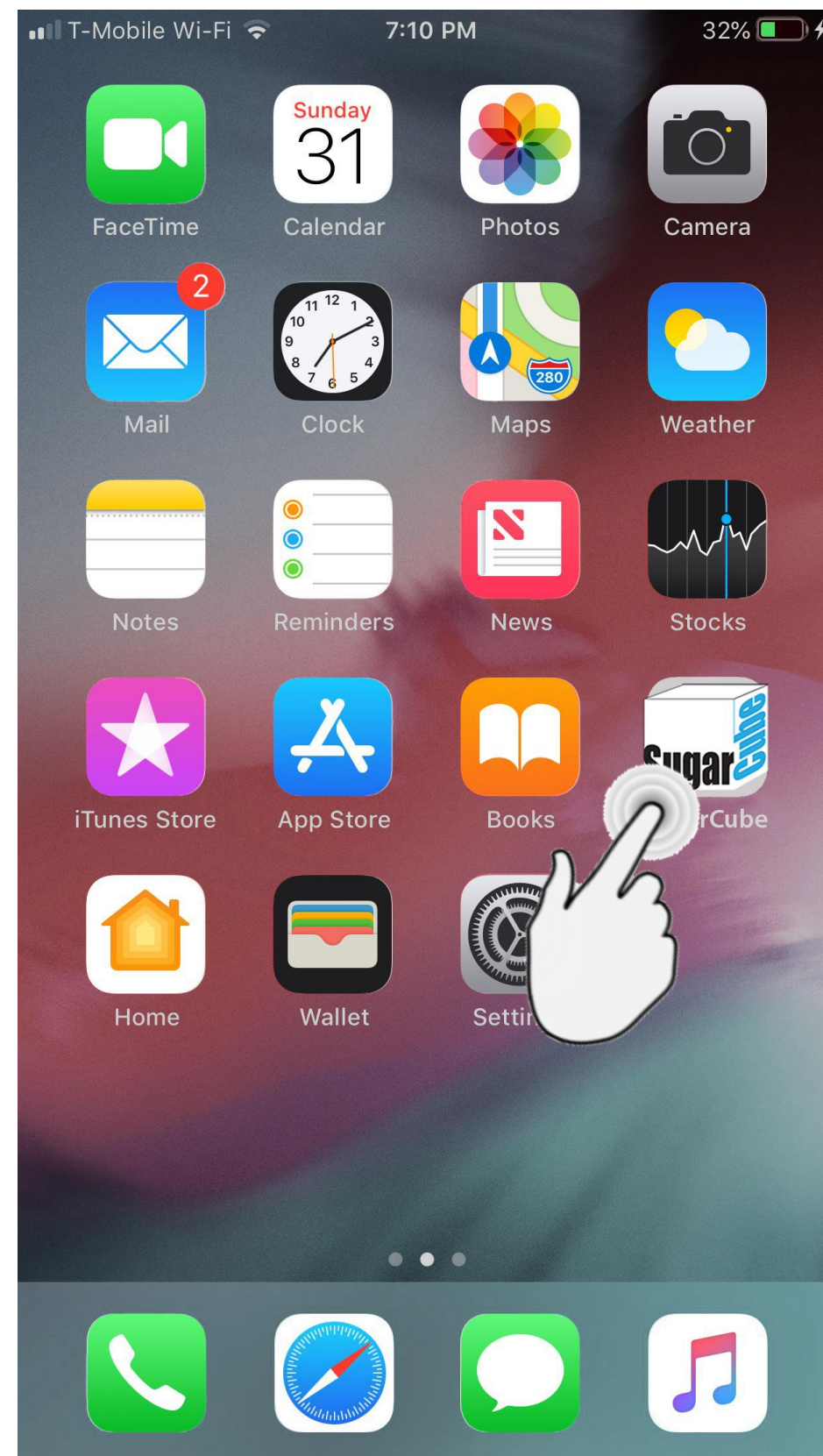


(8) Select 'SugarCube_nnnn', (whatever number appears after 'SugarCube_' is your device's serial number)

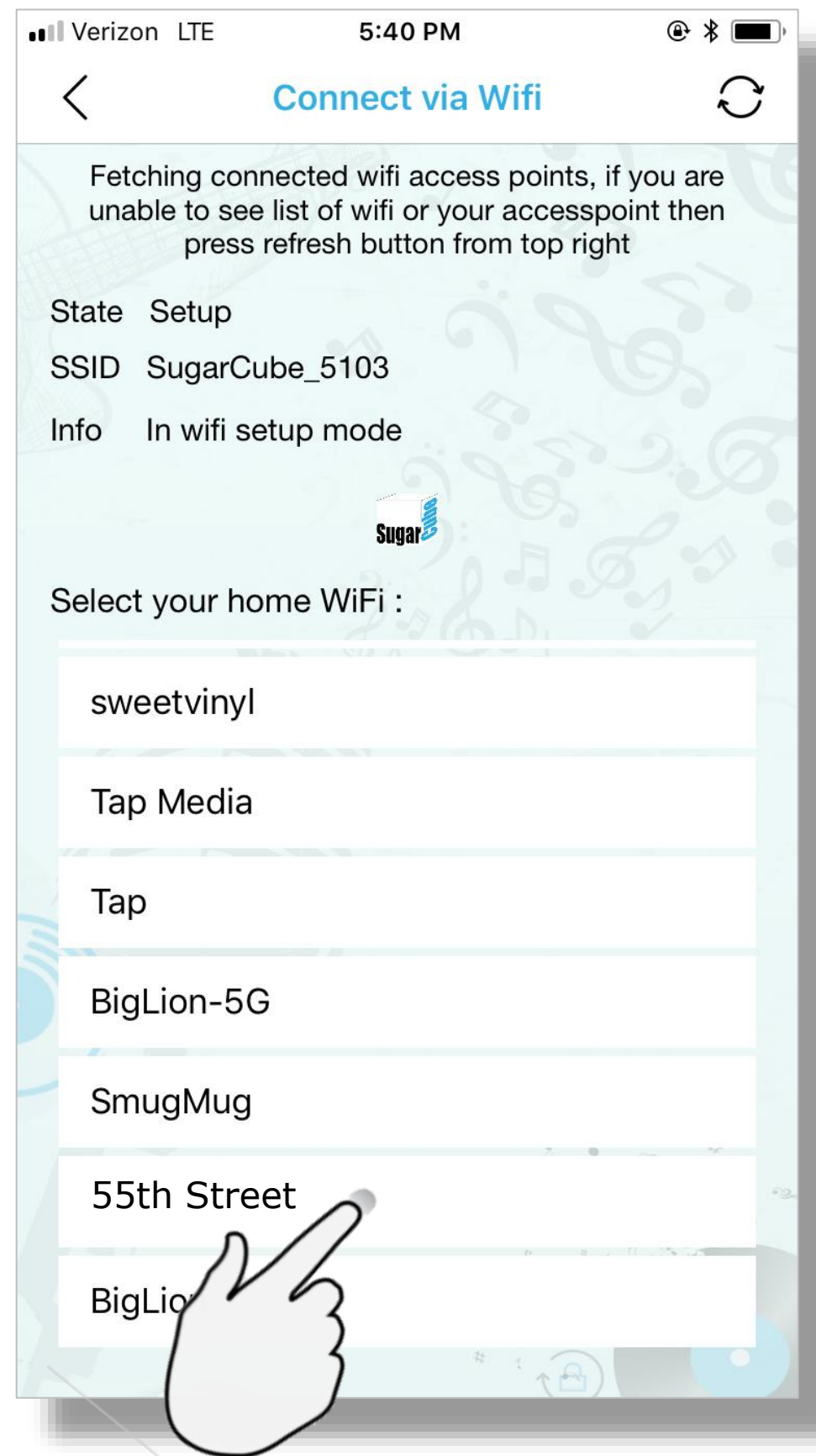


(9) With the 'SugarCube_nnnn' selected, your SugarCube is now temporarily acting as an access point.

3.3.9 Connecting App with SugarCube on WiFi



(10) Relaunch the SugarCube app

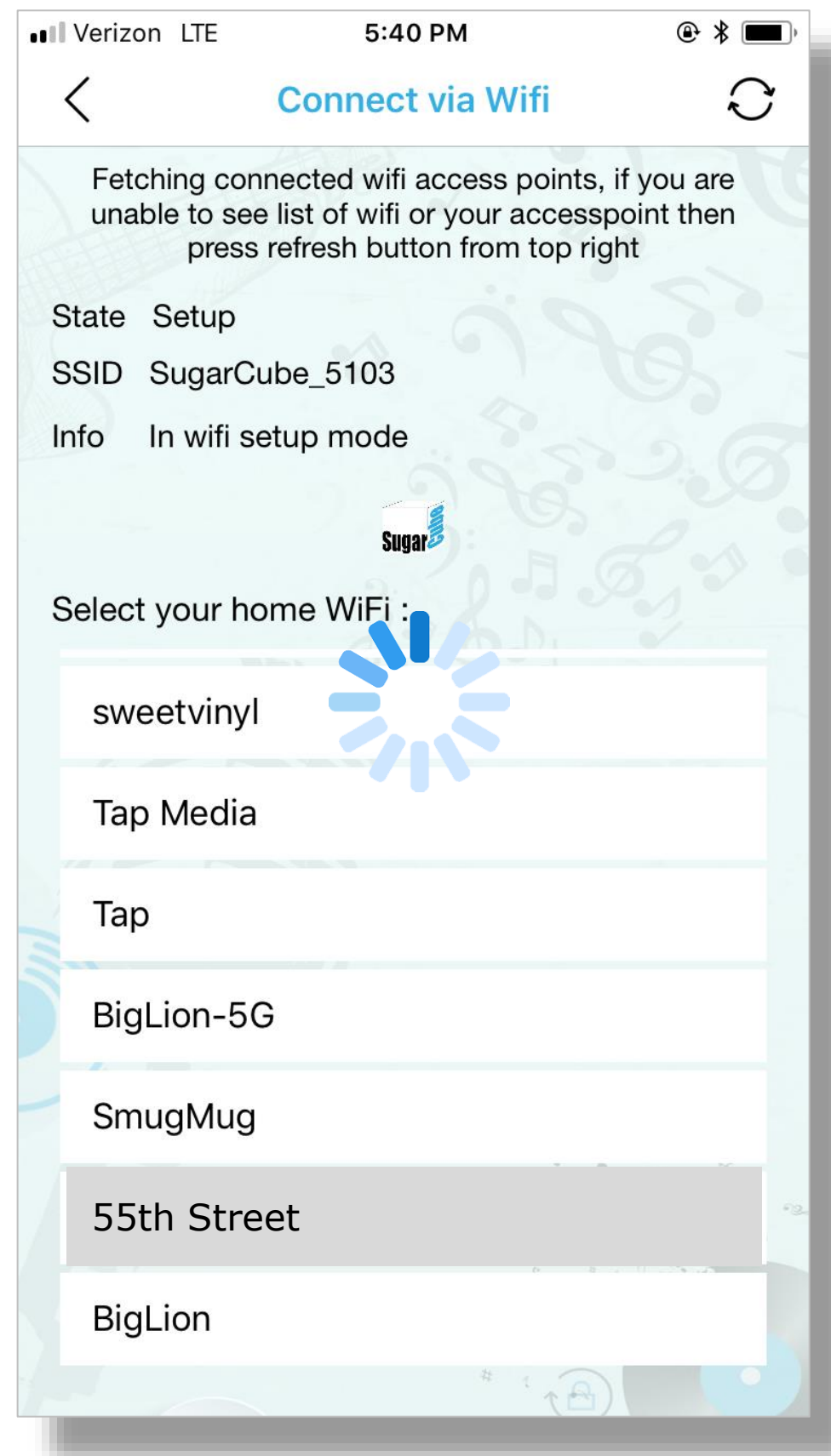


(11) Select the WiFi network that you wish to use

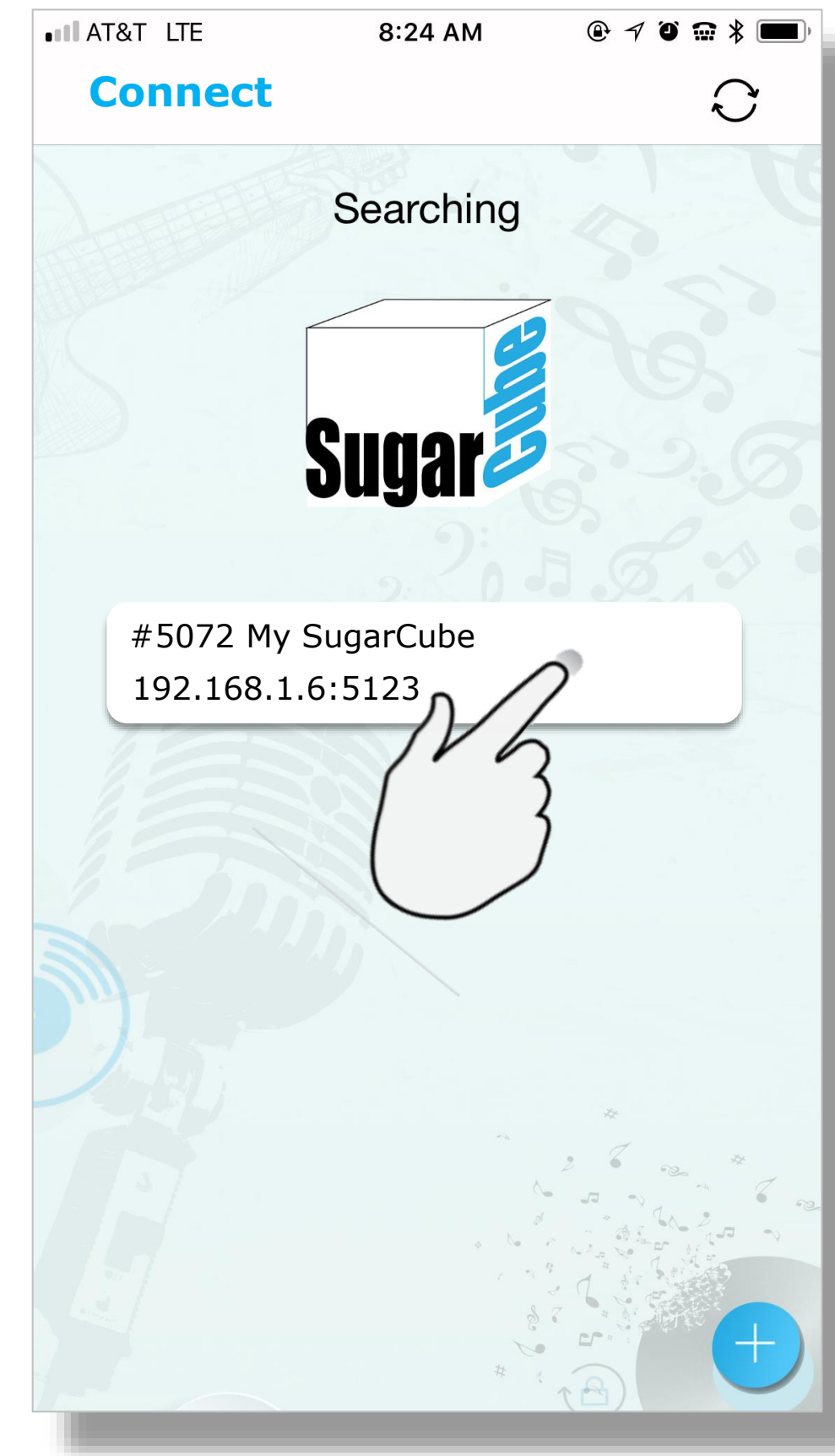


(12) Enter the WiFi password for your network

3.3.10 Connecting App with SugarCube on WiFi



(13) The app will now complete the connection to your WiFi network and also do a little housekeeping.



(14) The app will then display the SugarCube(s) found on your network. Select the appropriate device.

3.3.11 Connecting App with SugarCube on WiFi



(15) When asked to 'Pair', enter '**1111**' (or you can press the front panel knob of the SugarCube and read the code from the display.



(16) You will then see the 'Control' screen and you are done, congratulations!

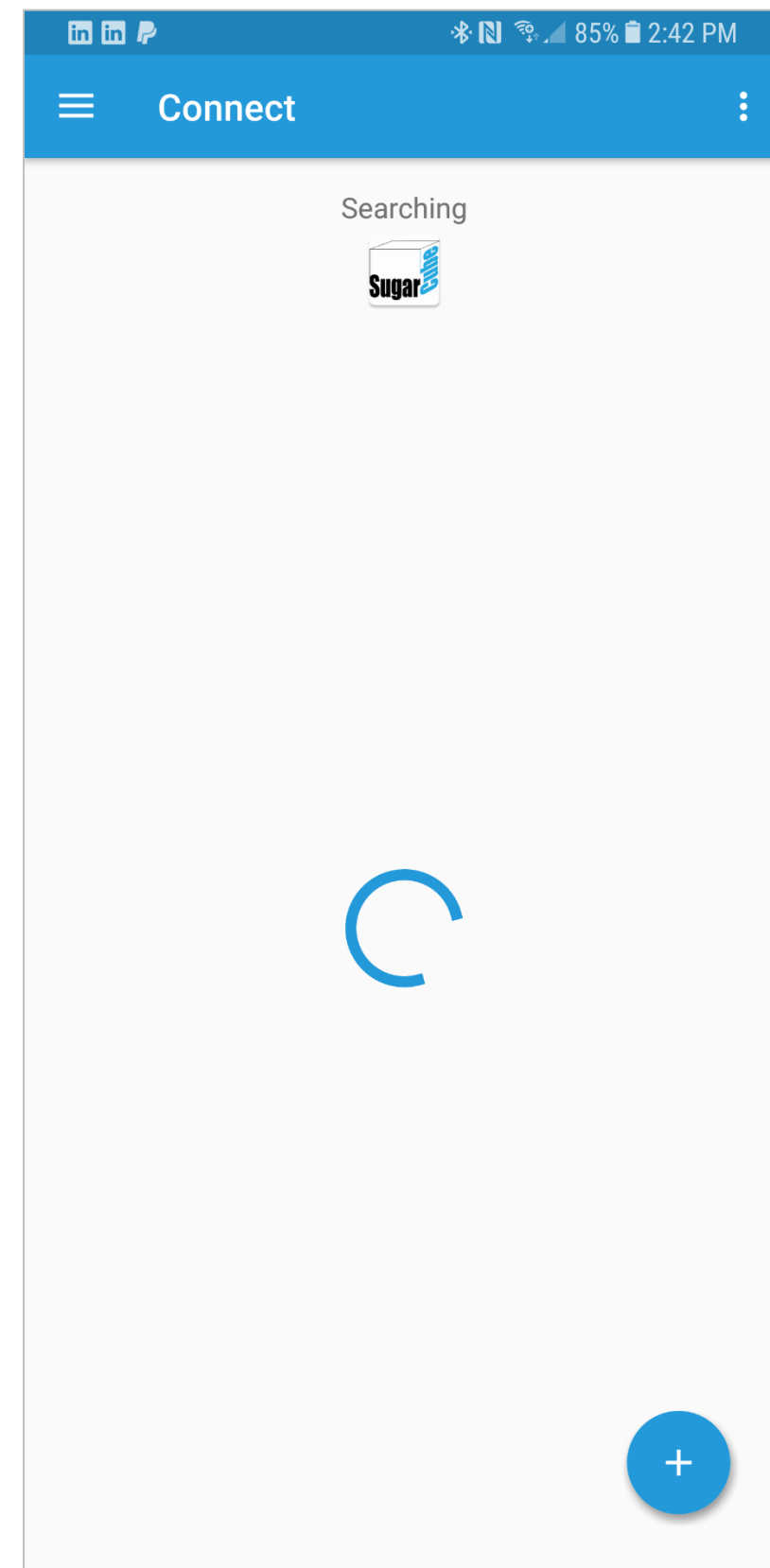


3.4 Using **Android App** with SugarCube

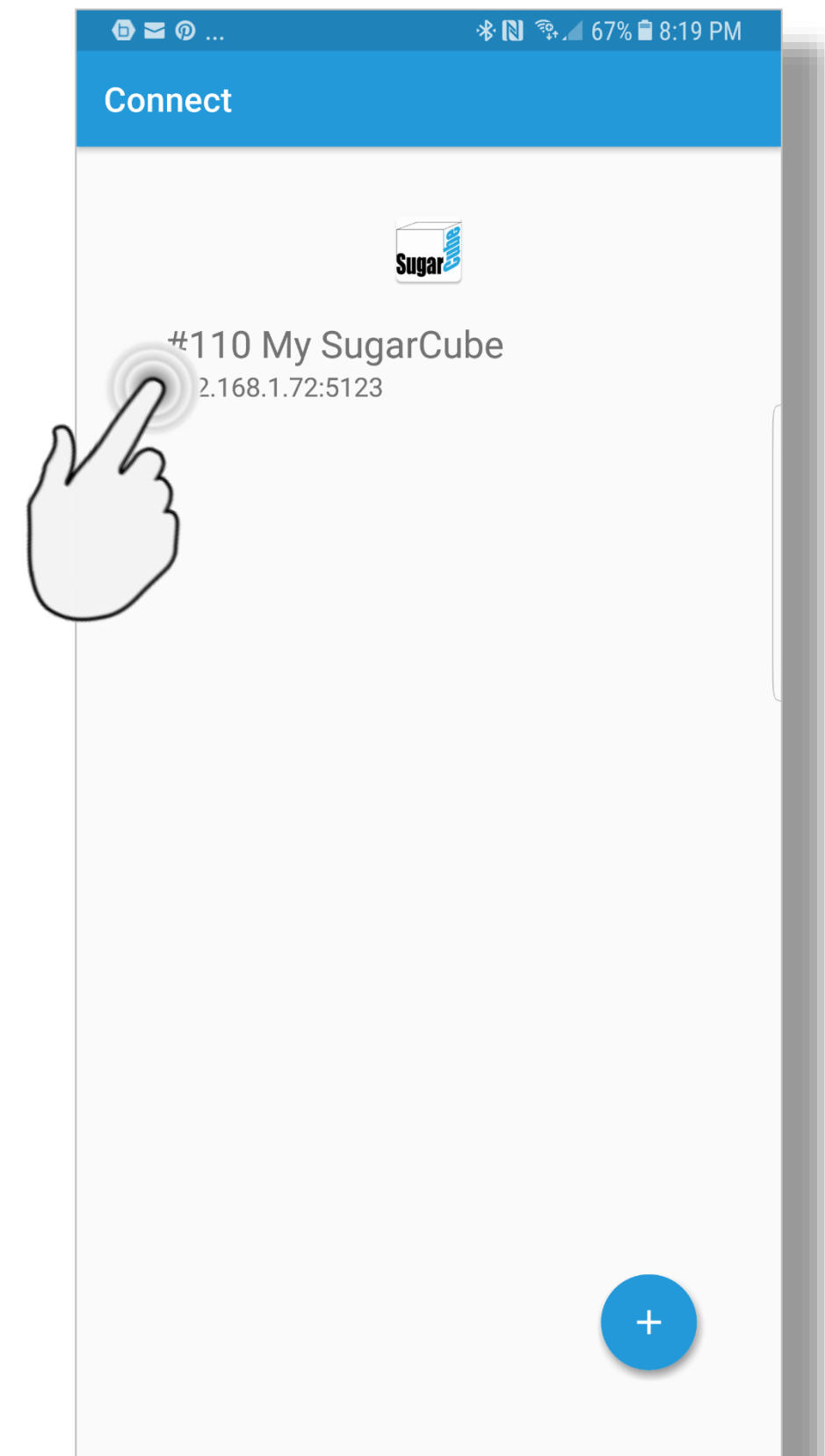
3.4.1 Connecting App with SugarCube on **Wired Ethernet**



(1) App is loading

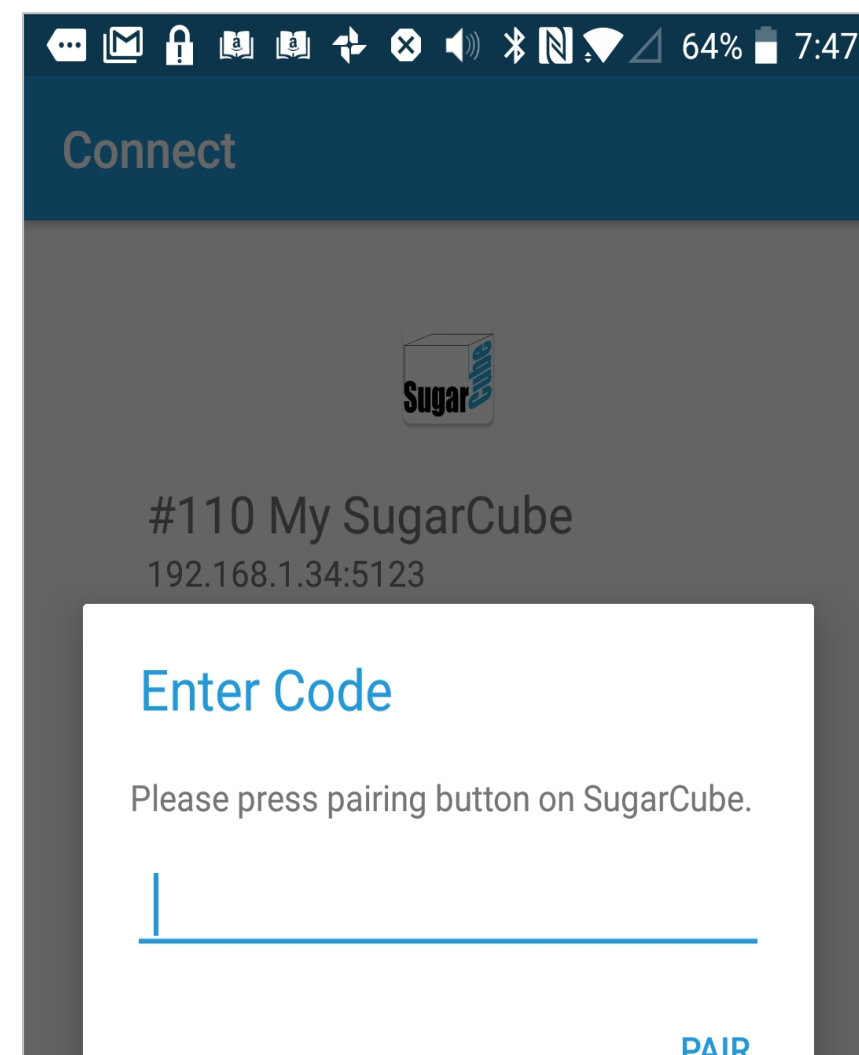


(2) App is searching for SugarCubes on the LAN



(3) App has found a SugarCube and will now initiate a connection.

3.4.2 Connecting App with SugarCube on Wired Ethernet

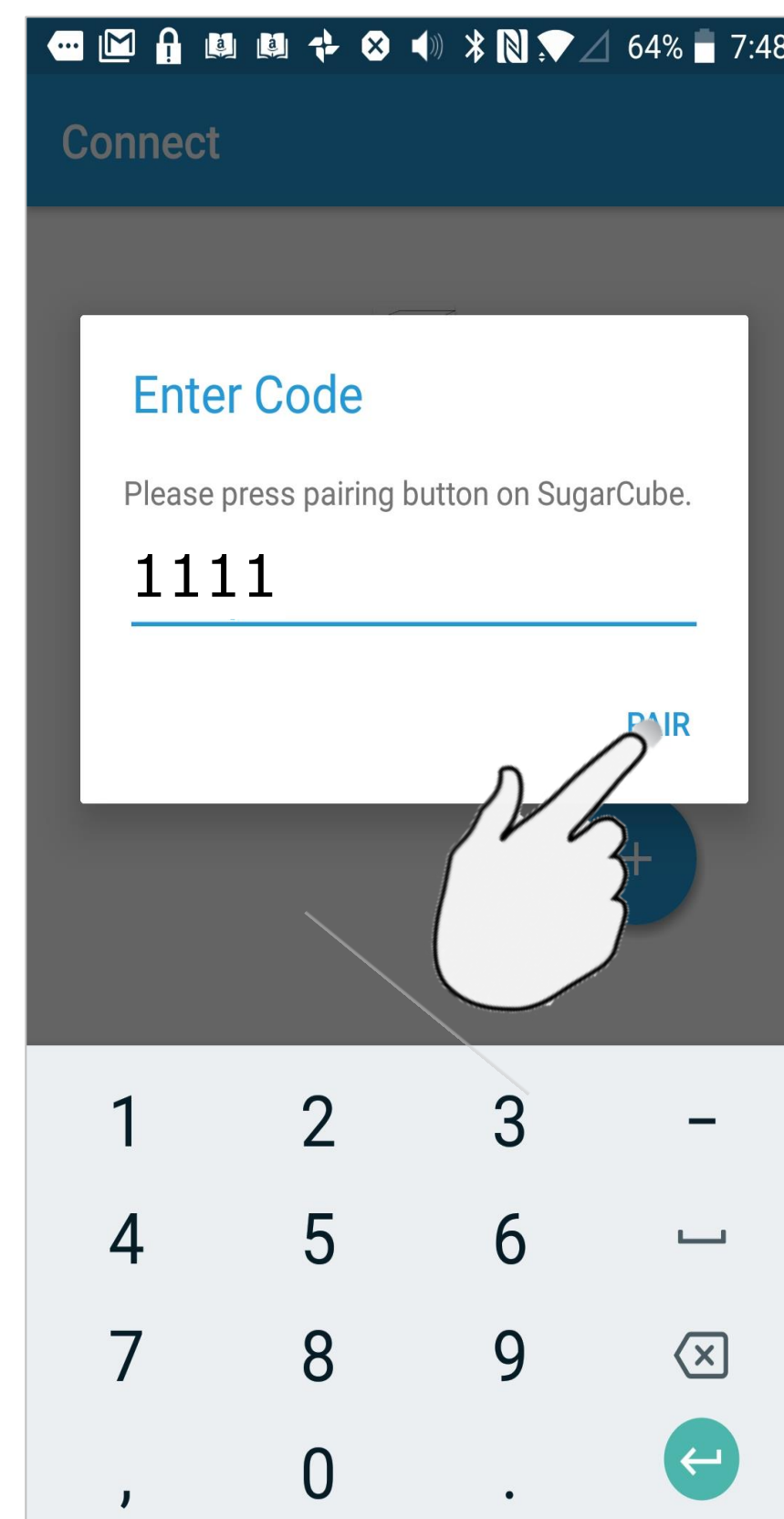


For pairing, enter:

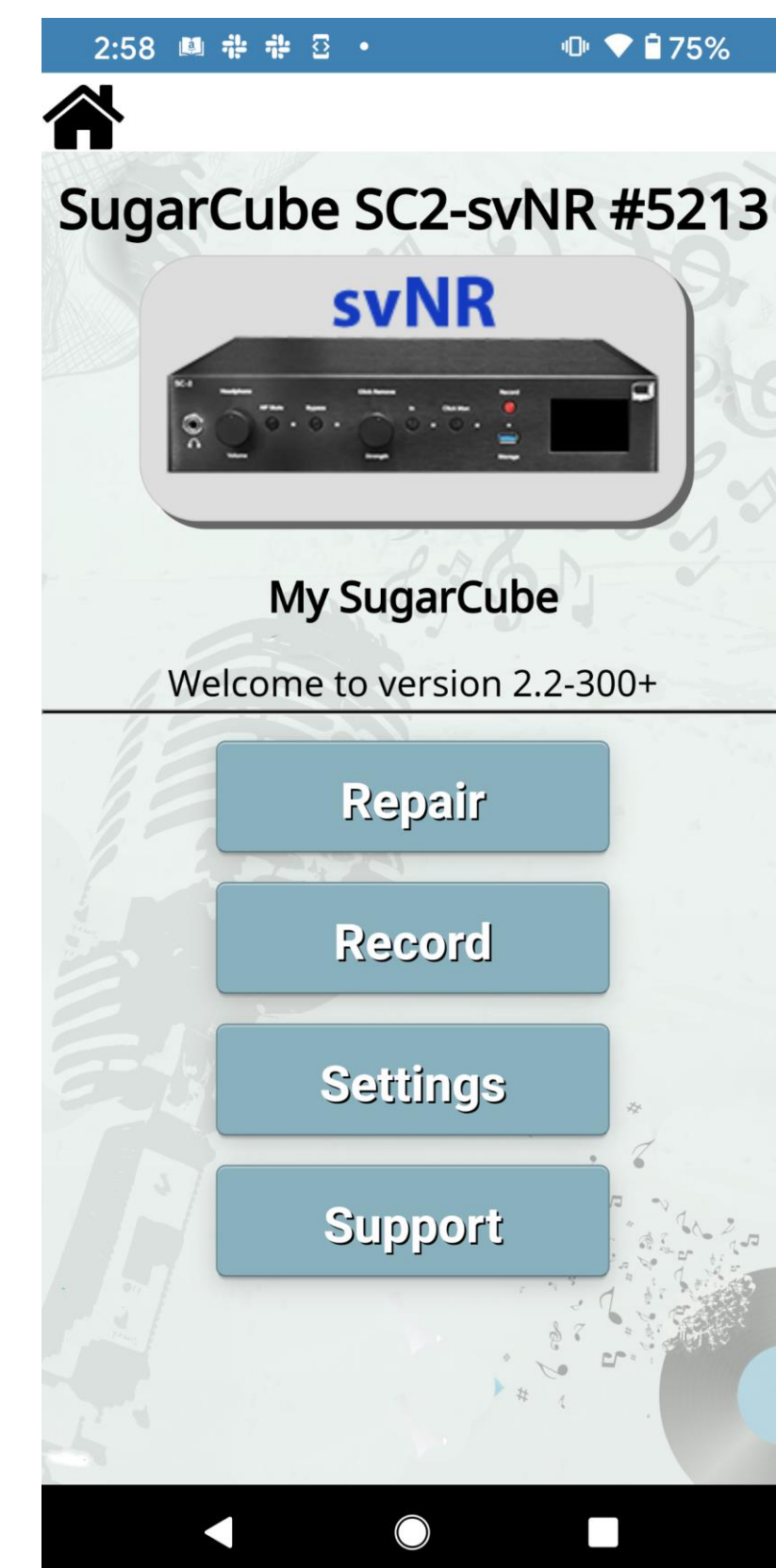
1 1 1 1

as a pairing code
(you may still get a code
from your SugarCube,
if you wish).

(4) The app needs you to enter a pairing code, press the knob on the front panel.

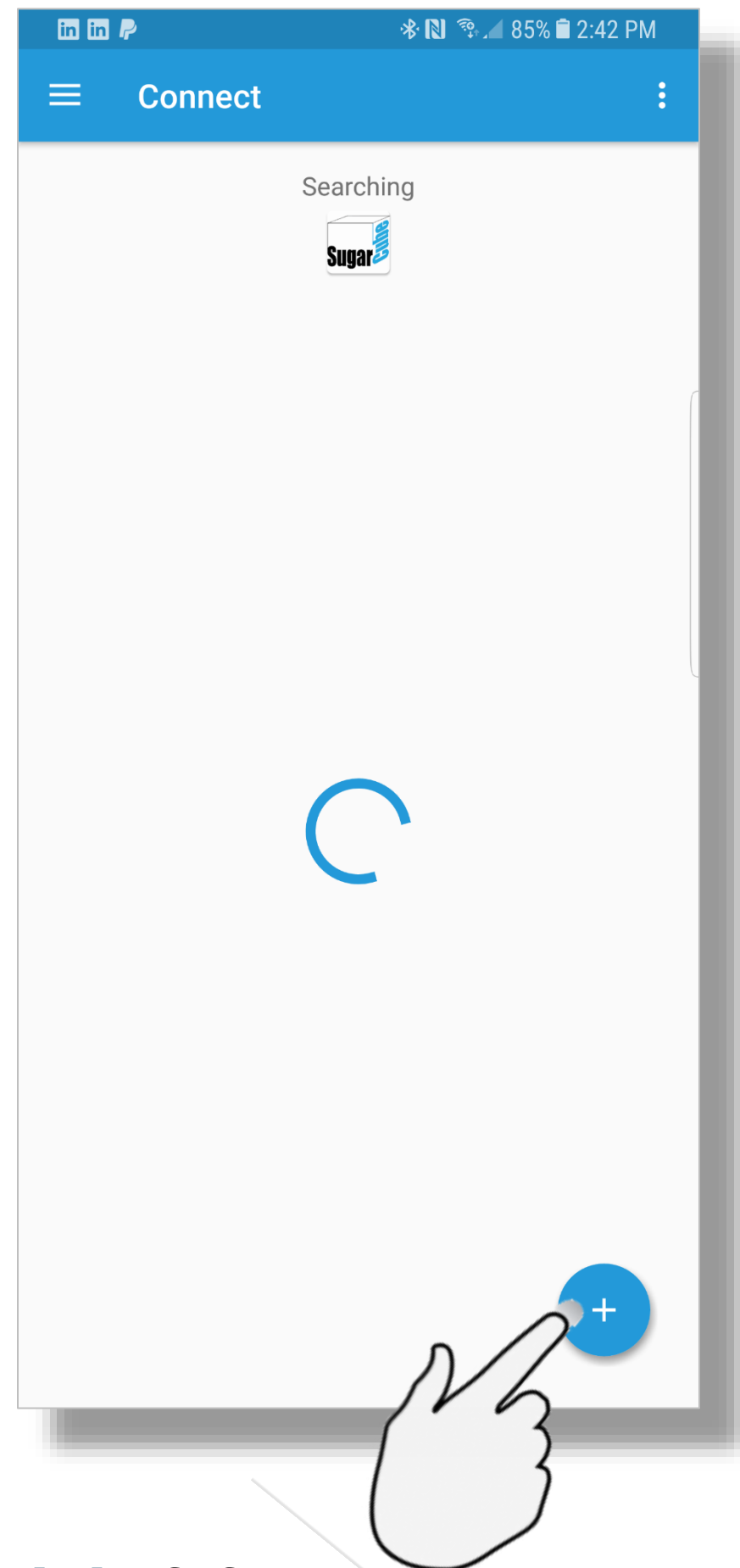


(5) App needs you to enter the pair code '**1111**' or what is displayed on the front panel display then touch '**PAIR**'.

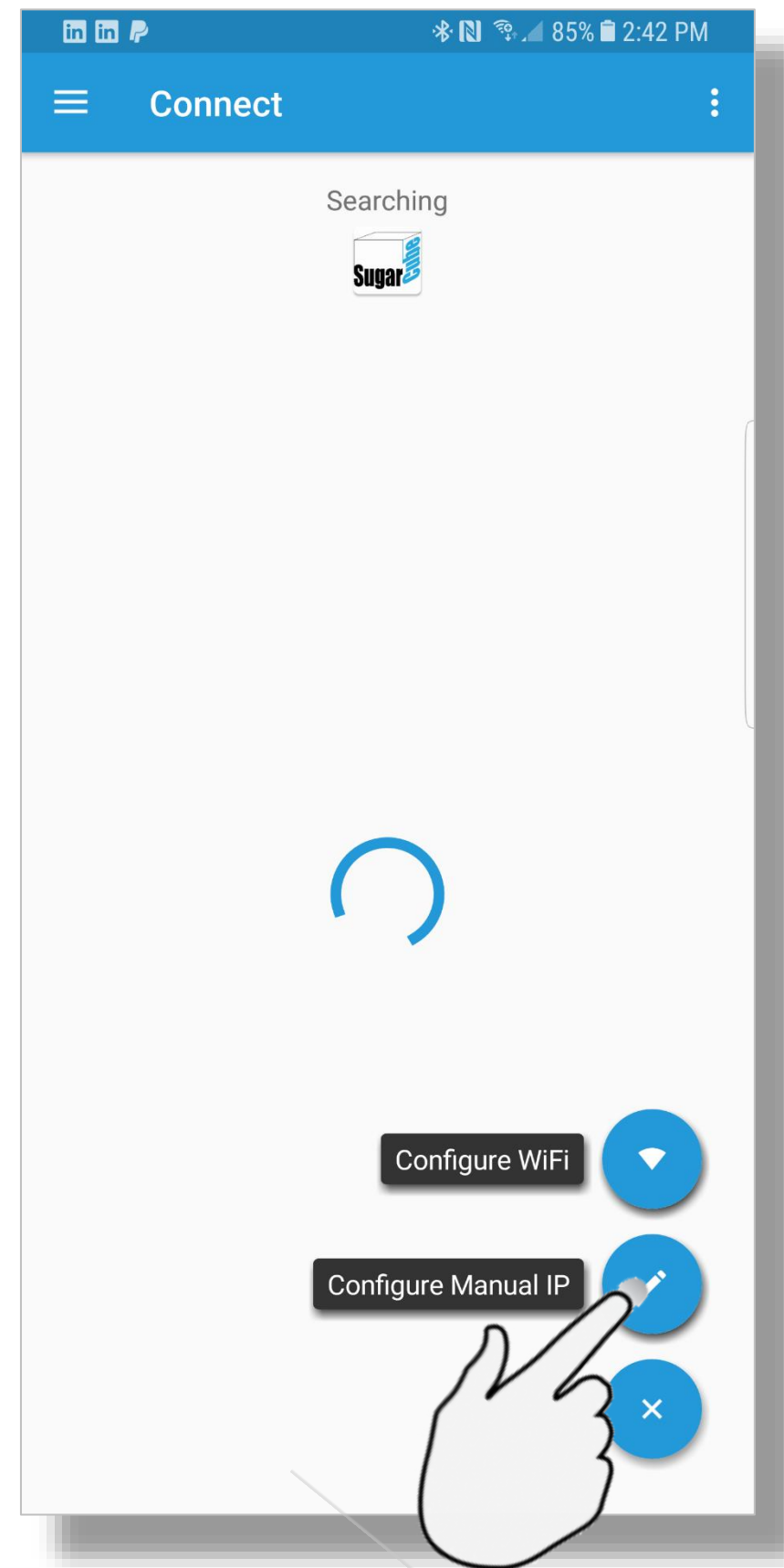


(6) App is linked to SugarCube and ready for use.

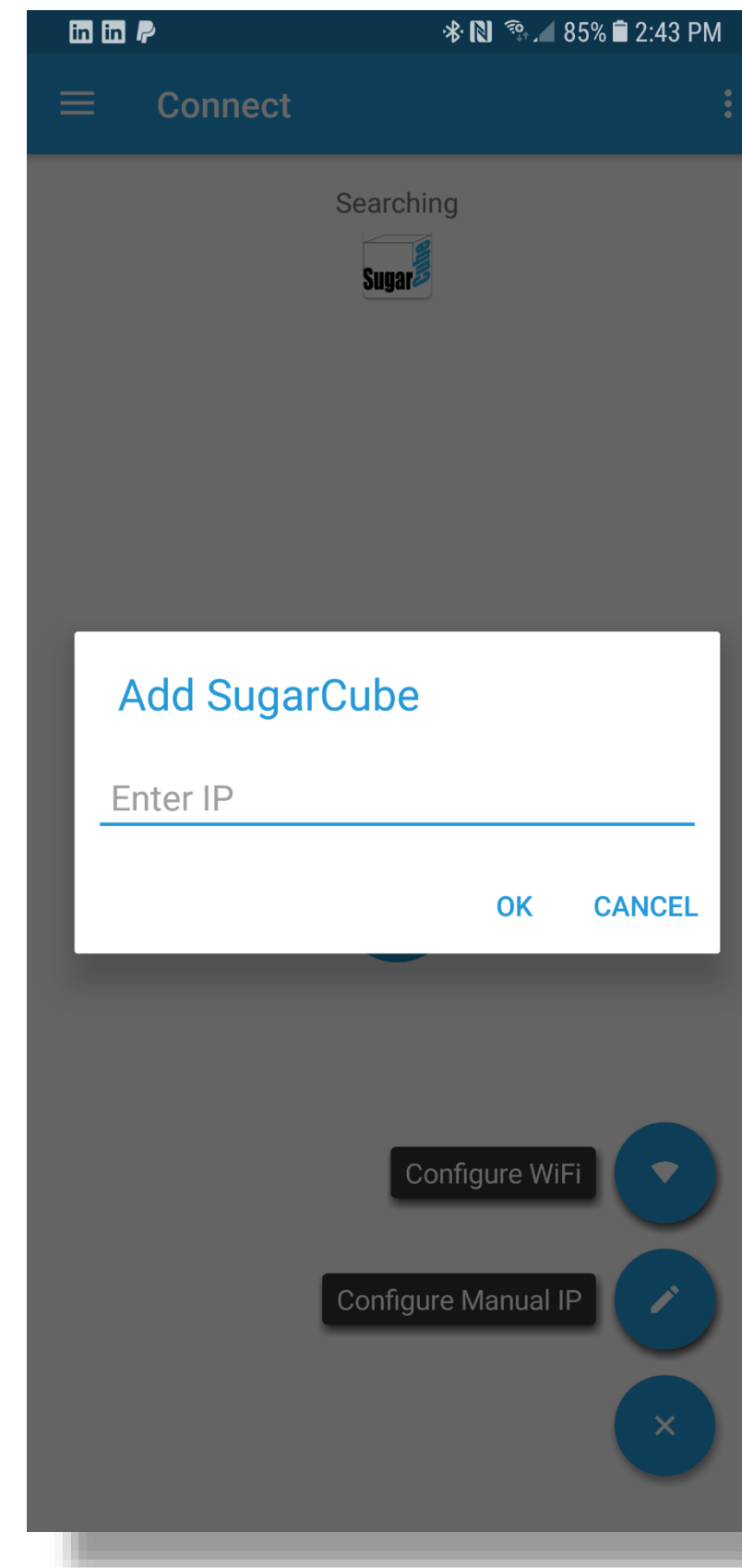
3.4.3 Connecting App with SugarCube Using IP Address



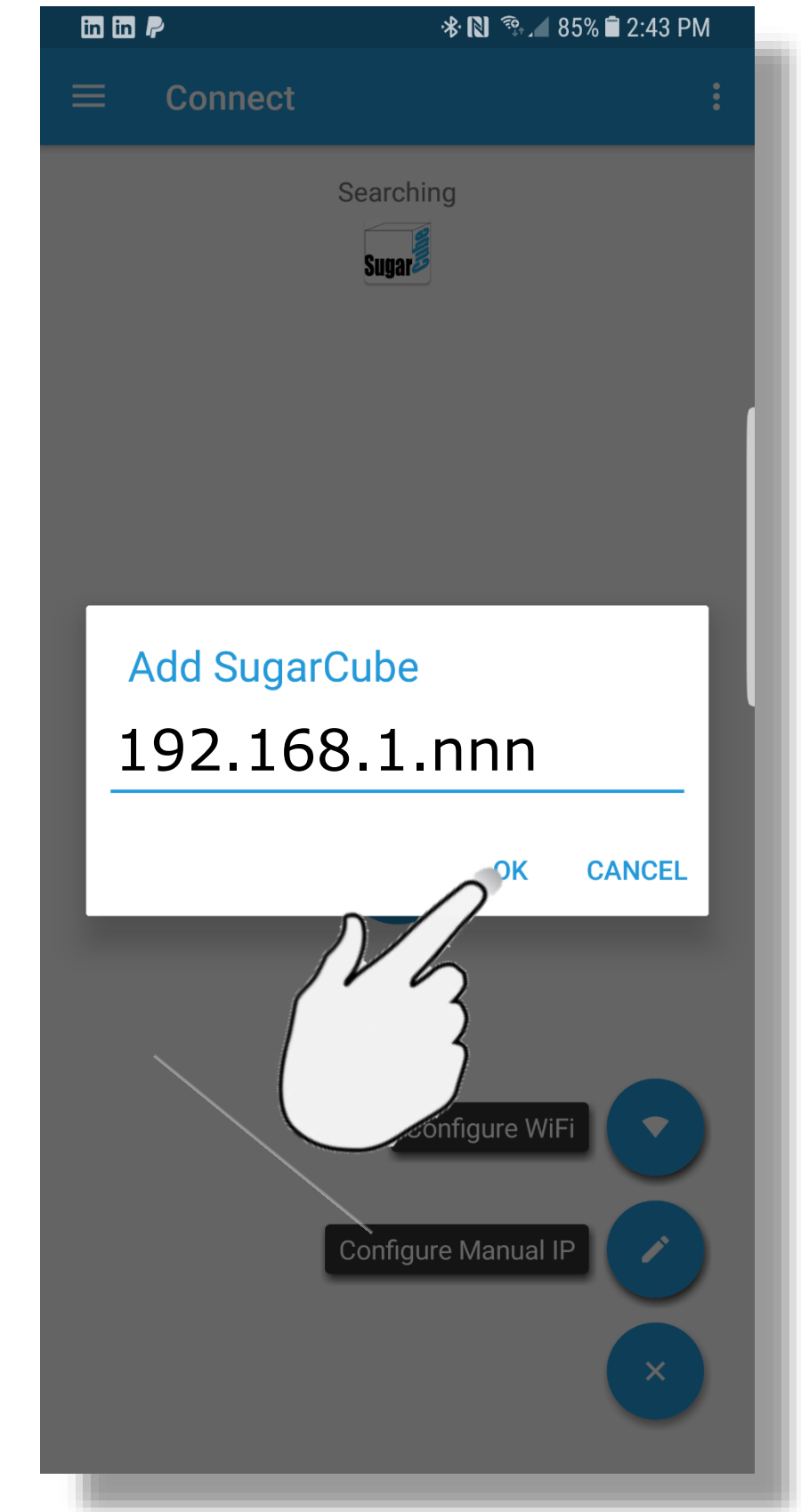
(1) If, for any reason, you wish to manually connect to your SugarCube touch the menu button in the lower right of the screen.



(2) Select 'Configure Manual IP'.

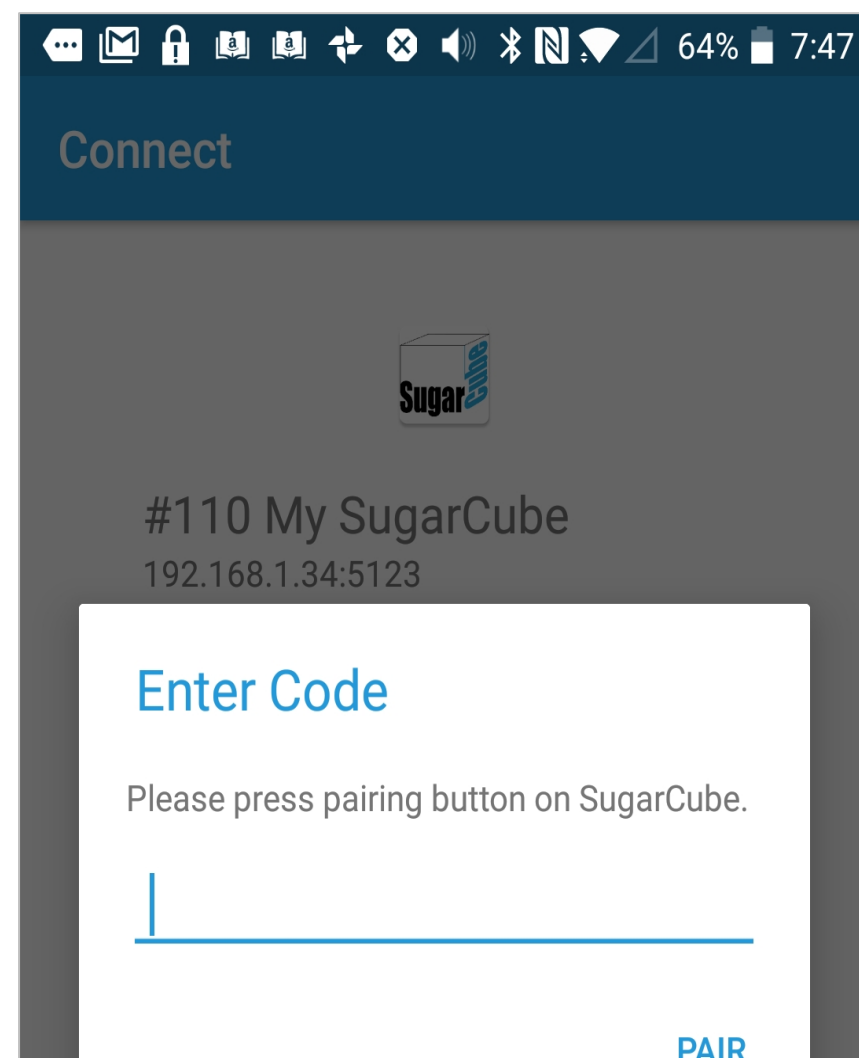


(3) You will then see a window to enter the IP address of the SugarCube. This address can be seen by pressing and holding 'Bypass' for a few seconds.



(4) Enter the IP address as displayed on the front panel display.

3.4.4 Connecting App with SugarCube Using IP Address

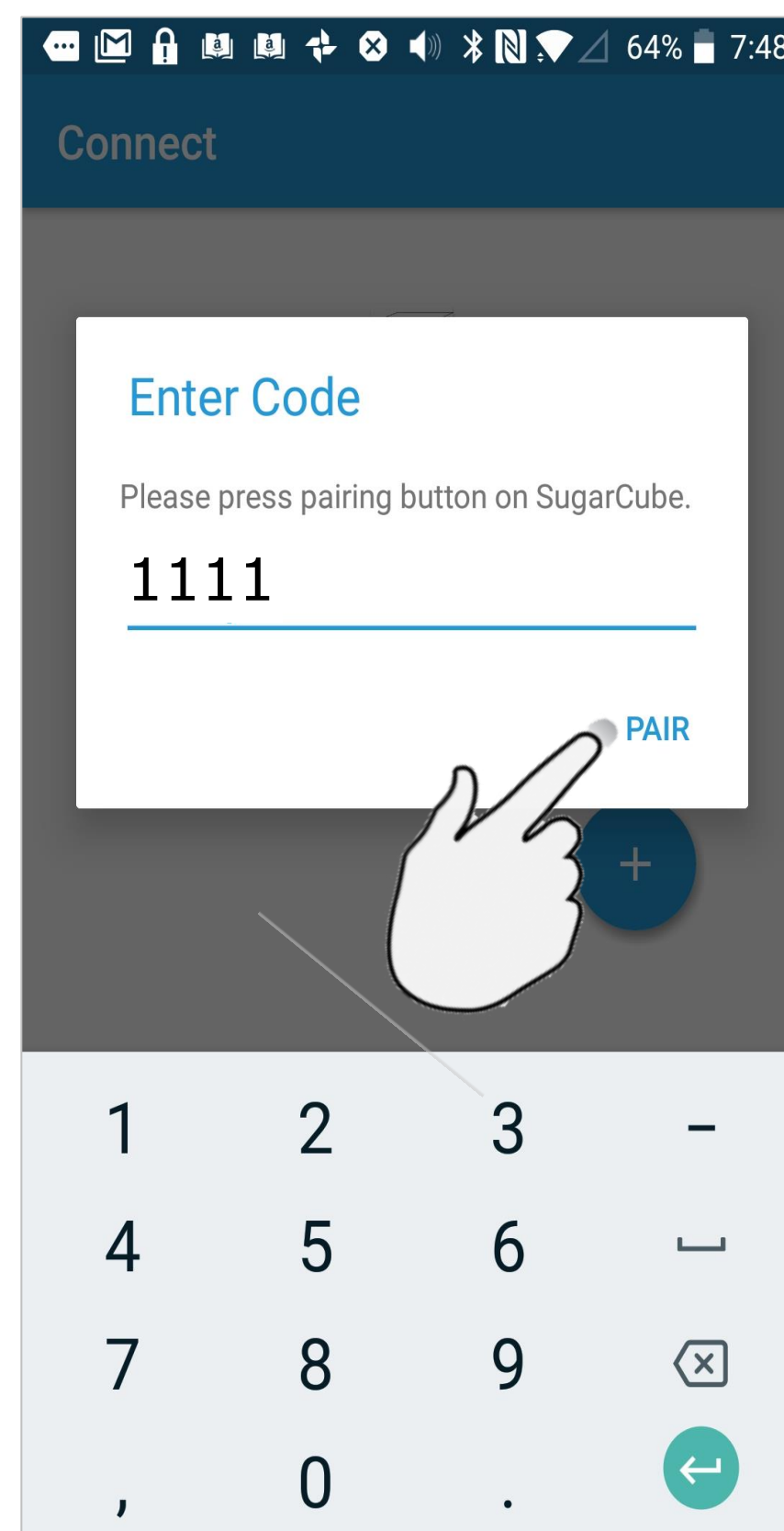


For pairing, enter:

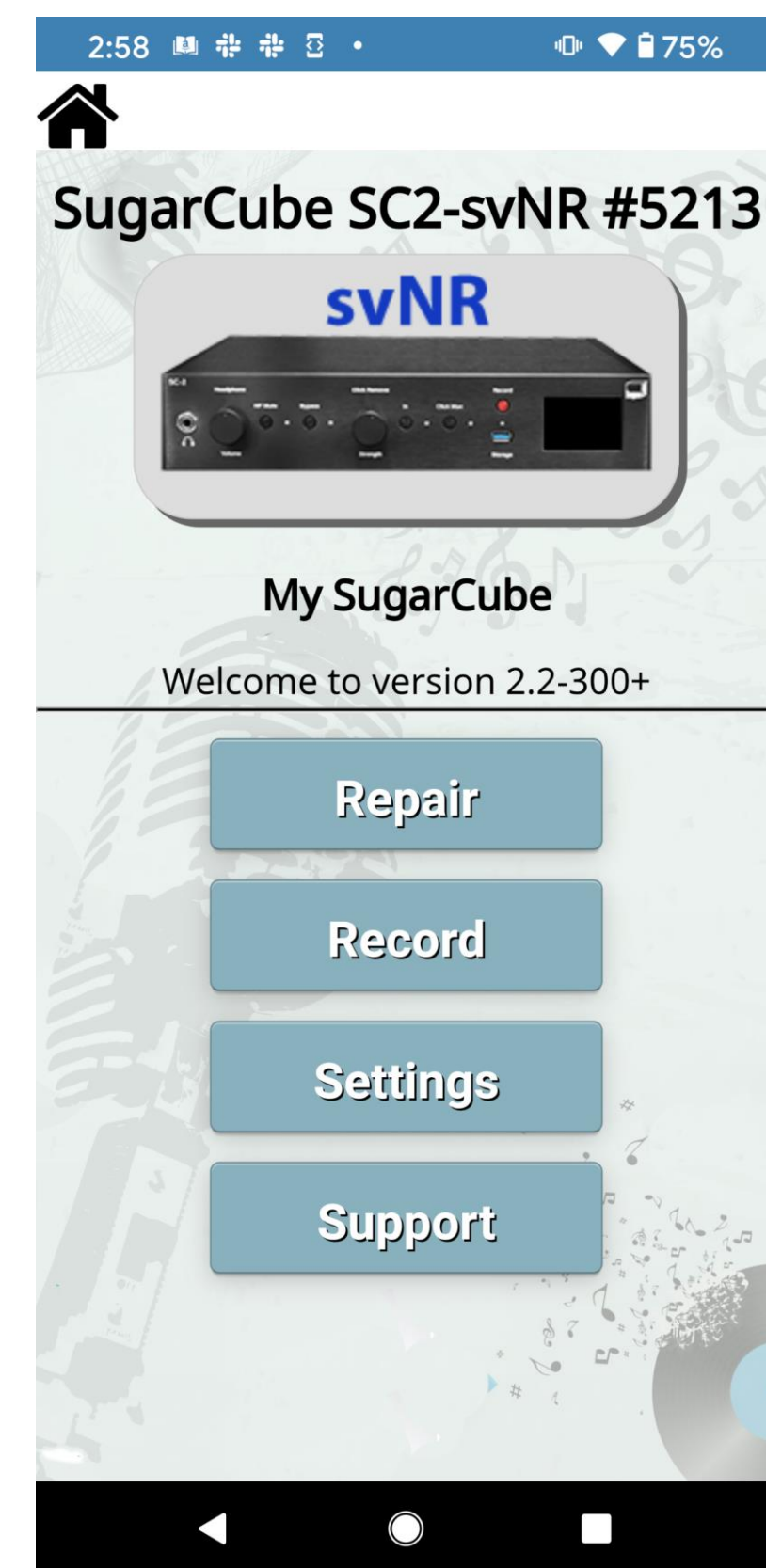
1 1 1 1

**as a pairing code
(you may still get a code
from your SugarCube,
if you wish).**

(5) The app need you to press the knob on the front panel.



(6) App needs you to enter the pair code displayed on the front panel display then touch 'PAIR'.



(7) App is linked to SugarCube and ready for use.

3.4.5 Setting up Android to control SugarCube with **WiFi**

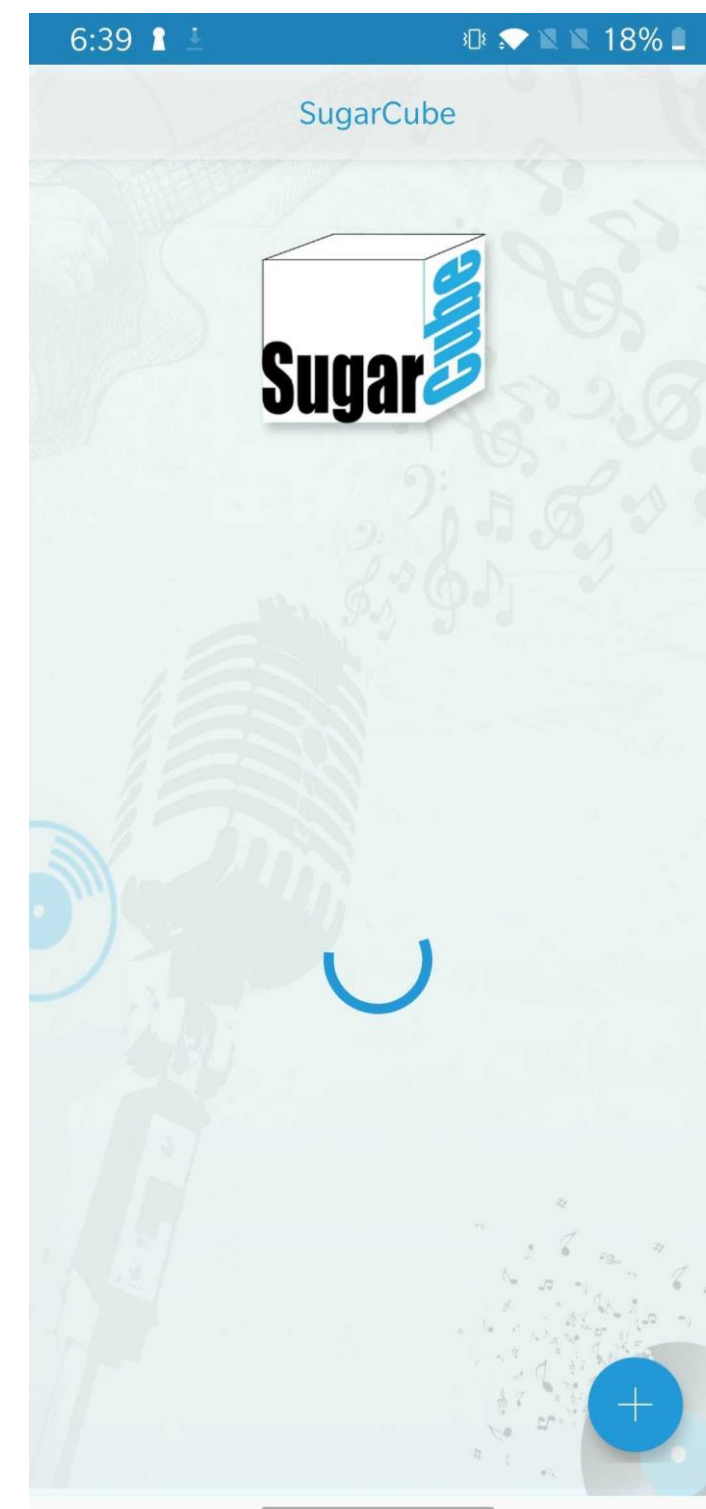
You will need:

1. An Android phone or tablet.
2. To connect with WiFi on your Android device on the same network that you plan to use with your SugarCube with WiFi or wired Ethernet.
3. Download the Android App from the Google Play Store.
(Look for: SugarCube by SweetVinyl.)
4. To plug in the WiFi dongle supplied with your SugarCube or a wired Ethernet cable from a port on your router.

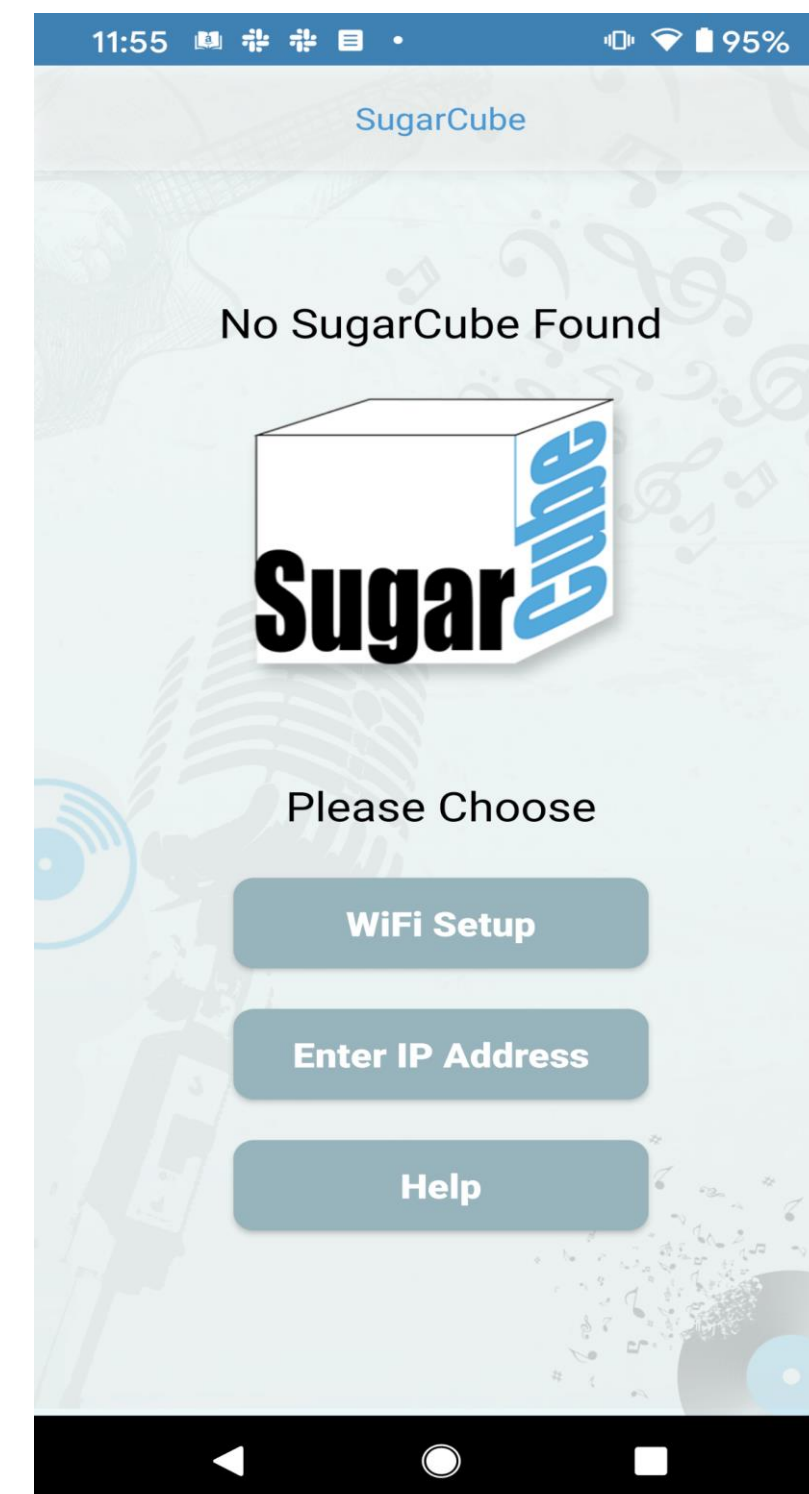
3.4.6 Connecting App with SugarCube on WiFi



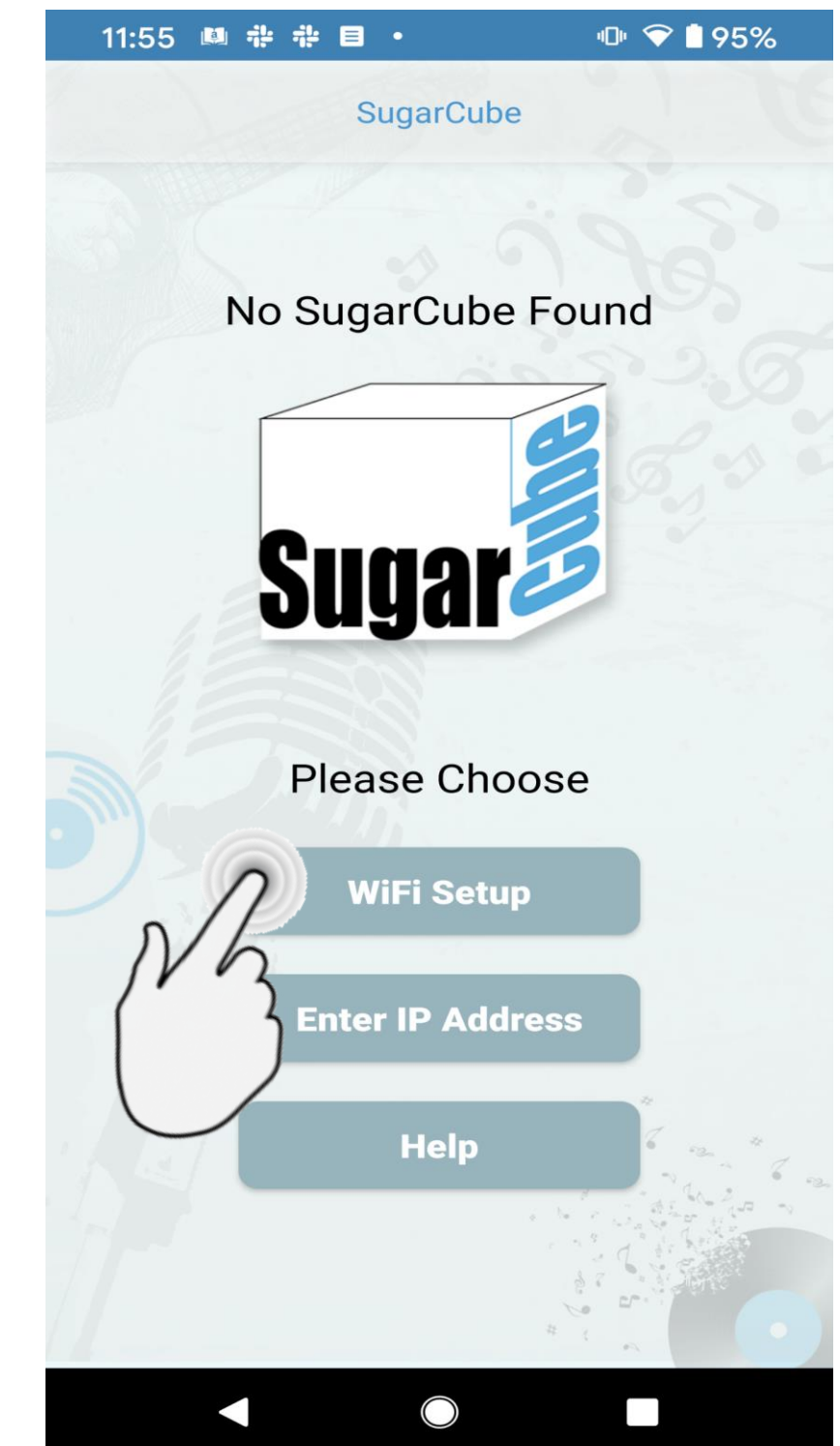
(1) App is loading



(2) App is searching for SugarCubes on the LAN

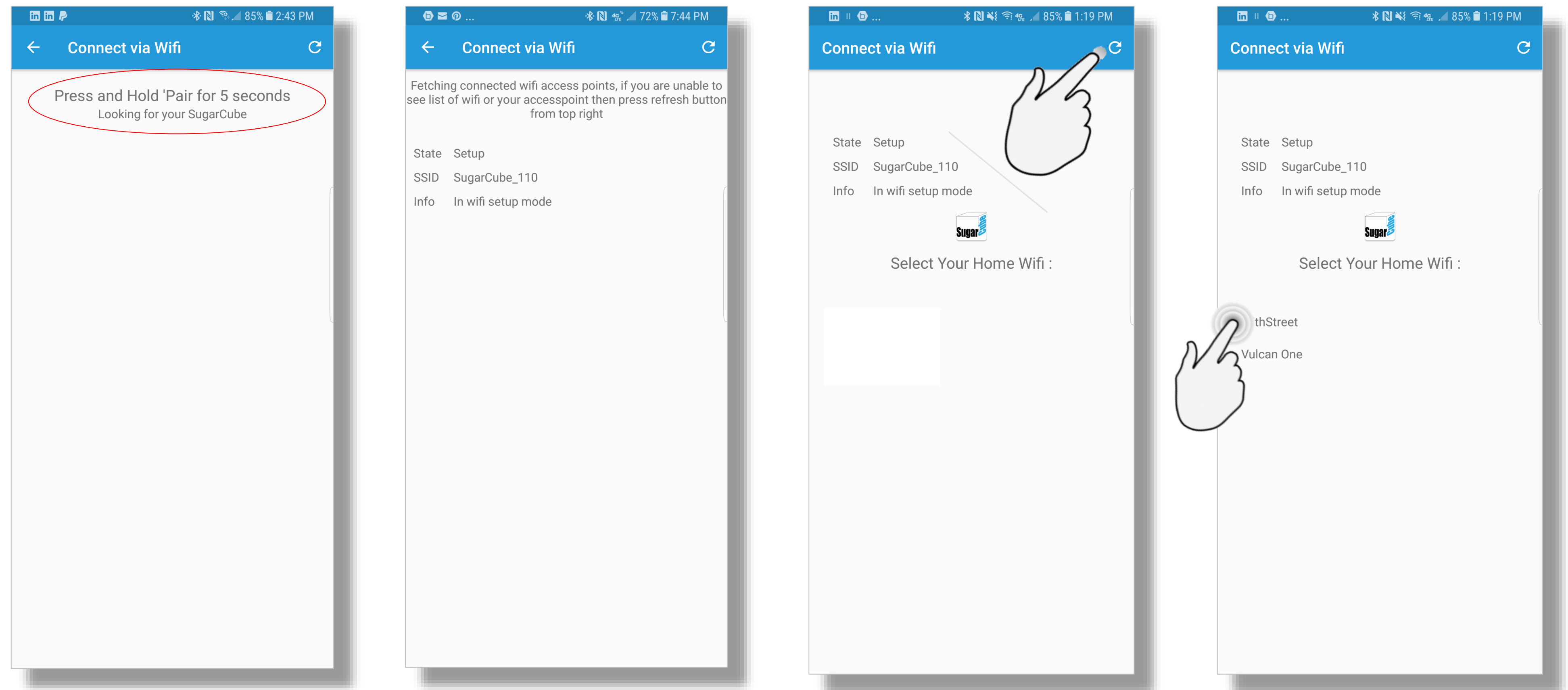


(3) No SugarCube is found so the app offers the option to run WiFi setup.



(4) User taps 'WiFi Setup' to start the process.

3.4.7 Connecting App with SugarCube on WiFi



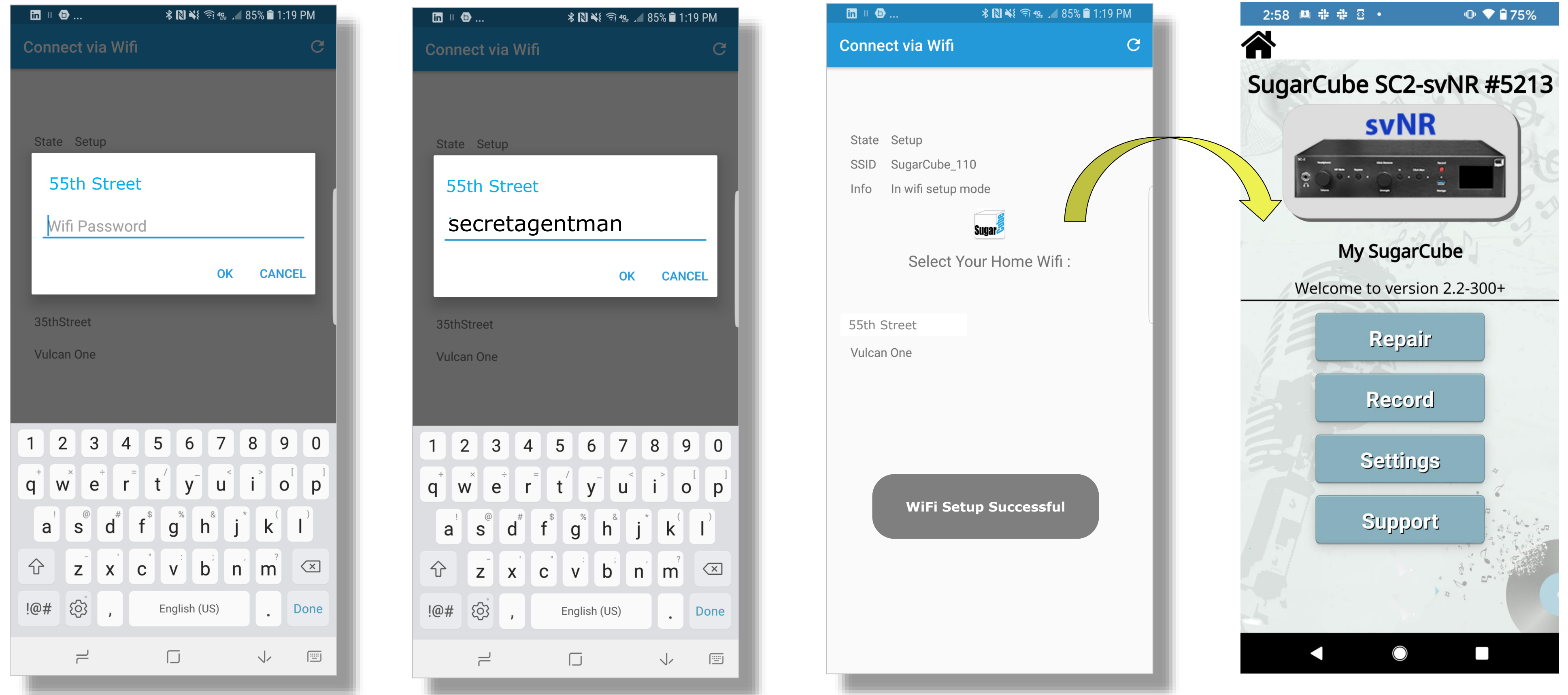
(5) Next, press and hold the pairing button (on the rear panel) for 5 seconds as instructed.

(6) After this, your SugarCube will be configured as an access point and display its SSID (e.g. SugarCube_110). In this screen the app has now connected to it automatically.

(7) Next, the SugarCube will find your available WiFi connections. This could take a minute but if nothing shows, please press the refresh (upper right).

(8) When you see your network, select the desired connection.

3.4.8 Connecting App with SugarCube on WiFi



(9) Next, enter the password for your WiFi network.

(10) The app has connected successfully with the SugarCube and will now jump to the main screen.

(11) The connection process is complete and you are ready to go.

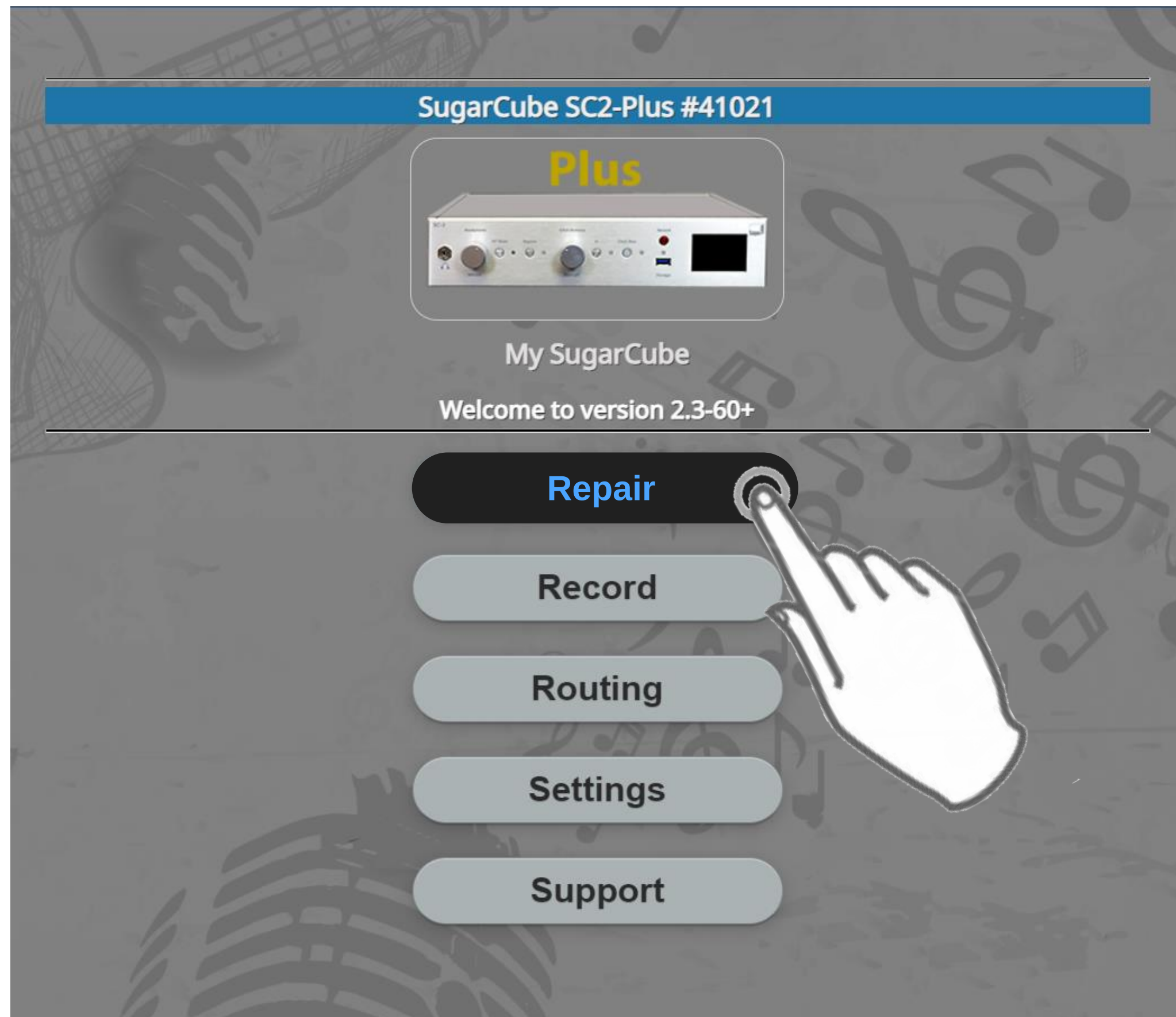
Section 4

Controlling SugarCube via a browser or app

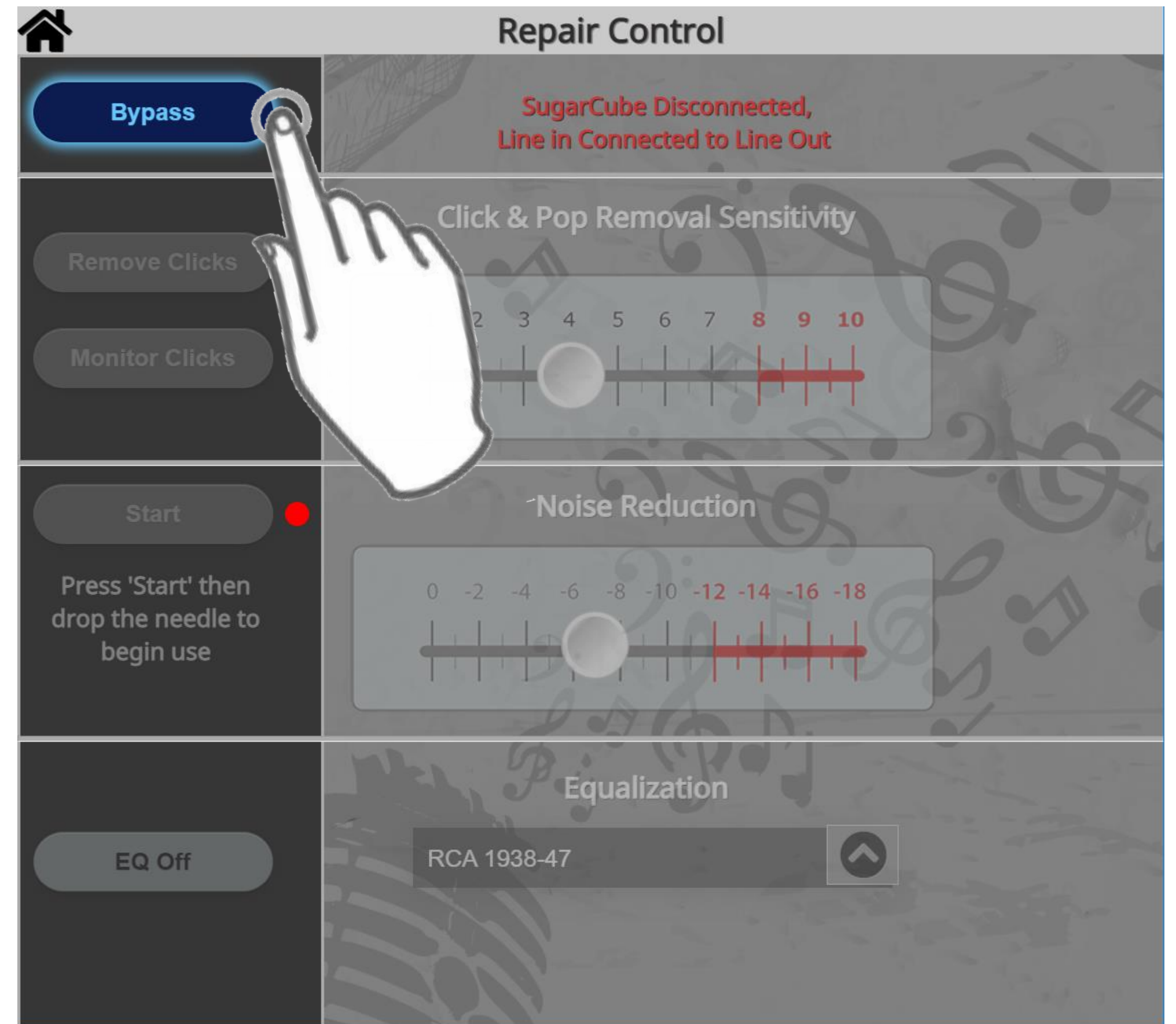
Chrome, Edge, or Safari
browser &
iOS or Android app



4.1 Home Page and Repair Control



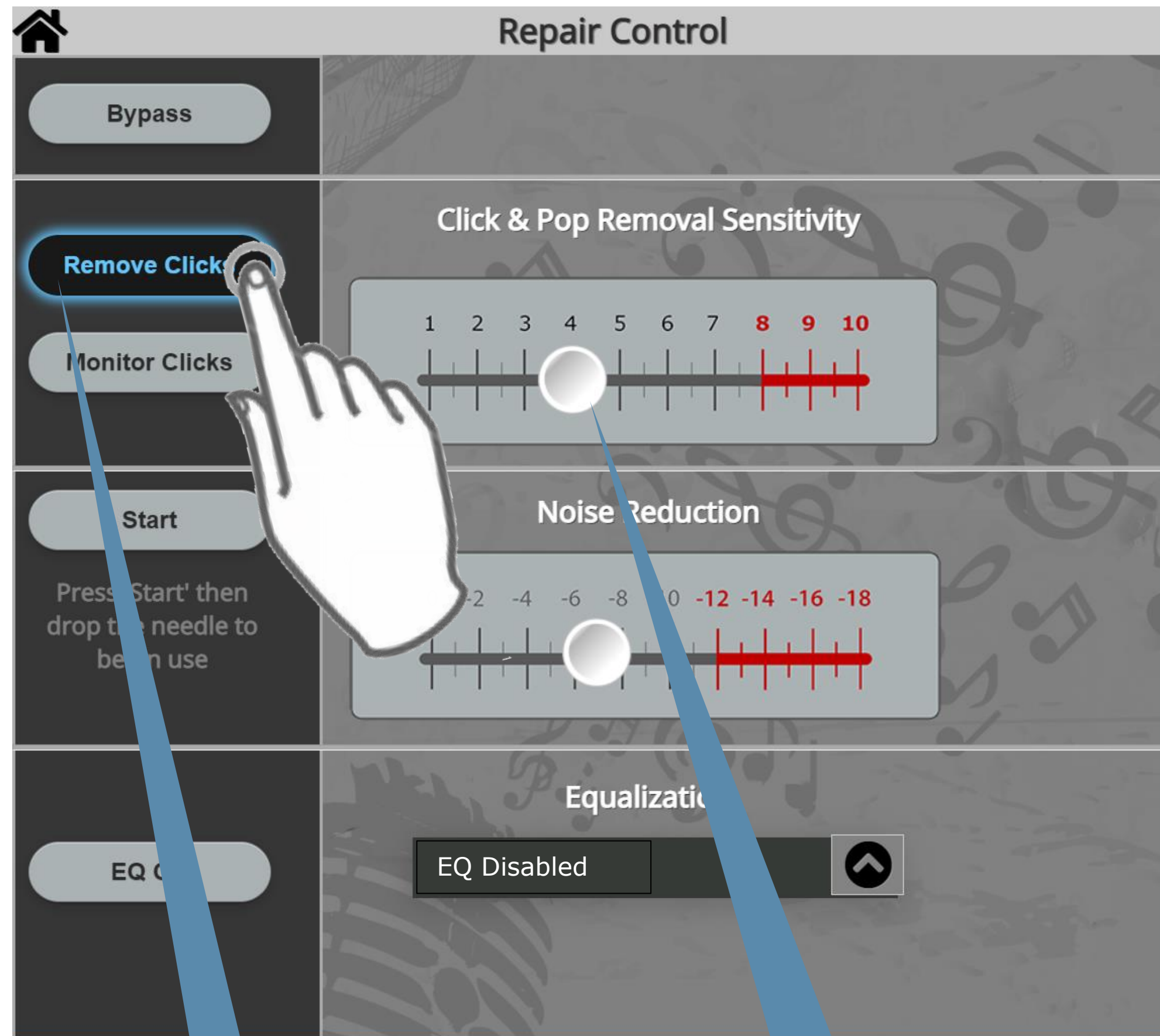
(1) Click on 'Repair'



(2) **Click and Pop** and **Bypass** functions available mirror the front panel controls. A change here is reflected on the front panel and a change on the front panel changes the above.

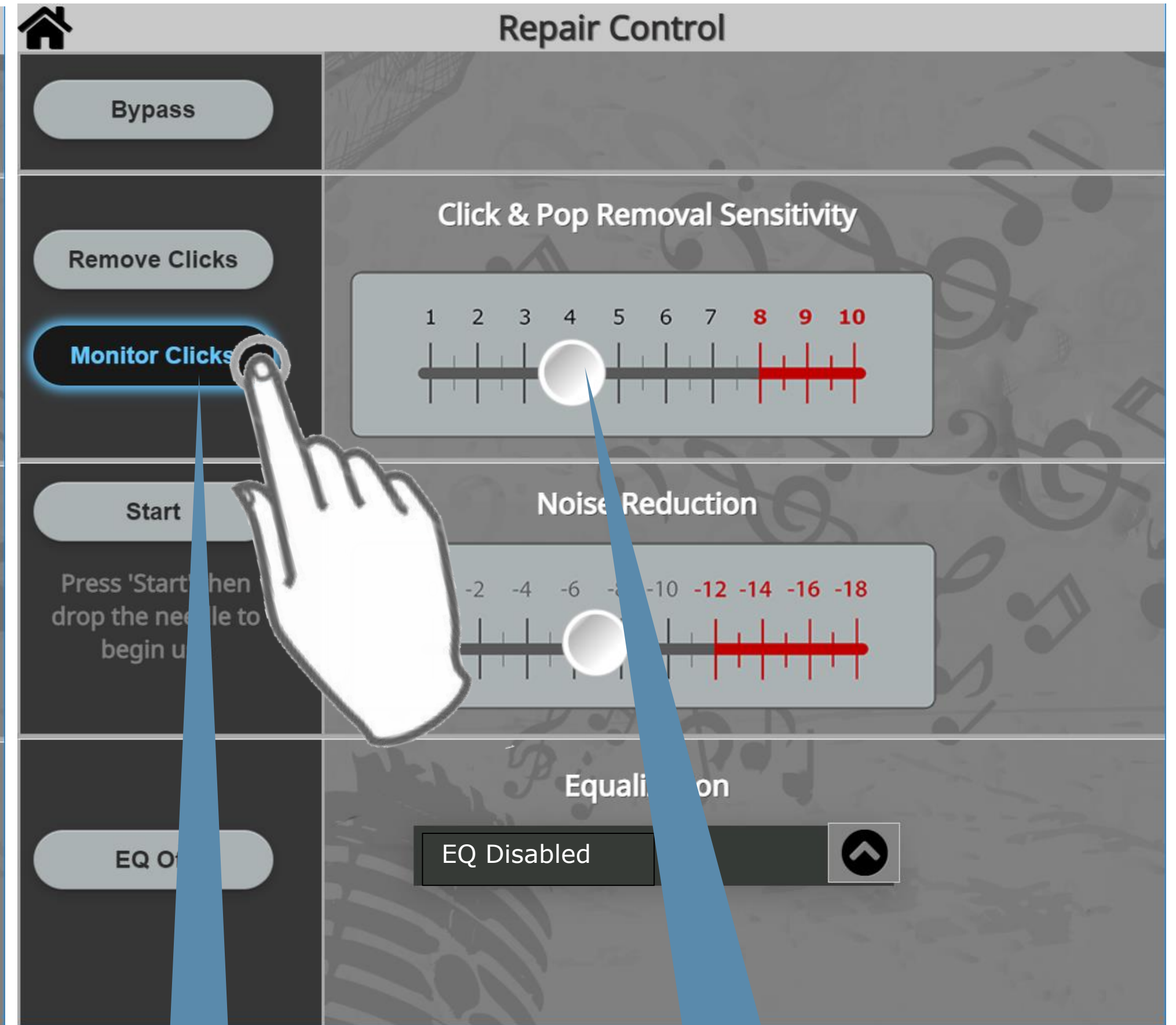
4.2 Repair Control Functions

Controlling Click and Pop Removal



Adjust the slider to listen to the result of setting 'Strength'.

The Click and Pop removal function has been engaged.

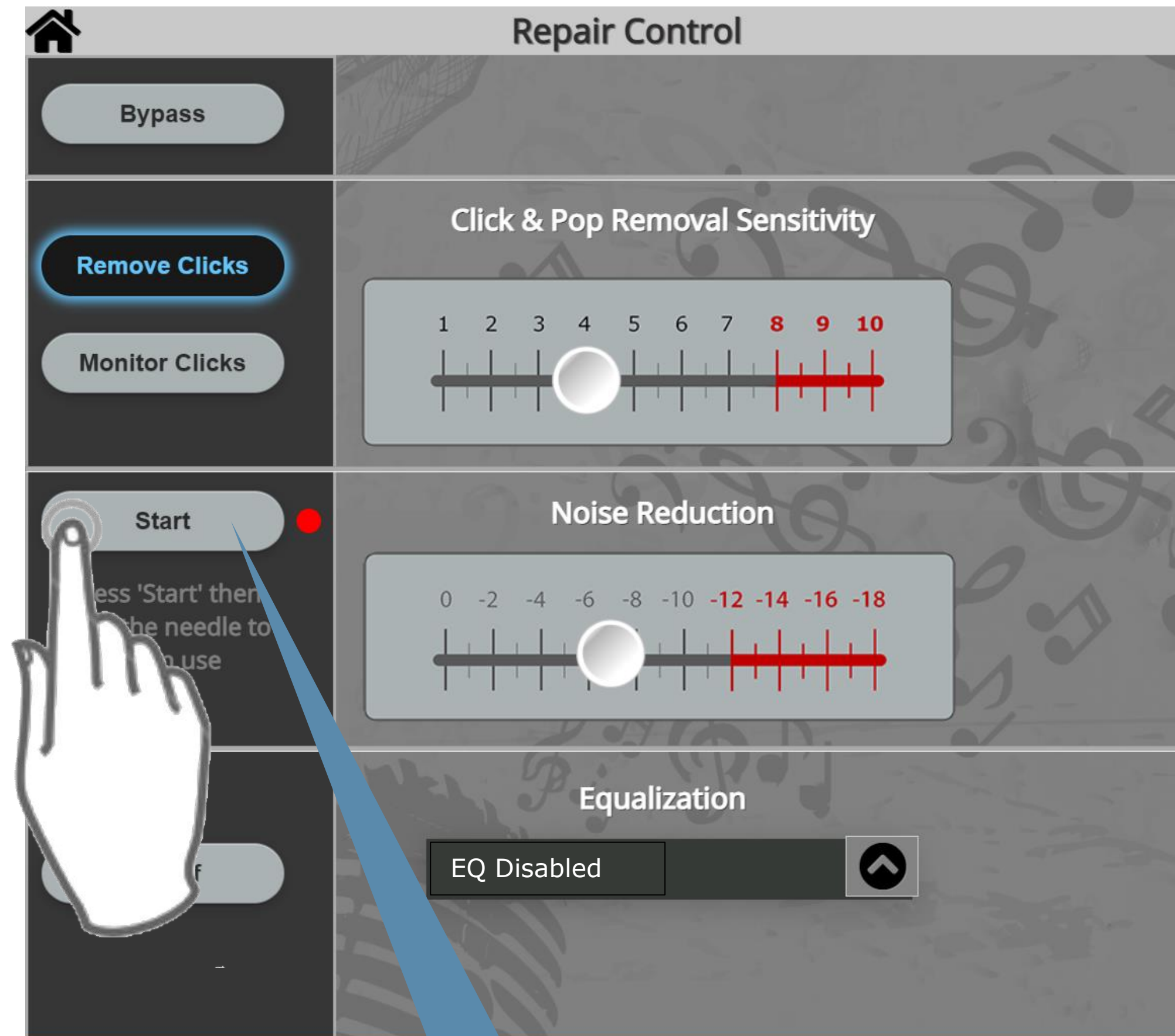


Monitor the clicks and pops that are being removed from the audio passing through the SugarCube.

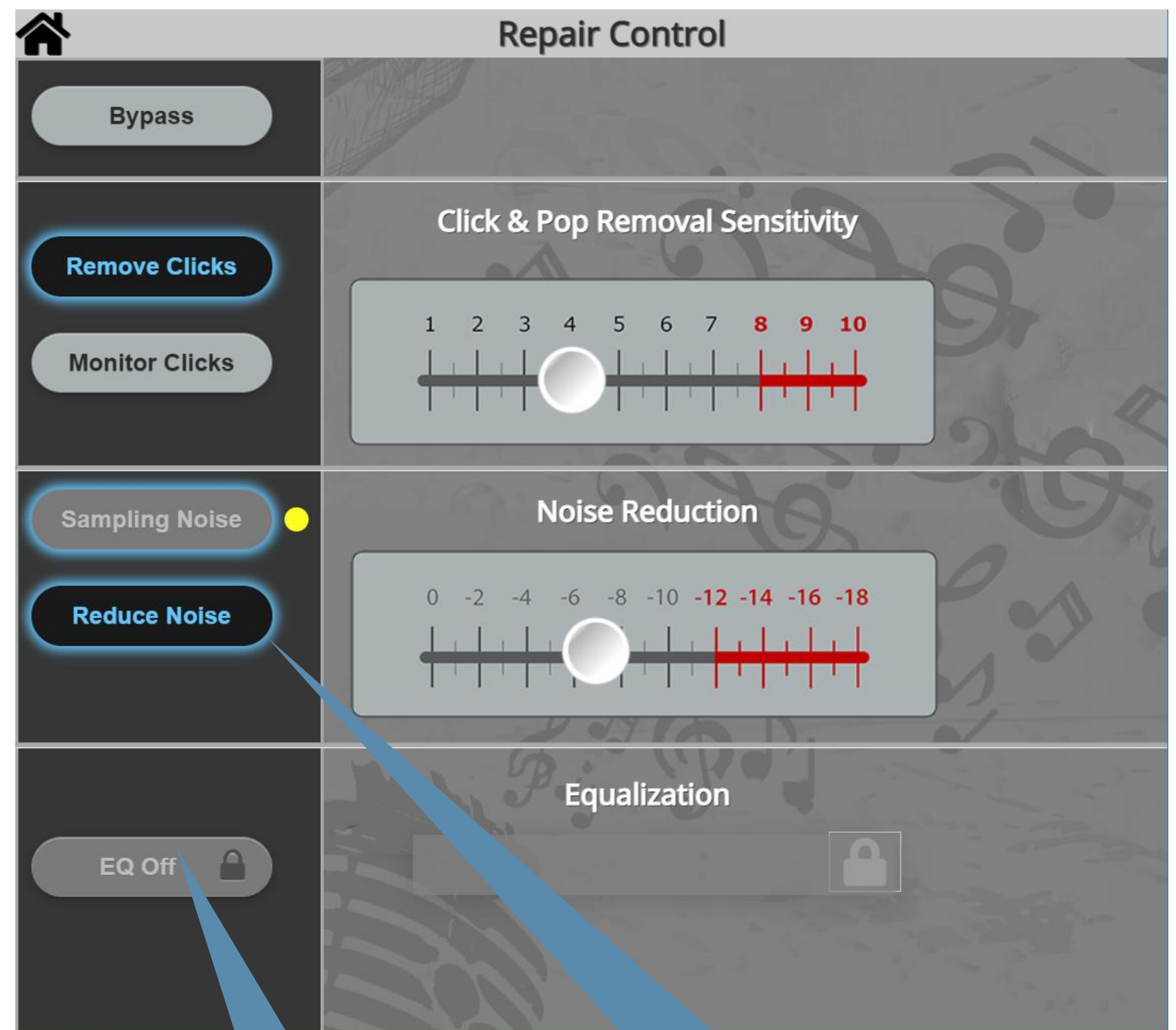
A setting of '3' or '4' is all you need for all but the most damaged LPs

4.3 Noise Reduction Functions

Controlling Noise Reduction



Click on '**Start**' to initialize '**Noise Reduction**'

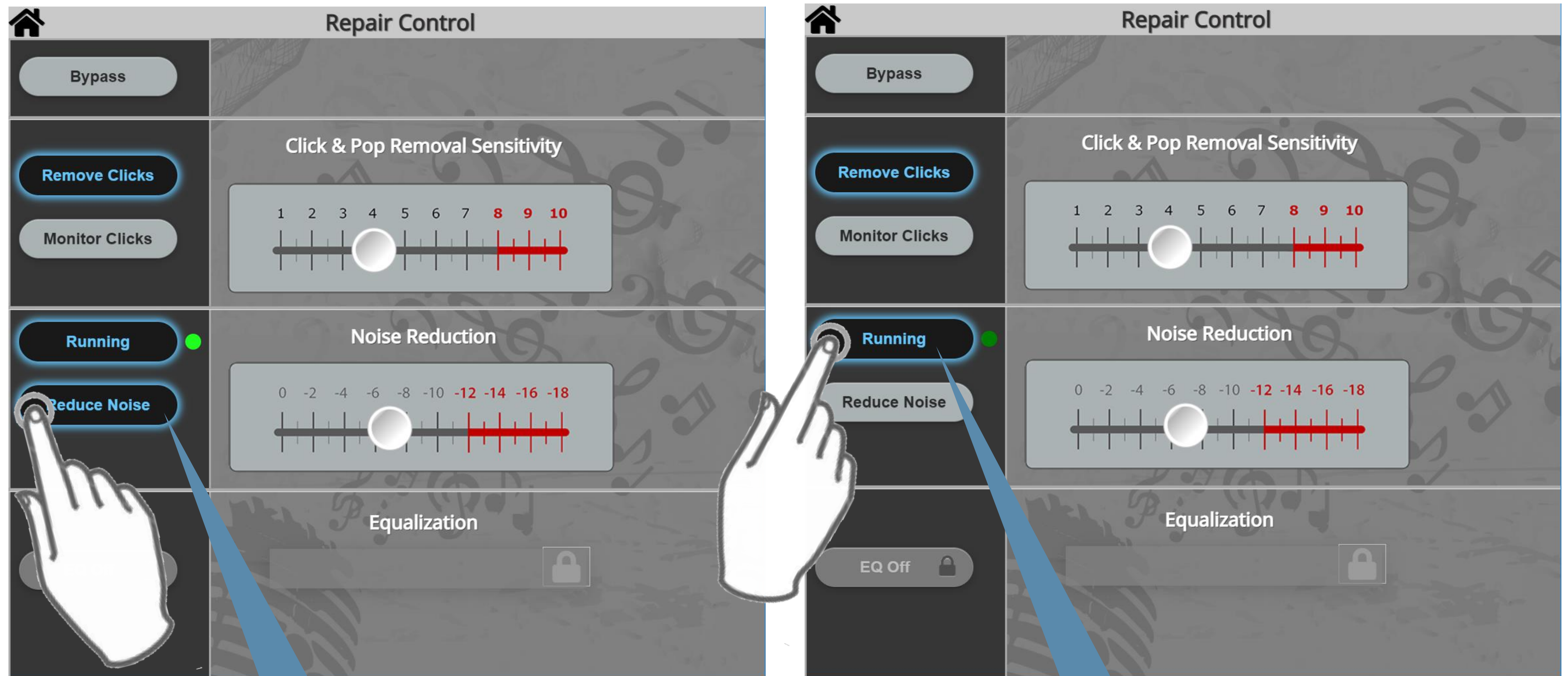


Note: **Record Equalization** is off (in this example) and cannot be started once NR is engaged.

'**Noise Reduction**'(NR) process samples noise at the lead-in to the record (for a few seconds)

4.4 Noise Reduction Functions

Controlling Noise Reduction



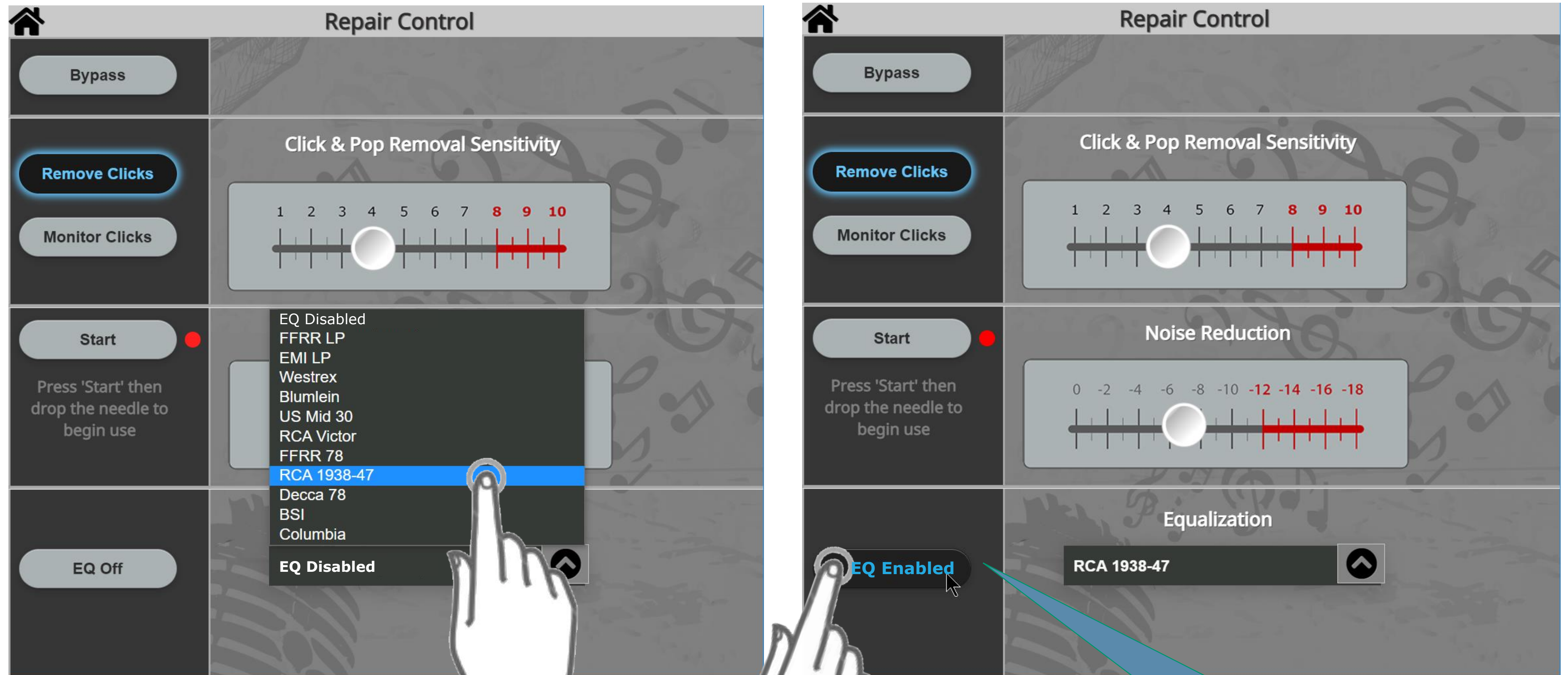
NR can be dis-engaged any time by clicking here.

Clicking on '**Running**' stops the NR process but you can leave it running indefinitely, it won't have an effect until you select **Reduce Noise** again.

NOTE: NR can be engaged or dis-engaged any time. While NR stays **running** it maintains the noise fingerprint initially sampled.

4.5 Equalization (EQ) Compensation

Controlling Playback EQ Curves

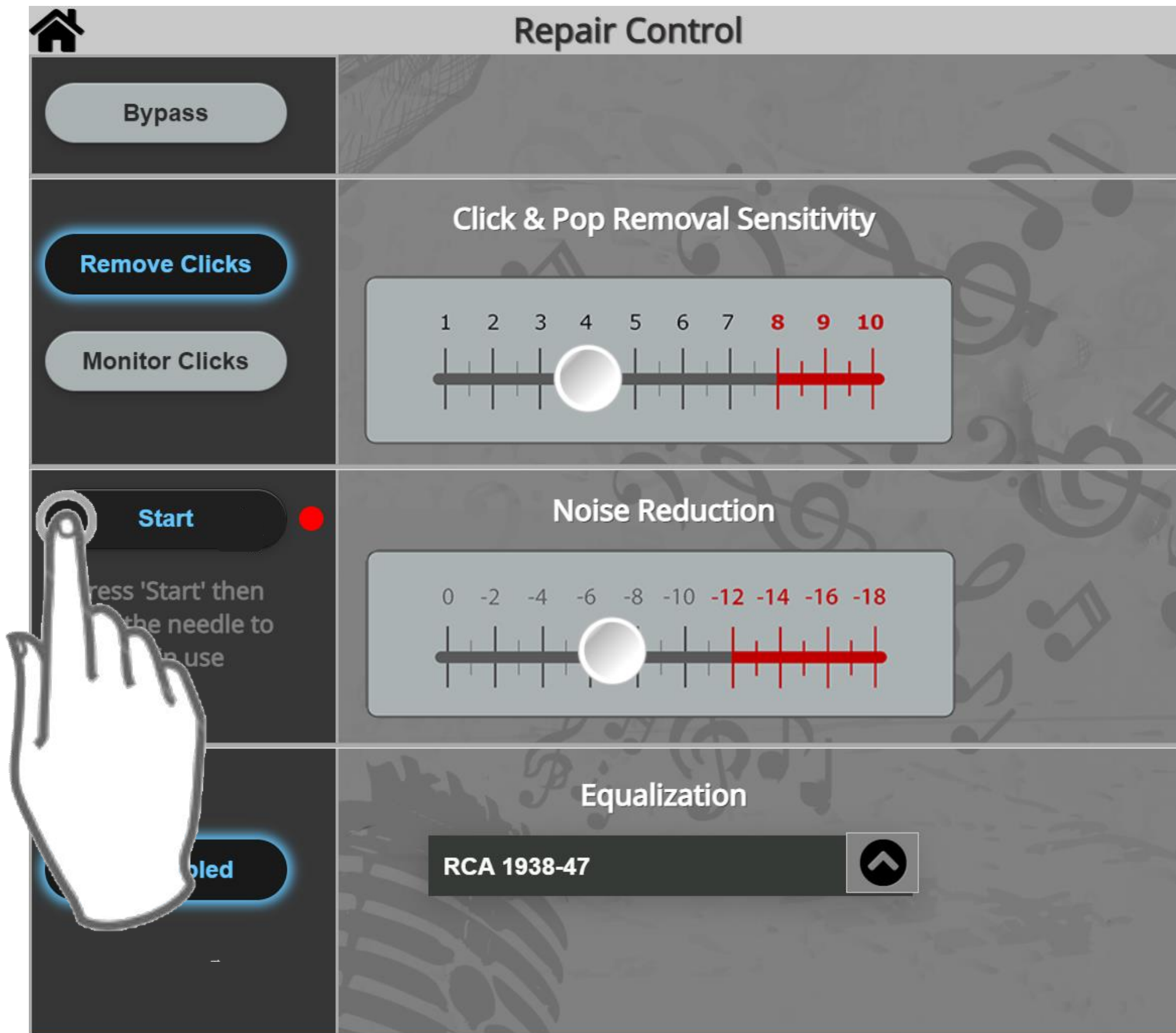


When NR is not running, an EQ curve can be chosen.

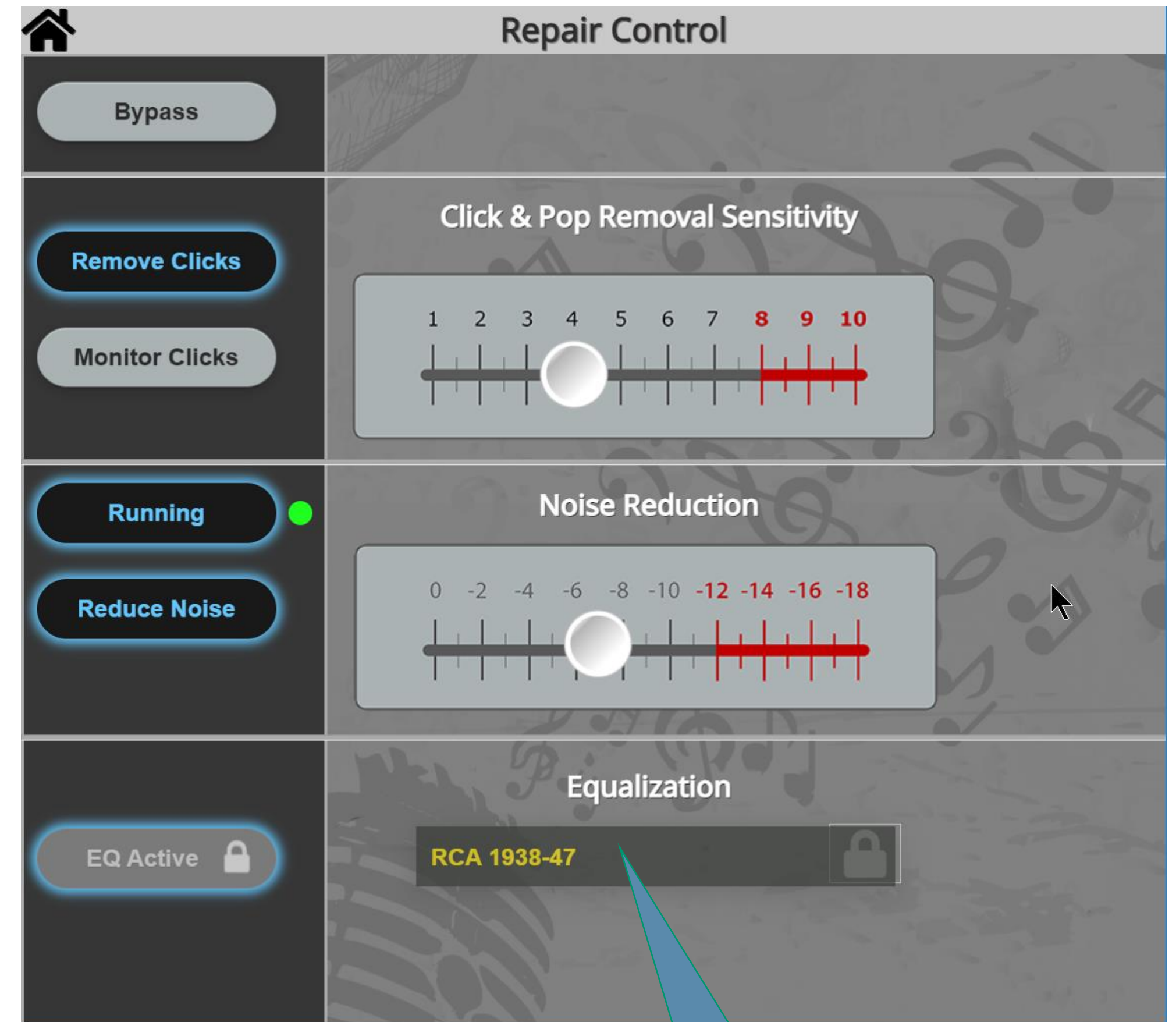
Clicking on EQ will enable it for the next time NR is started.

4.6 Equalization (EQ) Compensation

Controlling Playback EQ Curves



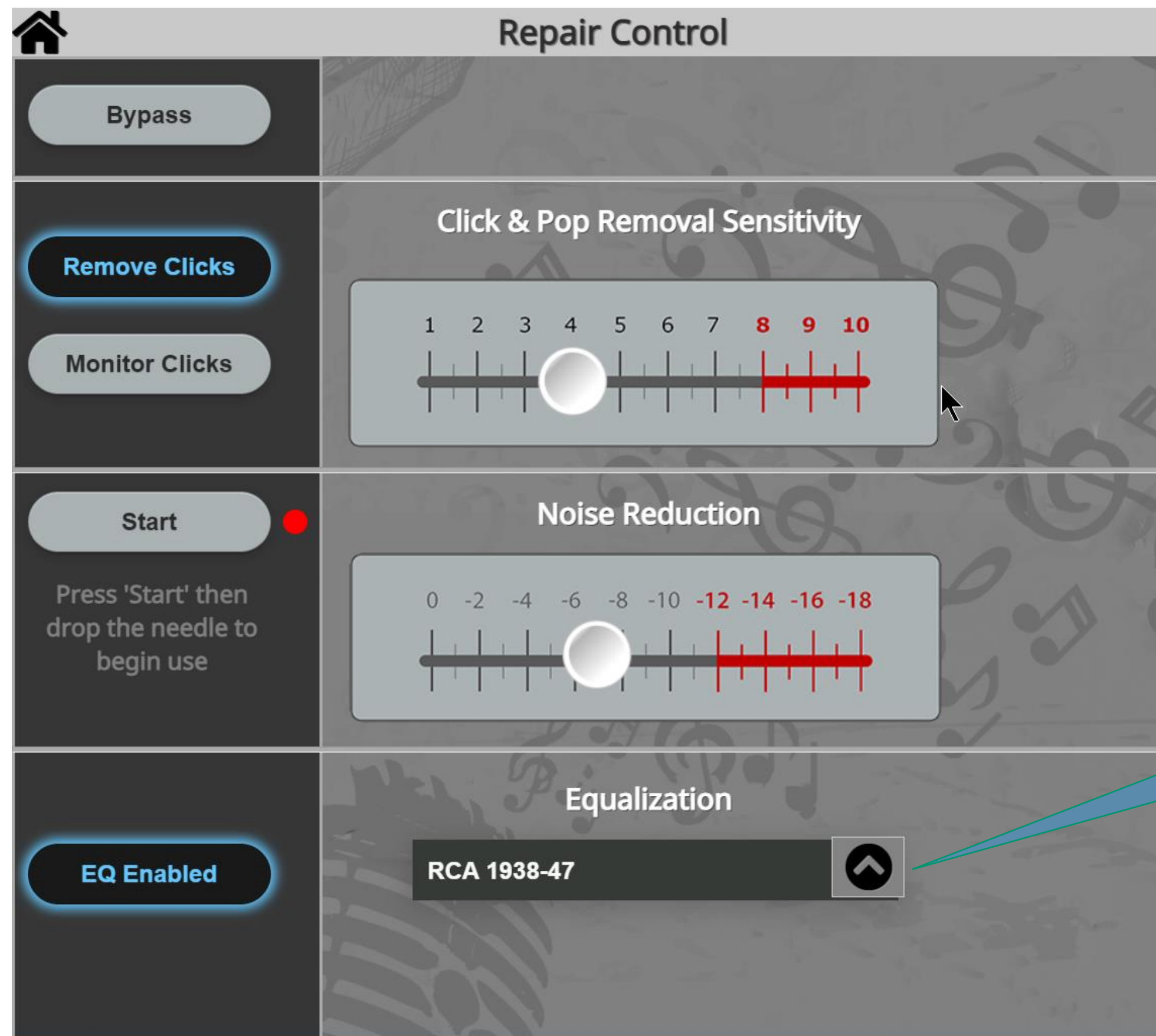
The chosen EQ curve will now be applied the next time NR is started.



While NR is running, EQ cannot be changed.

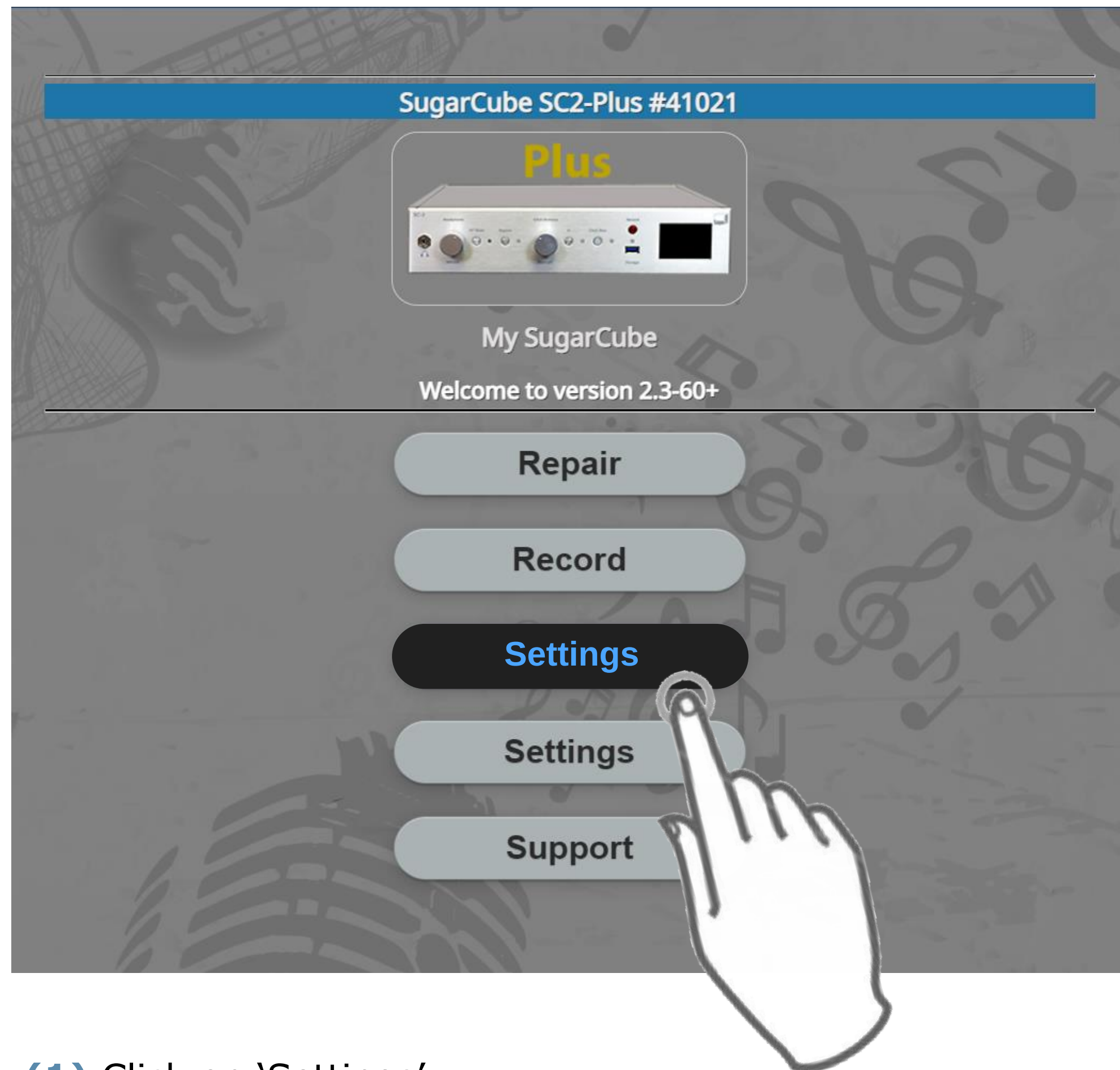
4.7 Equalization (EQ) Compensation

Controlling Playback EQ Curves

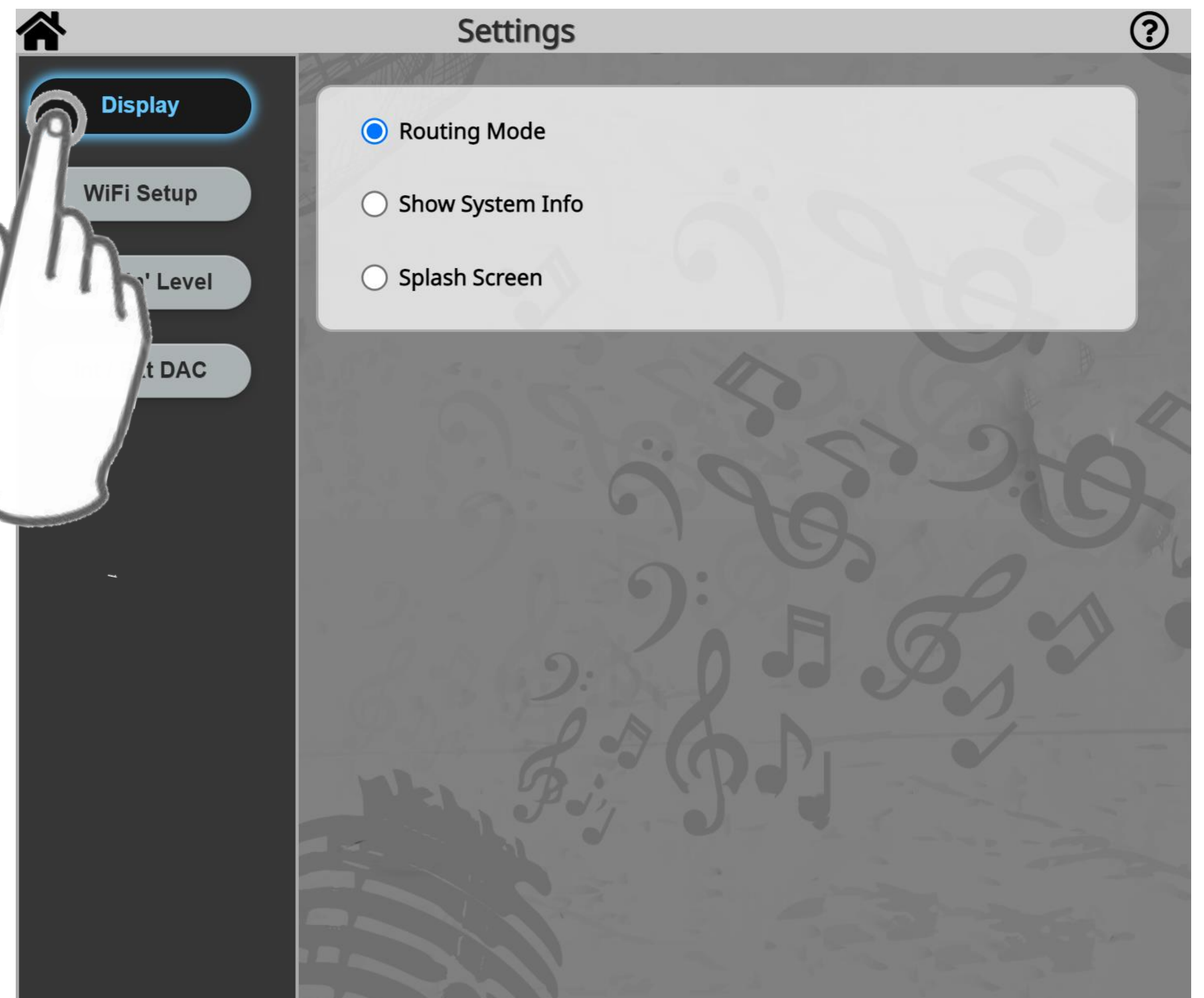


With NR not running, even though EQ is Enabled, it is not applied to the audio until NR is running again.

4.8 Settings and Display Functions

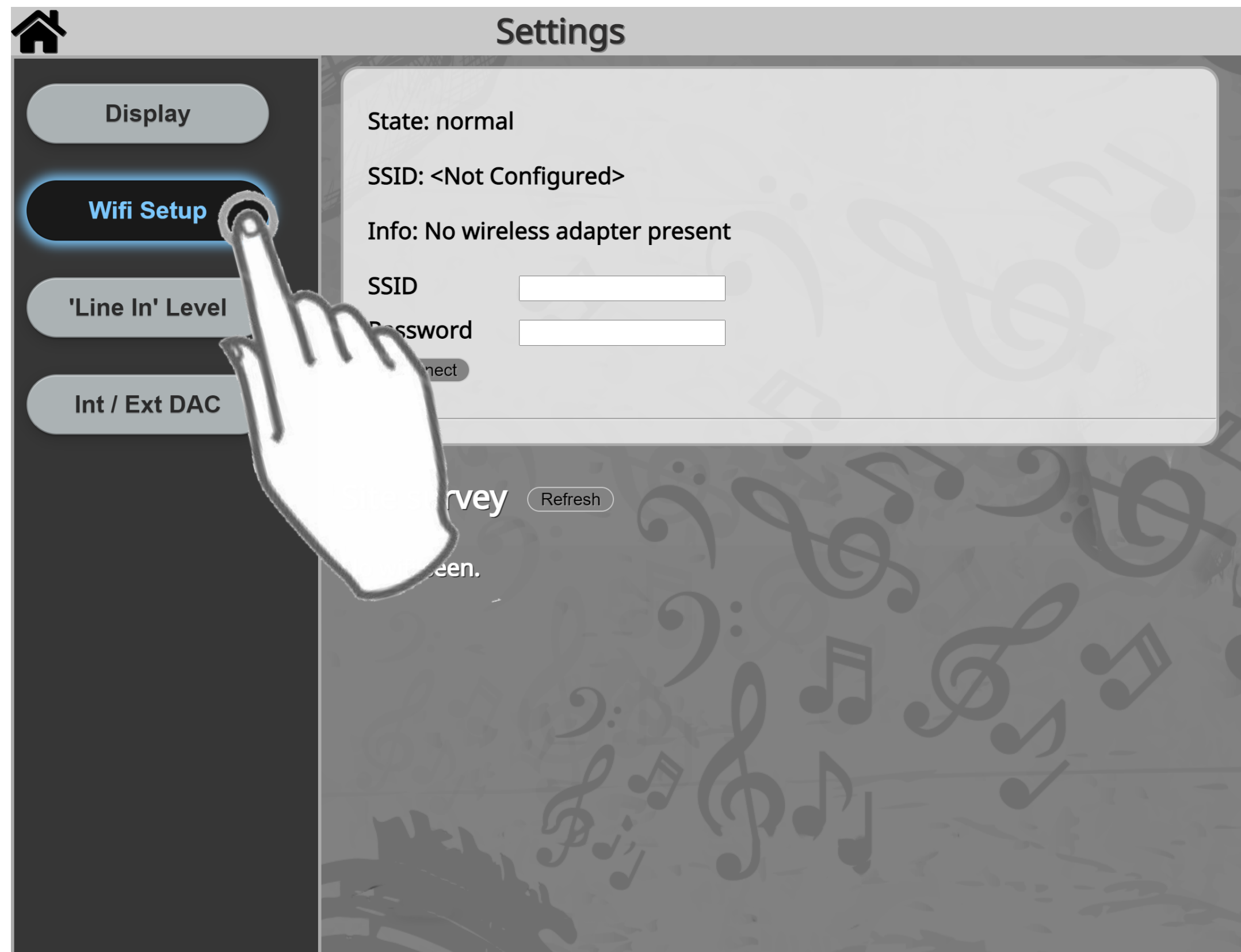


(1) Click on 'Settings'

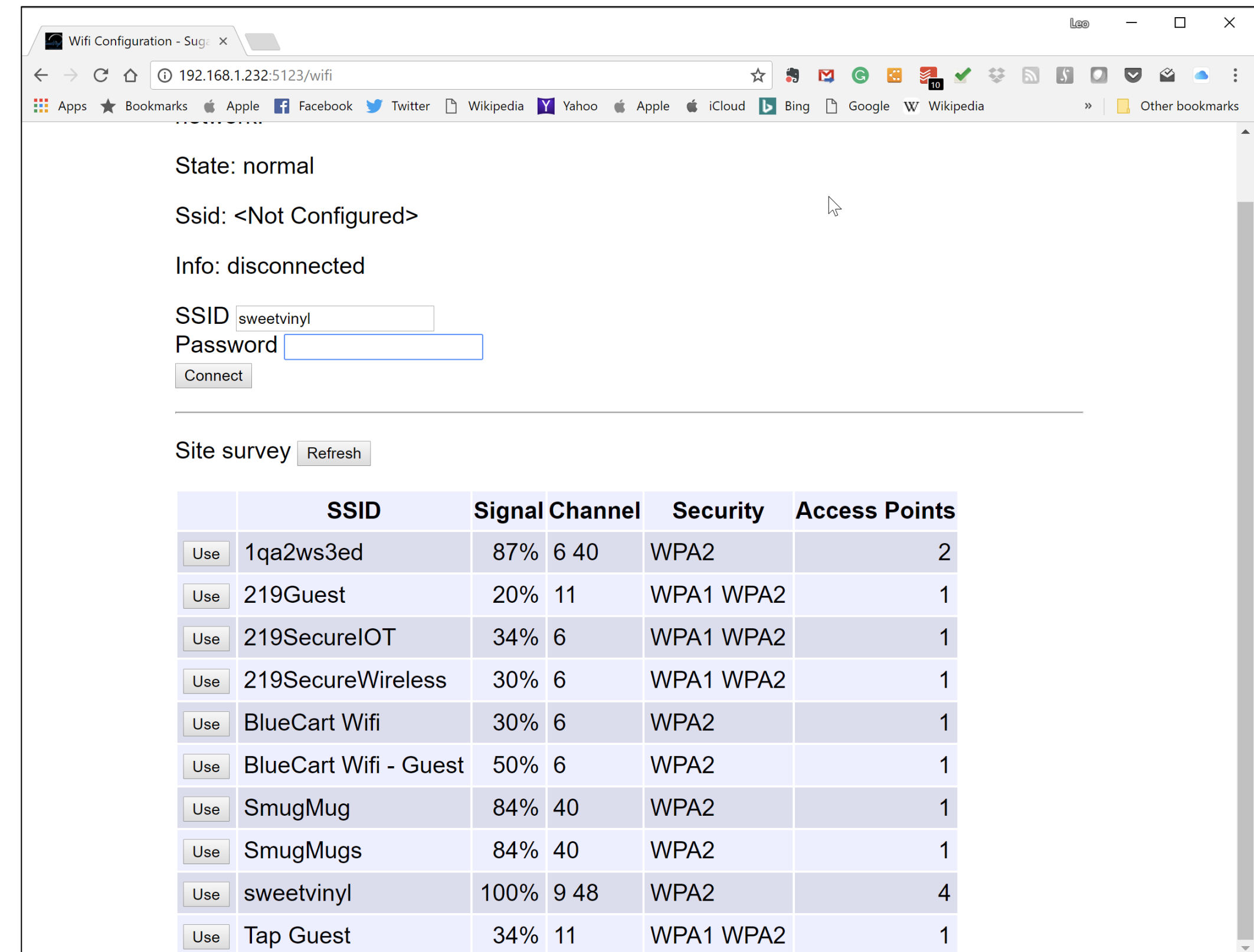


(2) Display choices include: **Routing Mode** to show audio routing information on the display; **Show System Info** such as **IP Address**, and **Software Version** (which can be obtained from the front panel by pressings and holding the '**Bypass**' button for about **five seconds**), and **Splash Screen** which is basically the model name.

4.9 Controlling the SugarCube via WiFi



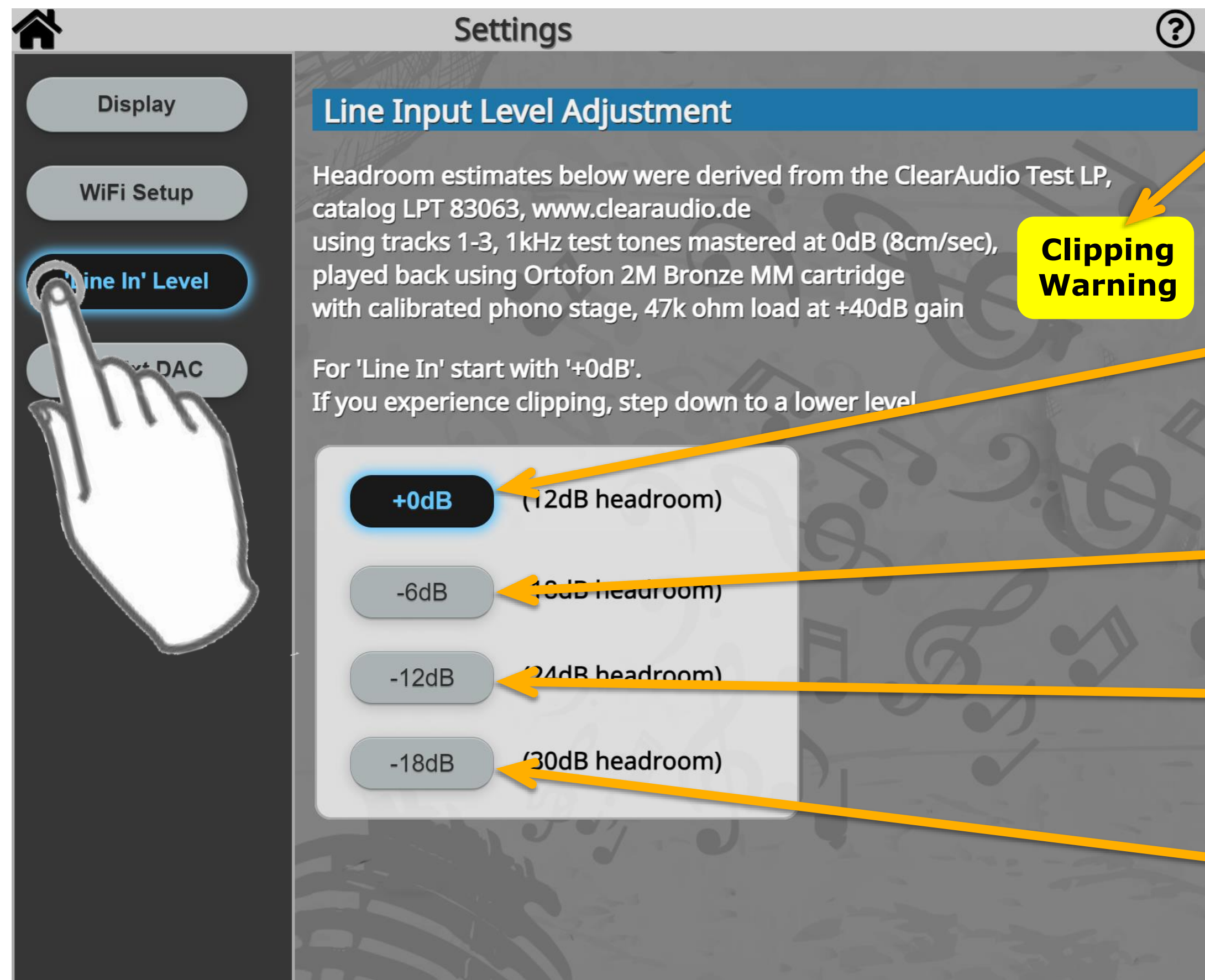
(1) Click on 'Wifi Setup'



(2) When the Wifi dongle is in the rear USB port available networks will appear and you can select the SSID and enter your password.

4.10 Setting the Line-Input Level Setting

PLEASE NOTE: unless you see a '**Clipping Warning**' or '**Clipping Alarm**' appear while playing music through your SugarCube, you do **NOT** need to change the audio setup.



NOTE: if this appears on startup or when, for example, you drop the needle on an LP, just touch it and it will go away.

This is the normal mode for the SC-Plus series. No need to attenuate the input, however, if the 'Clipping' alert is seen during use, go down to '-6dB'.

Input level is reduced by -6dB but added back on output. You now have 6 dB more headroom for listening or recording.

Input level is reduced by -12 dB but added back on output.

Input level is reduced by -18dB but added back on output

(1) Click on 'Line In Level.'

IMPORTANT NOTE: The output of the SugarCube will always equal the input. Changes in settings here will result in NO change in volume on your system.

4.10a Note on Clipping Alarm



Clipping Alarm

When music is playing and this message appears on any page of the interface, it means the audio has clipped and the Line In Level should be lowered one step, and maybe more.

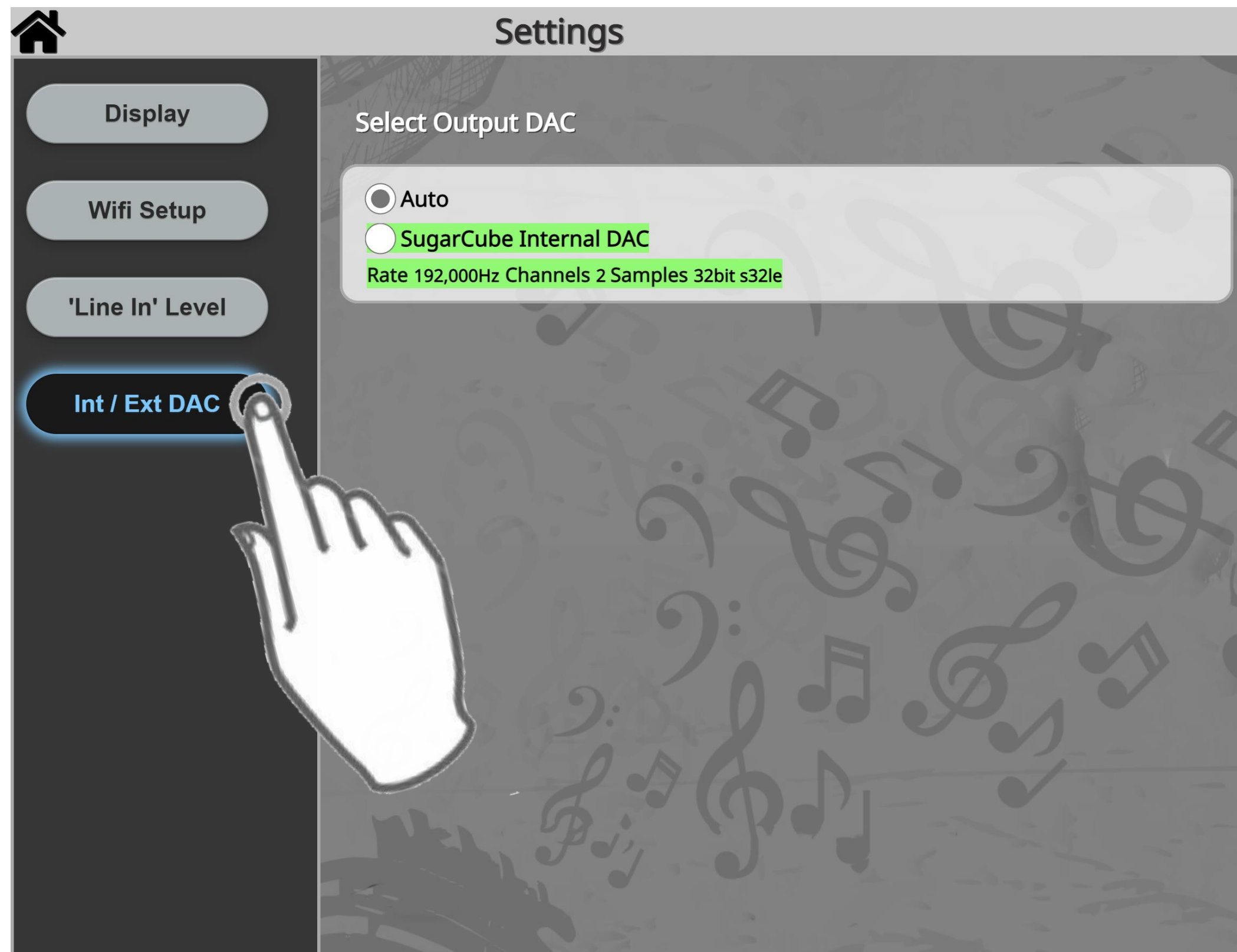
Touching or clicking on either message will clear the message until another related audio event should trigger it again.

Also, if you do not manually clear the messages, they will disappear after 20 minutes.

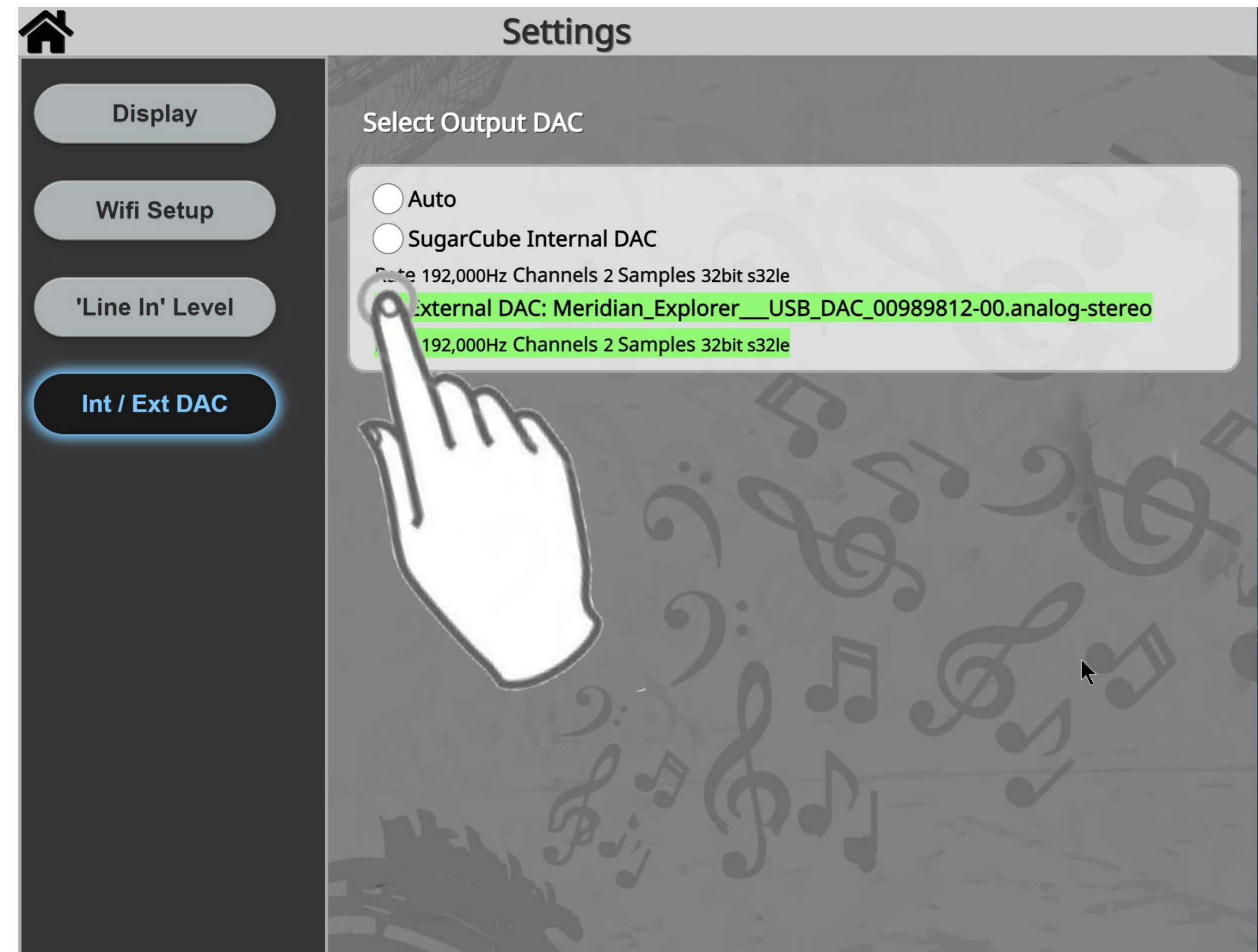
IMPORTANT NOTE: The output of the SugarCube will always equal the input. Changes in settings here will result in NO change in volume on your system.

4.11 Choosing an **External DAC** Output

Selecting an external DAC plugged into the SugarCube

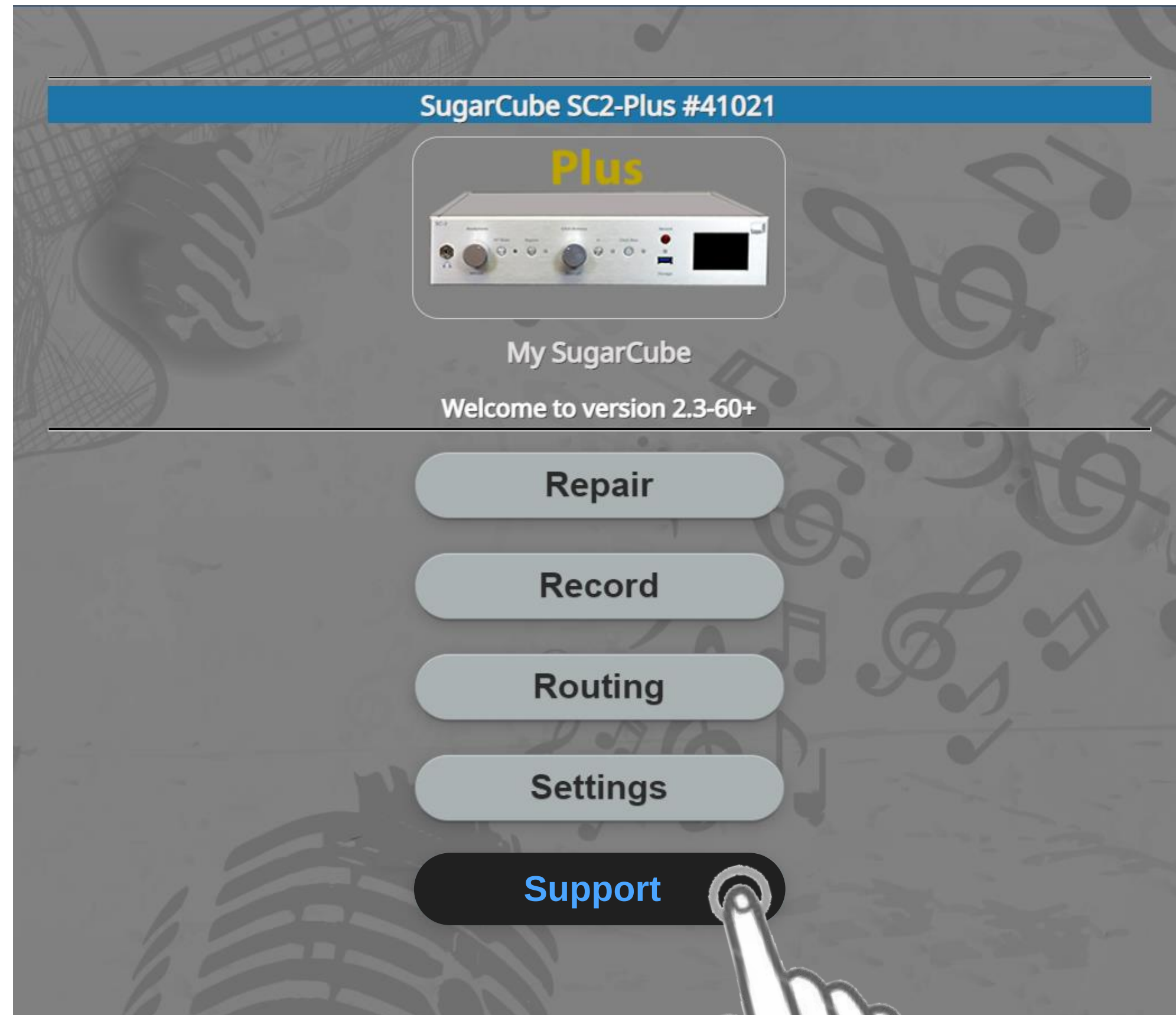


(1) Click on 'Int/Ext DAC.'

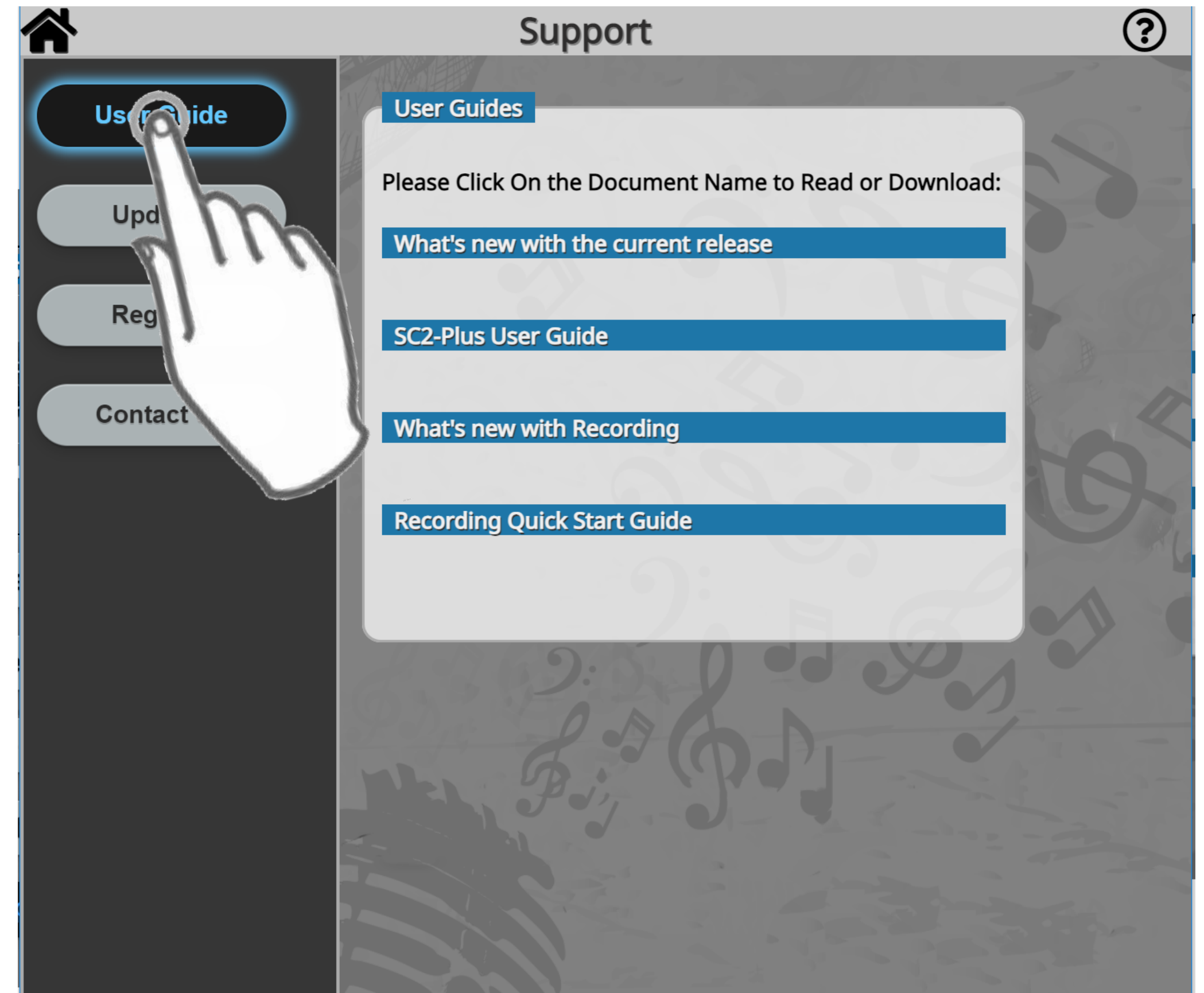


(2) When a USB DAC is inserted in one of the rear USB ports, it will appear here and can be selected instead of the internal DAC.

4.12 Support Menu & User Guide

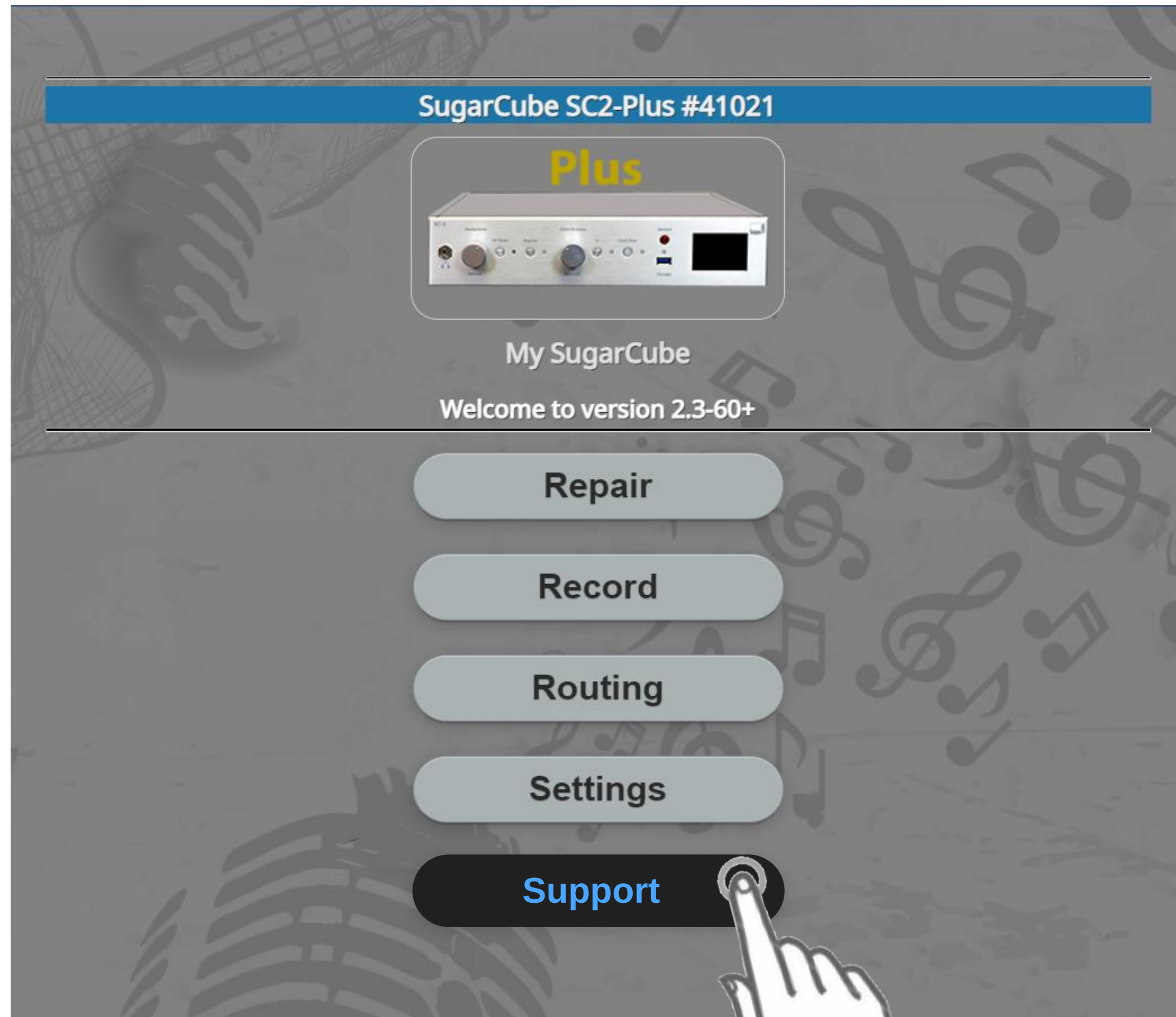


(1) Click on 'Support'

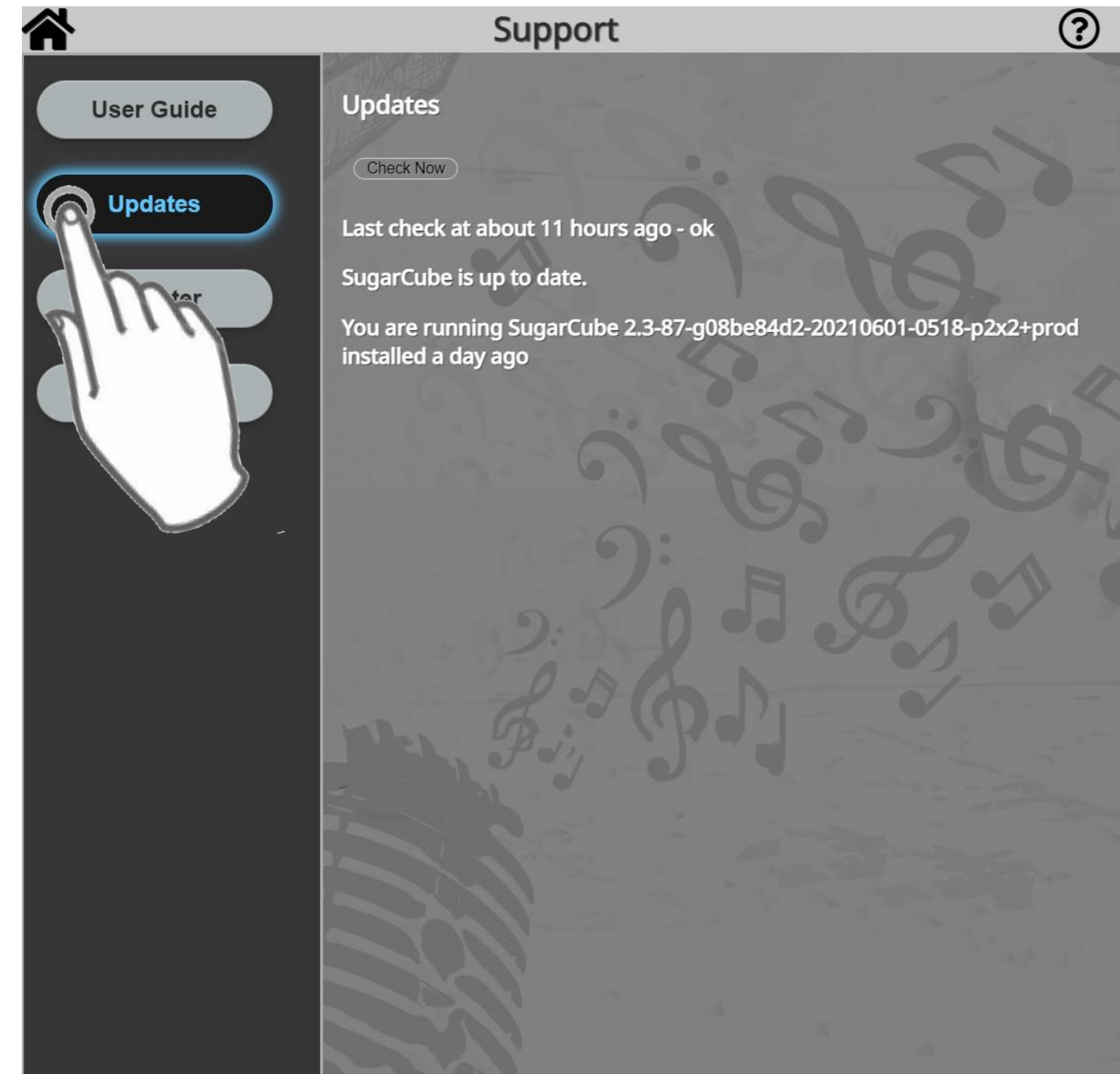


(2) The current User Guides for your model is available for download and/or viewing in the browser here.

4.13 Checking for Updates



(1) Click on 'Support.'



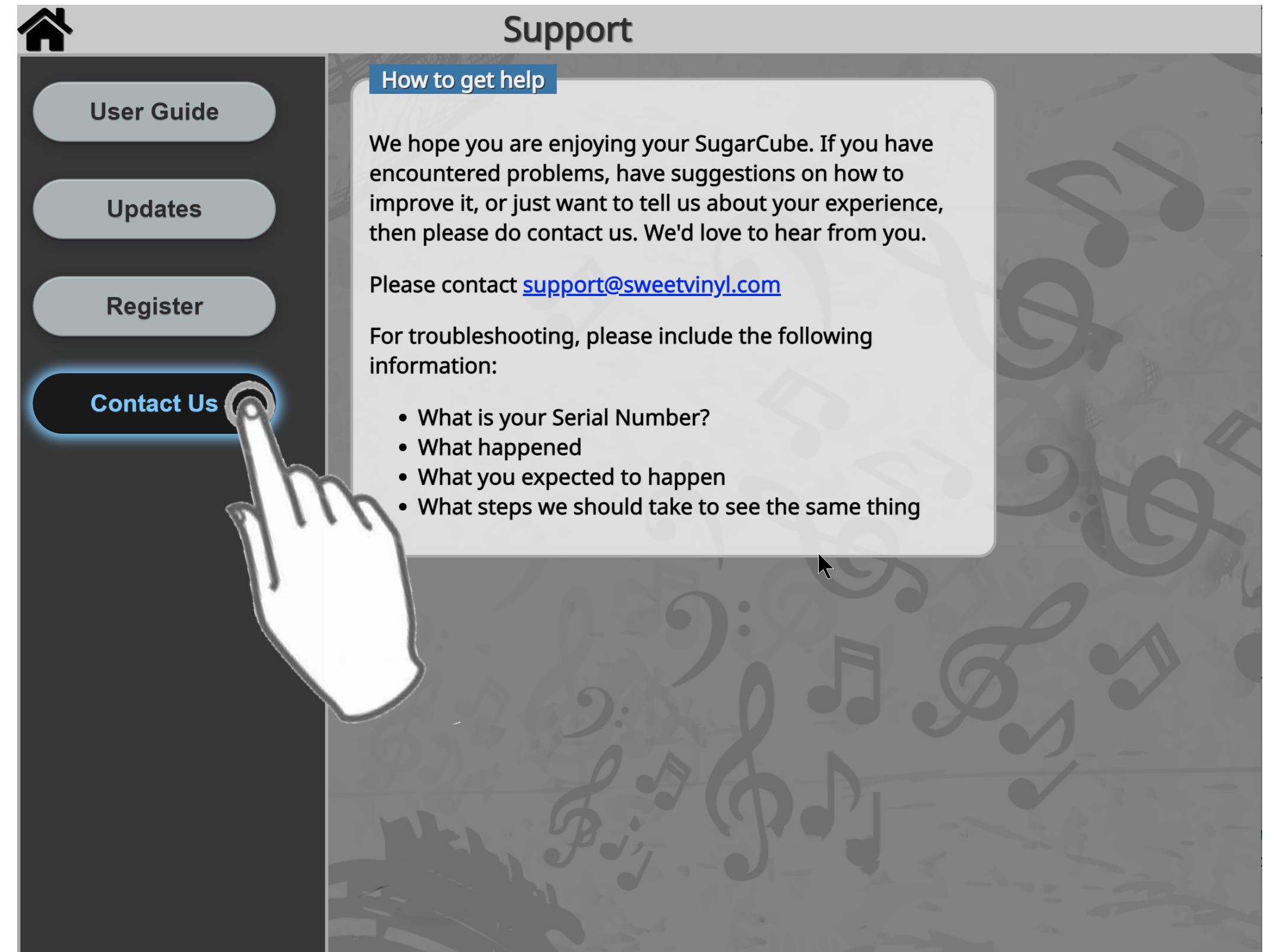
(2) Current SugarCube software version is displayed and updates can be checked. If updates are waiting can be updated here. Also shows last update and if an update is coming.

4.14 Register and Email Info



The screenshot shows a web interface titled 'Support' with a home icon in the top left. A sidebar on the left contains buttons for 'User Guide', 'Updates', 'Register', and 'Contact Us'. The 'Register' button is highlighted with a hand cursor. The main content area is titled 'Register your SugarCube:' and lists three bullet points: 'SweetVinyl registration for warranty purposes', 'Opt-in to newsletters about upcoming updates', and 'Opt-in to newsletters about new SweetVinyl products'. Below this, it says 'Newsletters are in English.' and 'Contact Information'. The form includes fields for 'First Name', 'Last Name', 'Email', 'Country', 'Purchased' (with a date picker set to 'mm/dd/yyyy'), and 'Where bought'. At the bottom, there is a 'Contact me about' section with three checkboxes: 'Updates to this SugarCube', 'Updates from SweetVinyl', and 'Selected third party products and services'. A 'Save' button is at the bottom right.

(1) Click on 'Register' and you can enter your information.



The screenshot shows the same 'Support' page, but the 'Contact Us' button in the sidebar is now highlighted with a hand cursor. The main content area is titled 'How to get help' and contains the following text: 'We hope you are enjoying your SugarCube. If you have encountered problems, have suggestions on how to improve it, or just want to tell us about your experience, then please do contact us. We'd love to hear from you.' Below this, it says 'Please contact support@sweetvinyl.com'. Further down, it says 'For troubleshooting, please include the following information:' followed by a list of four bullet points: 'What is your Serial Number?', 'What happened', 'What you expected to happen', and 'What steps we should take to see the same thing'.

(2) 'Contact Us' will take you to a screen where you can follow instructions to send an email to support@sweetvinyl.com.

Section 5

Connecting a Computer to your SugarCube with USB

MacOS, Windows,
or Music Server

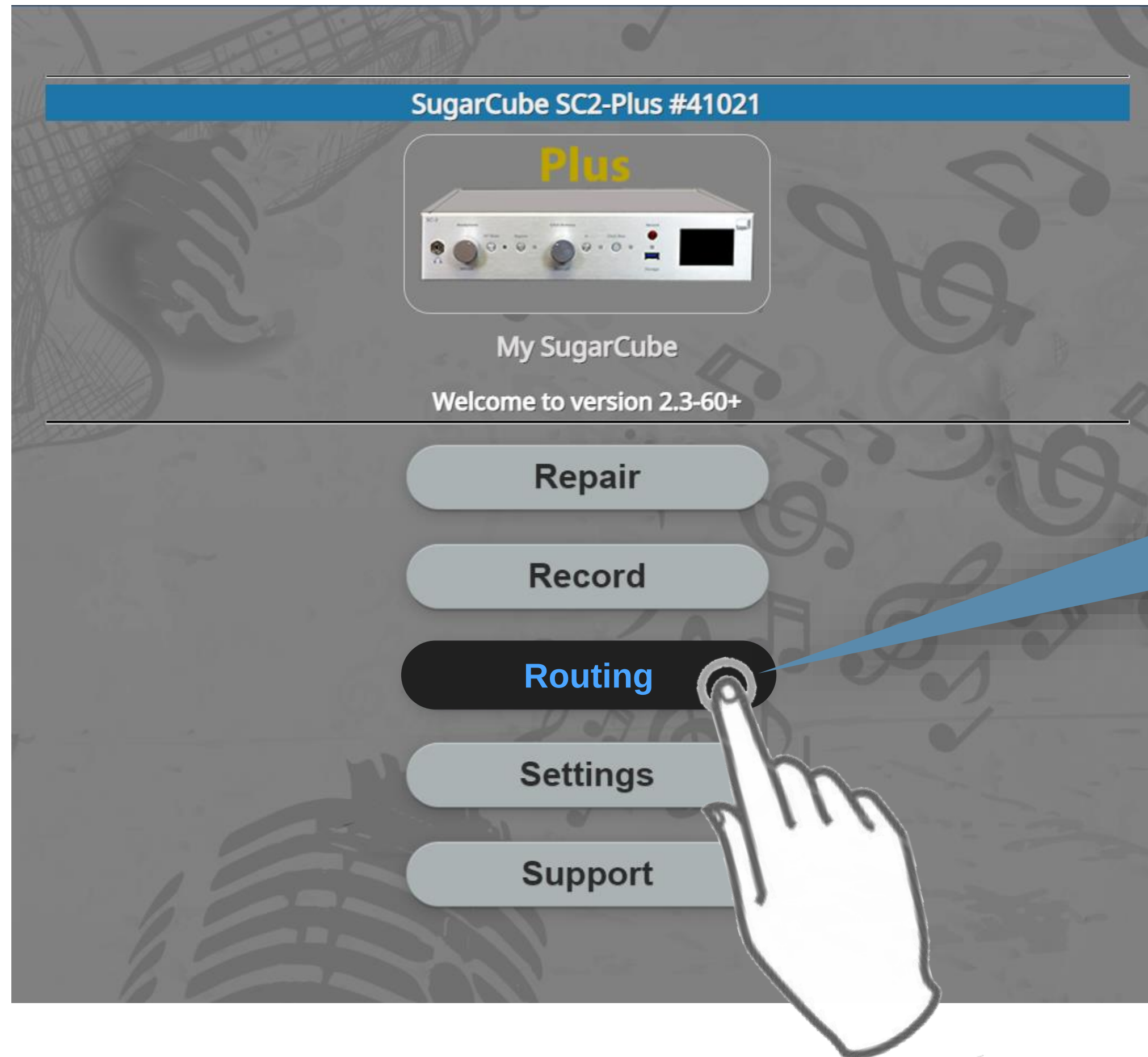


5.1 Connecting a Computer over USB



The SC2-Plus can be connected to a computer via the **USB-B** port. Plug one end into the USB port on the computer and the other to the **USB-B** port on the SugarCube. When connected, the computer will be able to control the **ADC** and **DAC** functions on the SugarCube. The computer can record the output of the **ADC** with any software available for recording or play digital files directly through the SugarCube.

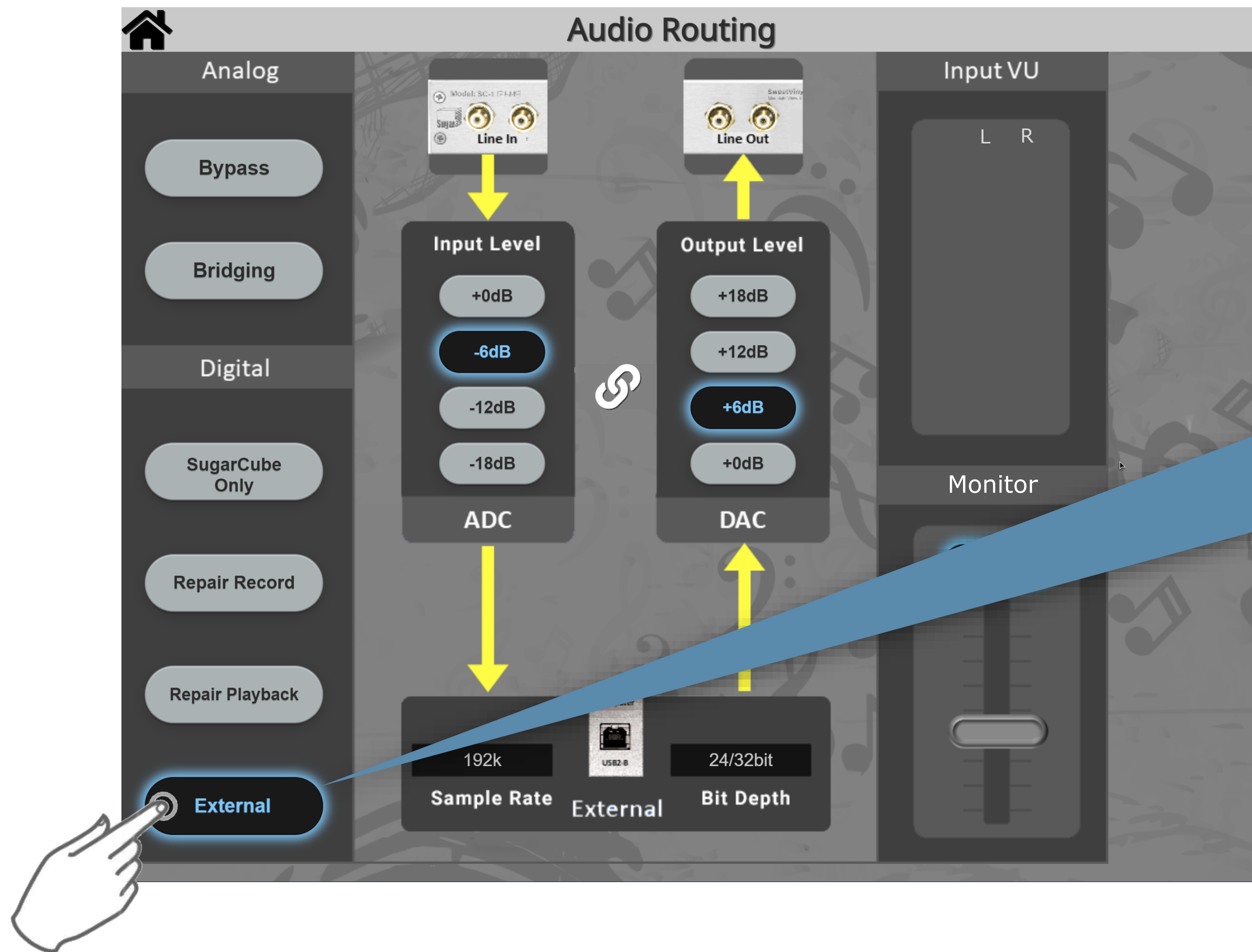
5.2 Audio Routing Menu to enter **External Mode**



Select the **Routing Menu** to manage the analog and digital signal paths.

To efficiently work with the SC2-Plus and a computer connected over USB you will want to open the **Routing Menu** page. The next pages show how the SugarCube, and an external computer, work together to provide the enhanced functionality.

5.3 Audio Routing Menu — External Mode



Step 1 - enter External mode that allows your computer to use the **ADC** and **DAC** at **any sample rate**.

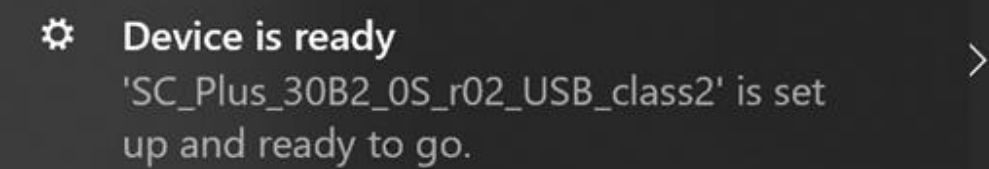
In **External** mode, the **SC2-Plus** acts like **any other** studio-quality **ADC/DAC** pair and will function at any standard sample rate and bit depth from 192k down to 44.1k and at 16/24/32 bit sample sizes.

Note: Please **press again** the button of **the current mode** of operation if the external device **loses lock** with the SugarCube (you will certainly know due to obvious digital noise.) This resets the internal timing and restores the audio.

5.4 Configuring Windows to control the SugarCube

1

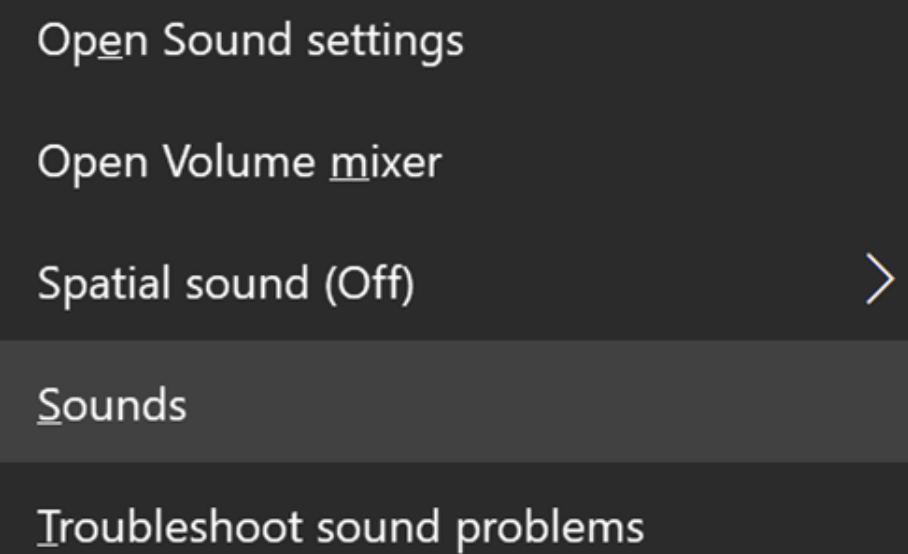
After connecting your Windows PC, the SugarCube should configure itself.



Device is ready
'SC_Plus_30B2_0S_r02_USB_class2' is set up and ready to go.

2

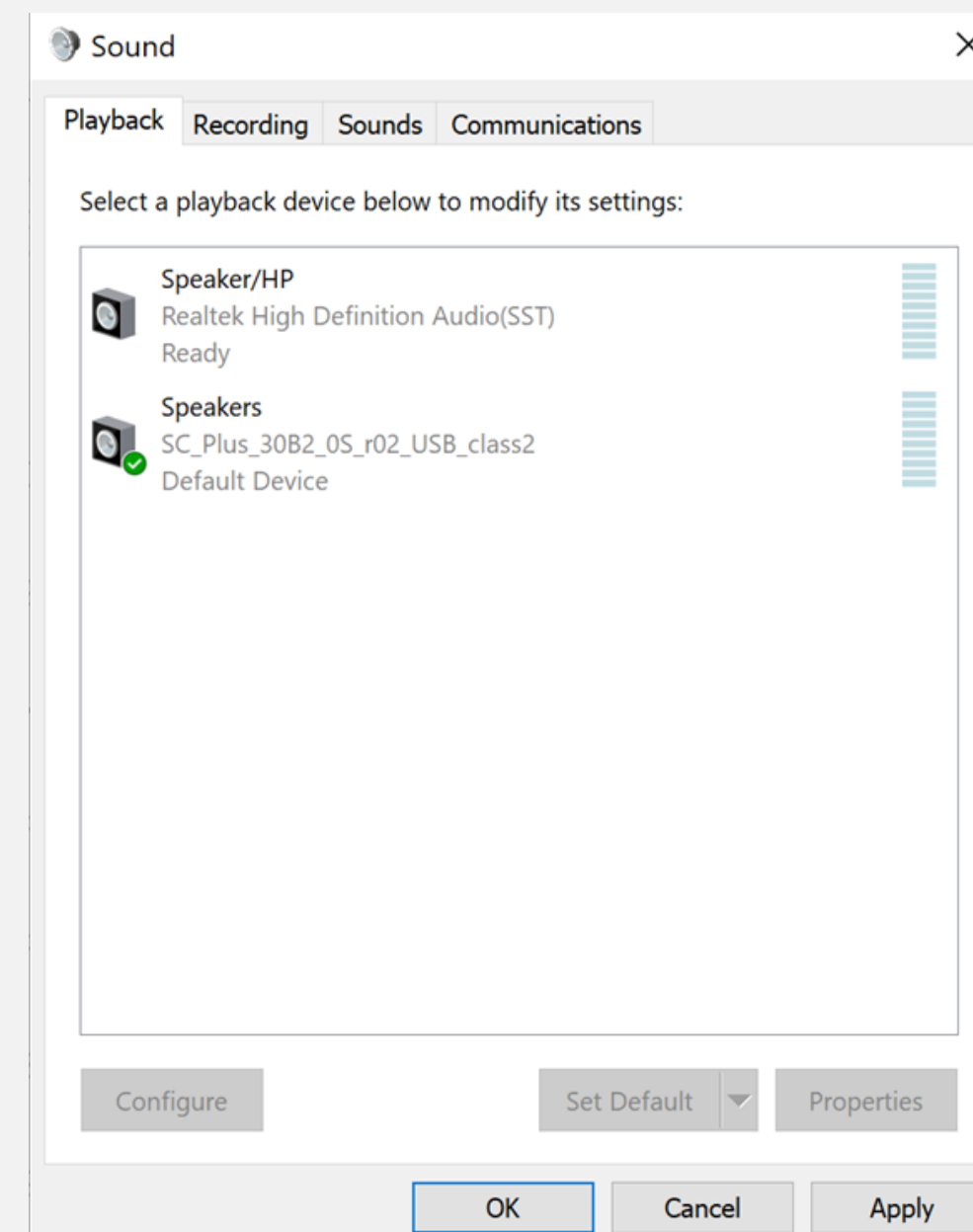
Drivers are built-in and you should be able to see and configure the SugarCube as a device under **Sounds** by right clicking on the speaker Icon in the System Tray (right end of the Task Bar.)



Open Sound settings
Open Volume mixer
Spatial sound (Off)
Sounds
Troubleshoot sound problems

3

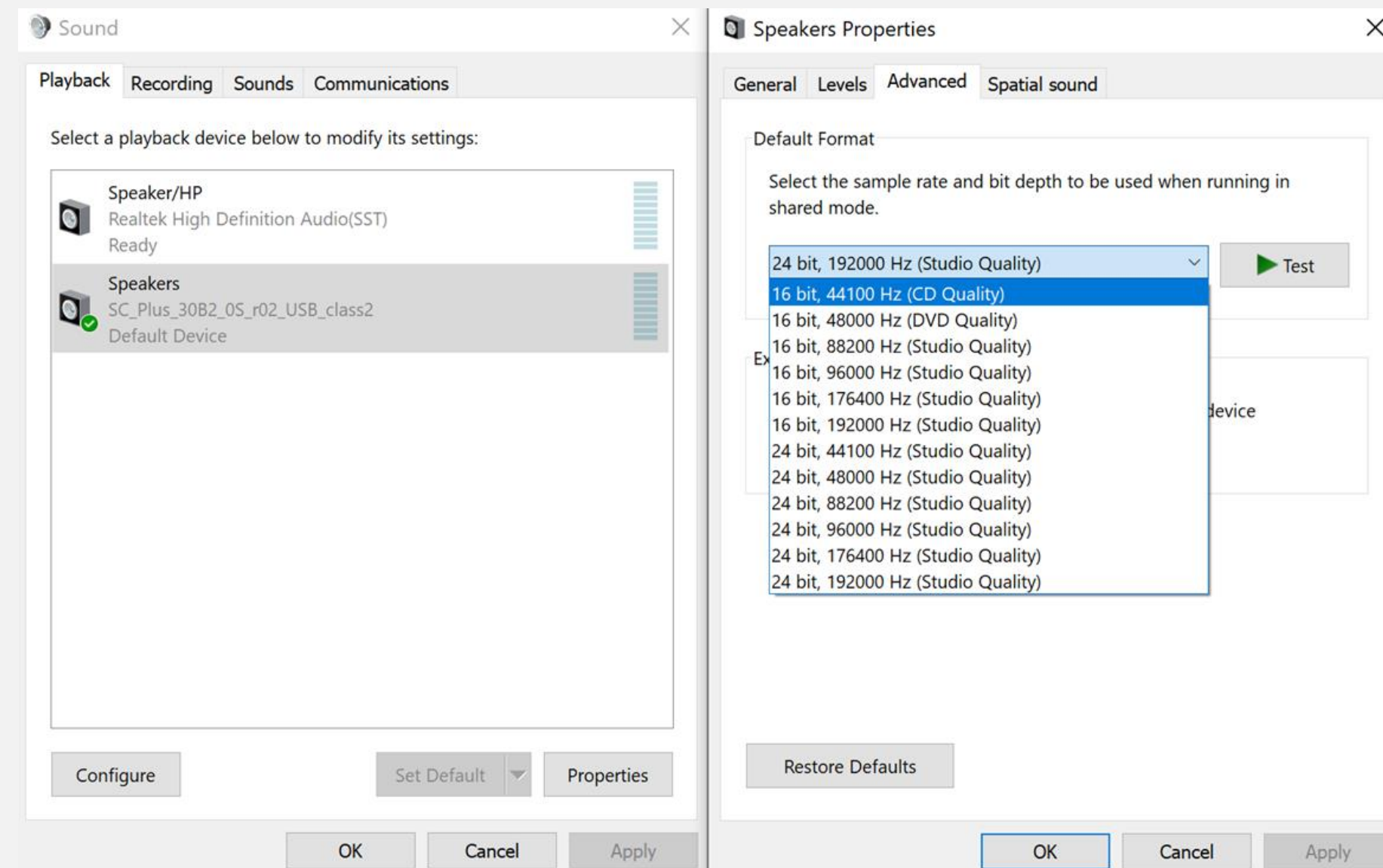
A Window like this will open:



5.5 Configuring Windows to control the SugarCube

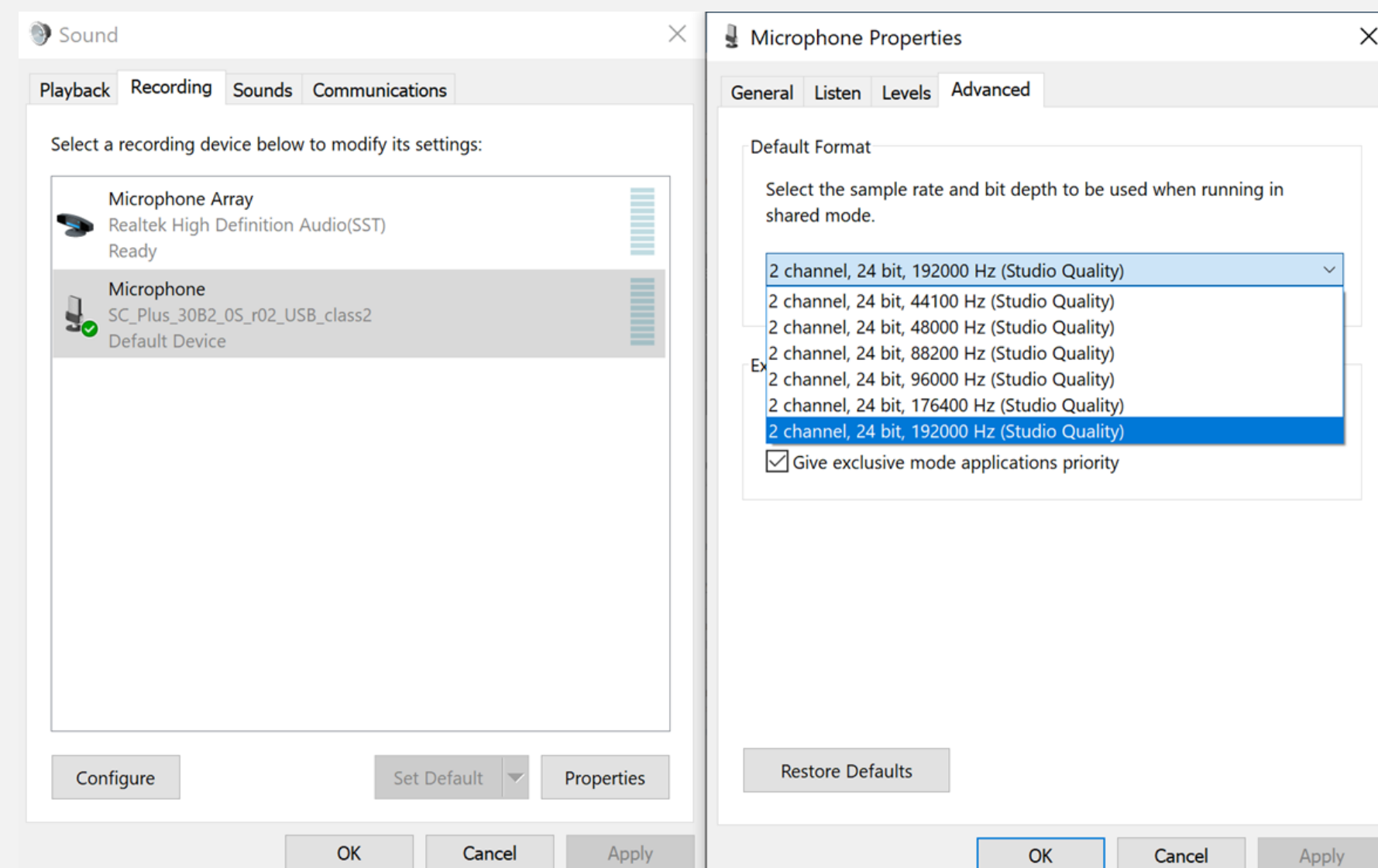
4

As a **Playback** device you will be able to configure the SugarCube **DAC** as a **Speaker** – by selecting the **SC_Plus** you can then select Properties and then the **Advanced** tab to select the sample rate and bit depth. You will want to set this to **24-bit, 192,000 Hz** if you want the SugarCube functions to work with the external source.



5

As a **Recording** device the SugarCube **ADC** will show up as a **Microphone** by selecting the SC_Plus you can then select **Properties** and then the **Advanced** tab and select the sample rate and bit depth. You will want to set this to **24-bit, 192,000 Hz** if you want the SugarCube functions to work with the external device.



5.6 Configuring MacOS to control the SugarCube

1

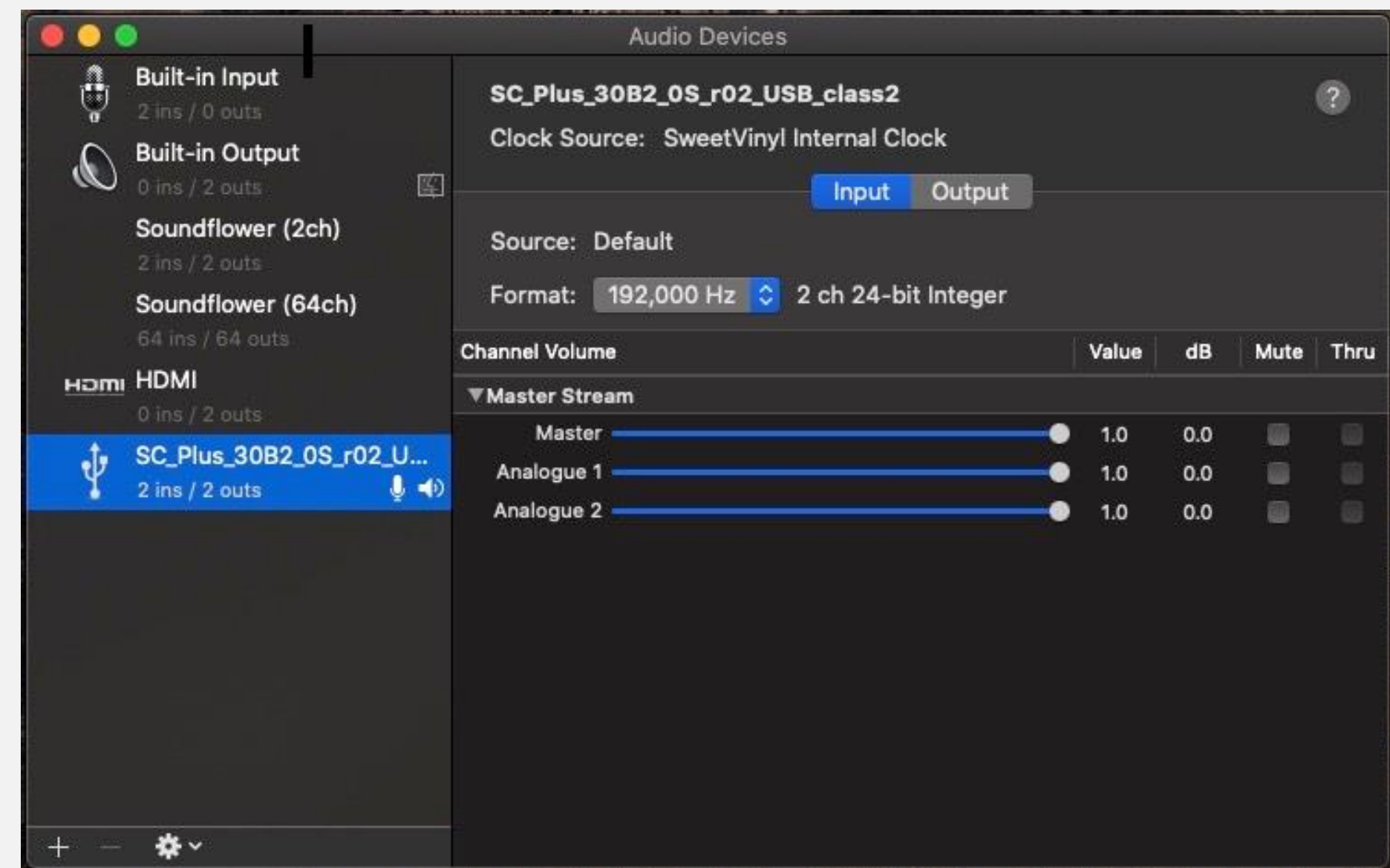
After connecting your MacOS computer, the SugarCube should configure itself.

2

Drivers are built-in and you should be able to see and configure the SugarCube as a device using the “**Audio Midi Setup**” app by searching for this or by going to the “**Utilities**” folder in “**Applications**”

3

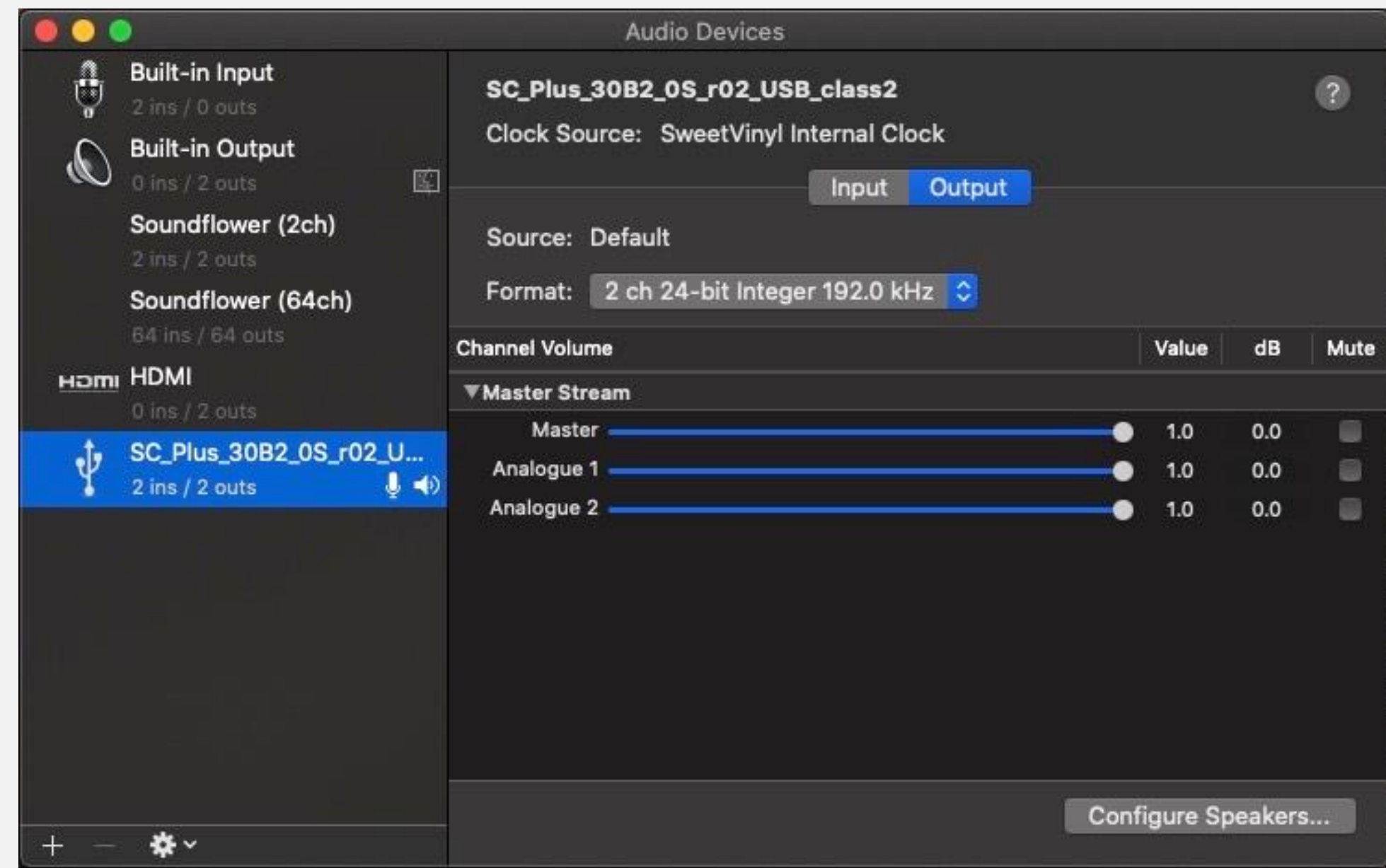
A Window like this will open and you will want to further select the “**Audio Setup**” section:



5.7 Configuring MacOS to control the SugarCube

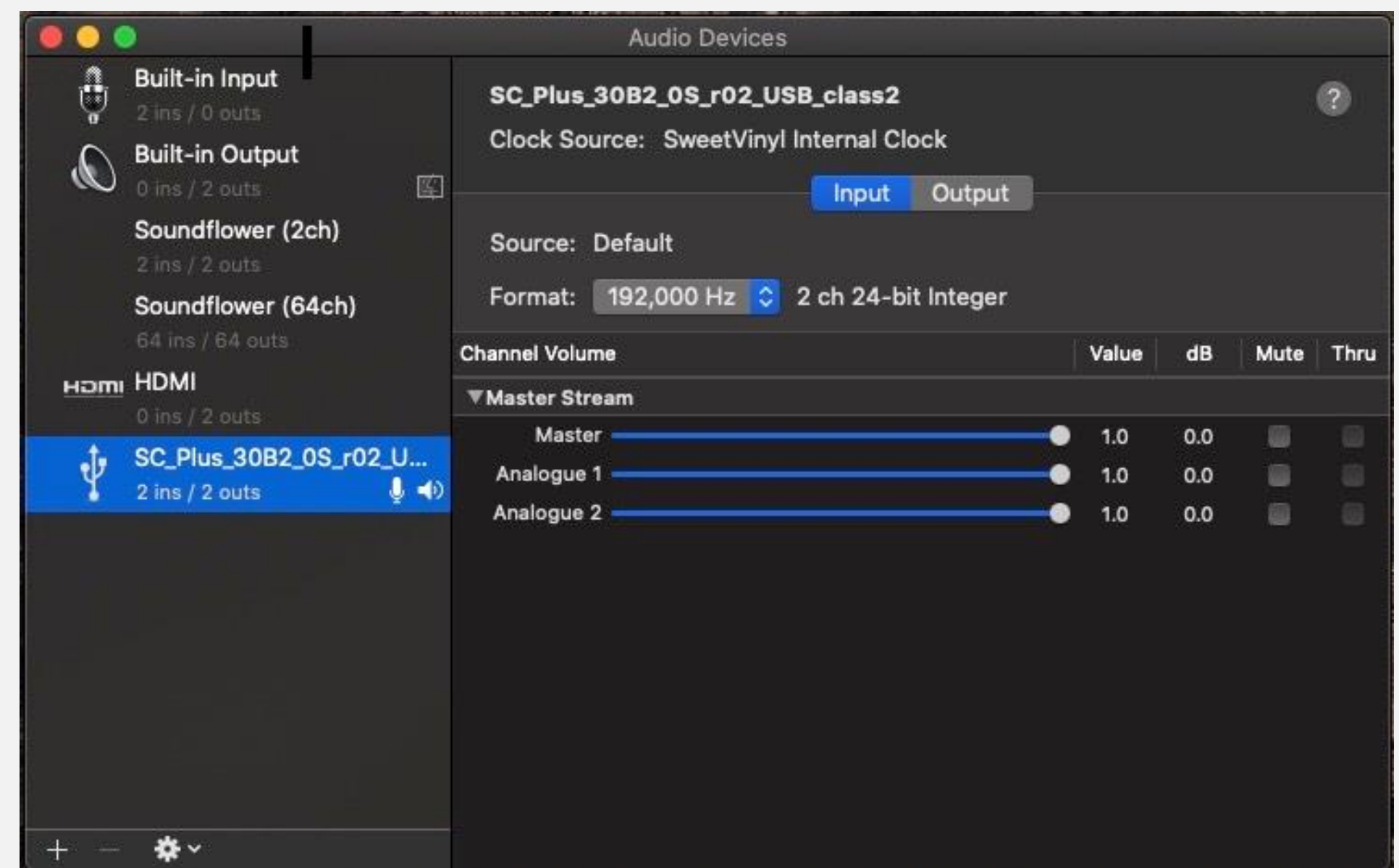
4

As an **Output** device you will be able to configure the **SugarCube DAC** – by selecting the **SC_Plus** you can the sample rate and bit-depth. You will want to set this to **24-bit, 192.0 kHz** if you want the SugarCube functions to work with a MacOS app.

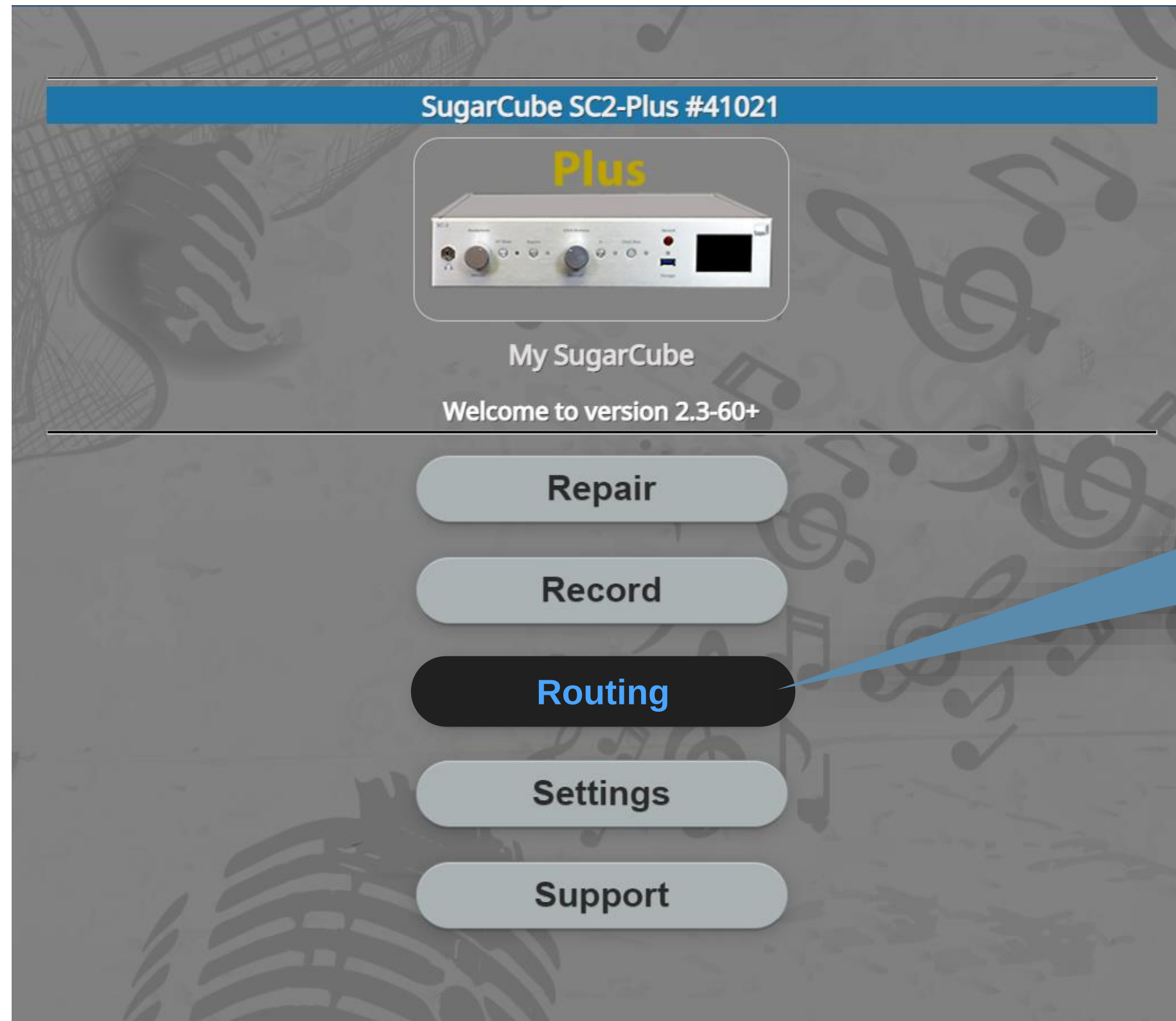


5

As an **Input** device the **SugarCube ADC** will appear and by selecting the **SC_Plus** you can select the sample rate and bitdepth. You will want to set this to **192,000 Hz** if you want the SugarCube functions to work with a MacOS app. Also, note bit-depth is usually fixed to whatever is the output but if at 16bit, you will need to reset it to 24bit.



5.8 Audio Routing Menu



Select the **Routing Menu** to manage the analog and digital signal paths.

To efficiently work with the SC2-Plus and a computer connected over USB you will want to open the **Routing Menu** page. The next pages show how the SugarCube, and an external computer, work together to provide the enhanced functionality.

5.9 Audio Routing Menu — Control Layout

The screenshot displays the 'Audio Routing' control interface. On the left, a vertical menu lists 'Analog' and 'Digital' routing modes. The 'Analog' section includes 'Bypass' and 'Bridging' buttons. The 'Digital' section includes 'SugarCube', 'Repair Record', 'Repair Playback', and 'External' buttons. The main area shows 'Line In' and 'Line Out' ports at the top, connected by a yellow arrow. Below them are 'Input Level' and 'Output Level' sliders, both set to '+0dB'. A chain icon connects the two sliders. At the bottom, a 'Sample Rate' section shows '192k' and '24/32bit' options, with a 'USB2-B' icon. On the right, an 'Input VU' meter shows levels for 'L' and 'R' channels, and a 'Monitor' section with a 'Mute' button. Annotations with yellow arrows point to various elements: 'Line-In level setting controls the level of input audio to the ADC', 'External computer or music server connects to SugarCube via USB2.', 'The SC1-Plus appears as a combination ADC/DAC to the computer', 'Line-Out level setting controls the level of audio gain after the DAC', 'VU Meter shows level of input as applied to the ADC', and 'Level of audio sent to browser as a convenience monitor of input signal'.

Analog Routing Modes

- Bypass
- Bridging

Digital Routing Modes

- SugarCube
- Repair Record
- Repair Playback
- External

Line-In level setting controls the level of **input audio** to the **ADC**

External computer or music server connects to SugarCube via **USB2**.

The **SC1-Plus** appears as a combination **ADC/DAC** to the computer

Line-Out level setting controls the level of **audio gain after the DAC**

VU Meter shows level of input as applied to the ADC

Level of audio sent to browser as a convenience monitor of input signal

5.10 Audio Routing Menu — Bypass Mode



When **Bypass** is selected, **Line-In** connects directly to **Line-Out** thru an audiophile-grade mechanical relay.

In **Bypass** mode, **Line In** connects to **Line Out** via a mechanical relay; this can be selected on this menu or on the front panel. When in **Bypass** no signal is processed by the SugarCube nor by a connected computer.

5.11 Logical Flow — Bypass Mode

Line-In connects directly to Line-Out.



Bypass

Bridging

SugarCube

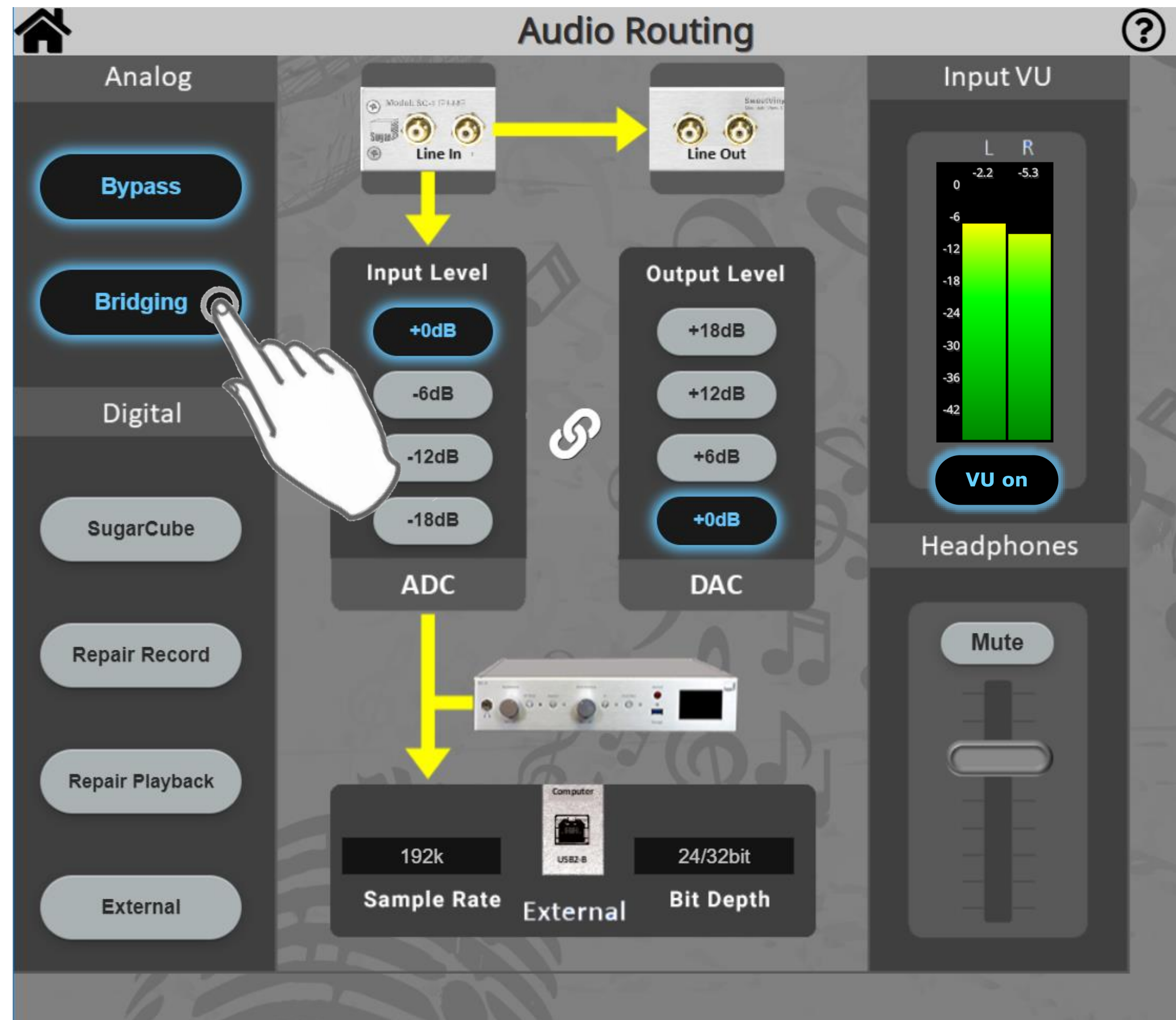
Repair
Record

Repair
Playback

External

In Bypass mode **Line-In** is connected to **Line-Out** via a mechanical relay, this can be selected on the menu or on the front panel. When in **Bypass**, no signal is processed by the SugarCube or a connected computer.

5.12 Audio Routing Menu — Bridging Mode



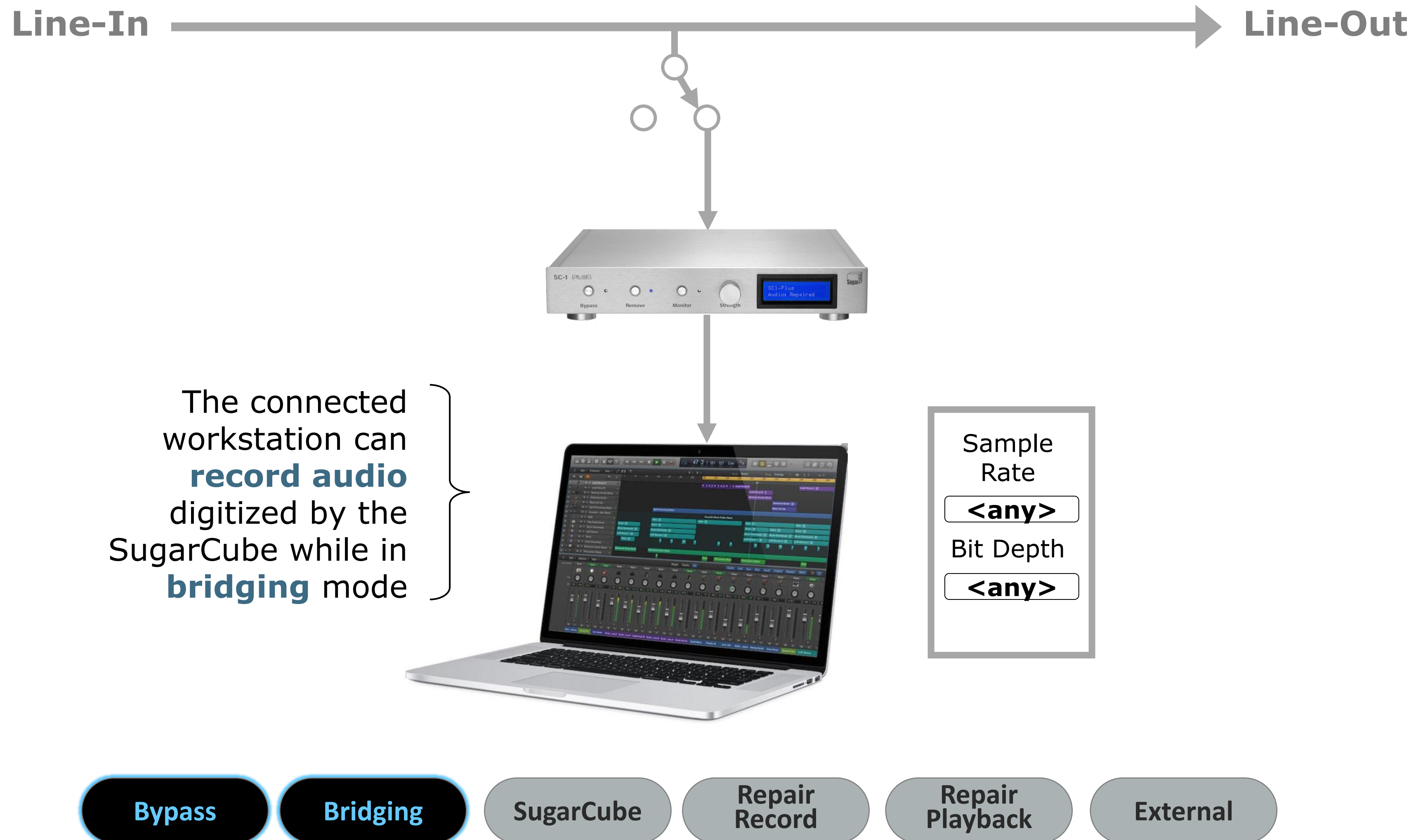
There is no audible affect as **Line-In** passes directly to **Line-Out**.

Bridging mode passes **Line-In** to the Analog-to-Digital Converter (**ADC**) which passes its digital output signal to the attached computer over **USB**, output only.

The **ADC** operates at 192k/24bits by default but the external computer can change to any standard sample rate and bit depth.

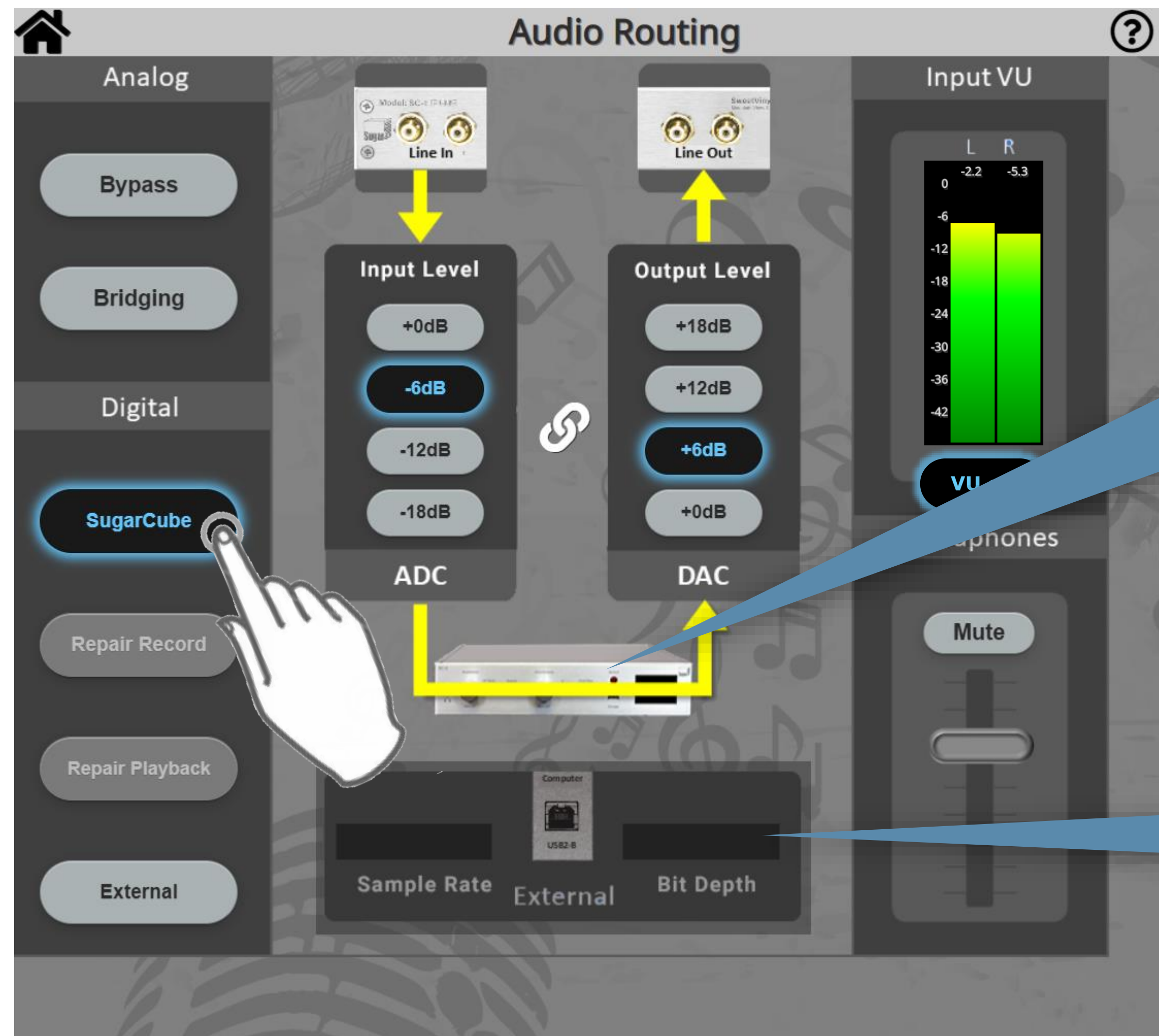
5.13 Logical Flow — Analog Bridging Mode

Line-In connects directly to Line-Out and SugarCube bridges the signal (with high impedance.)



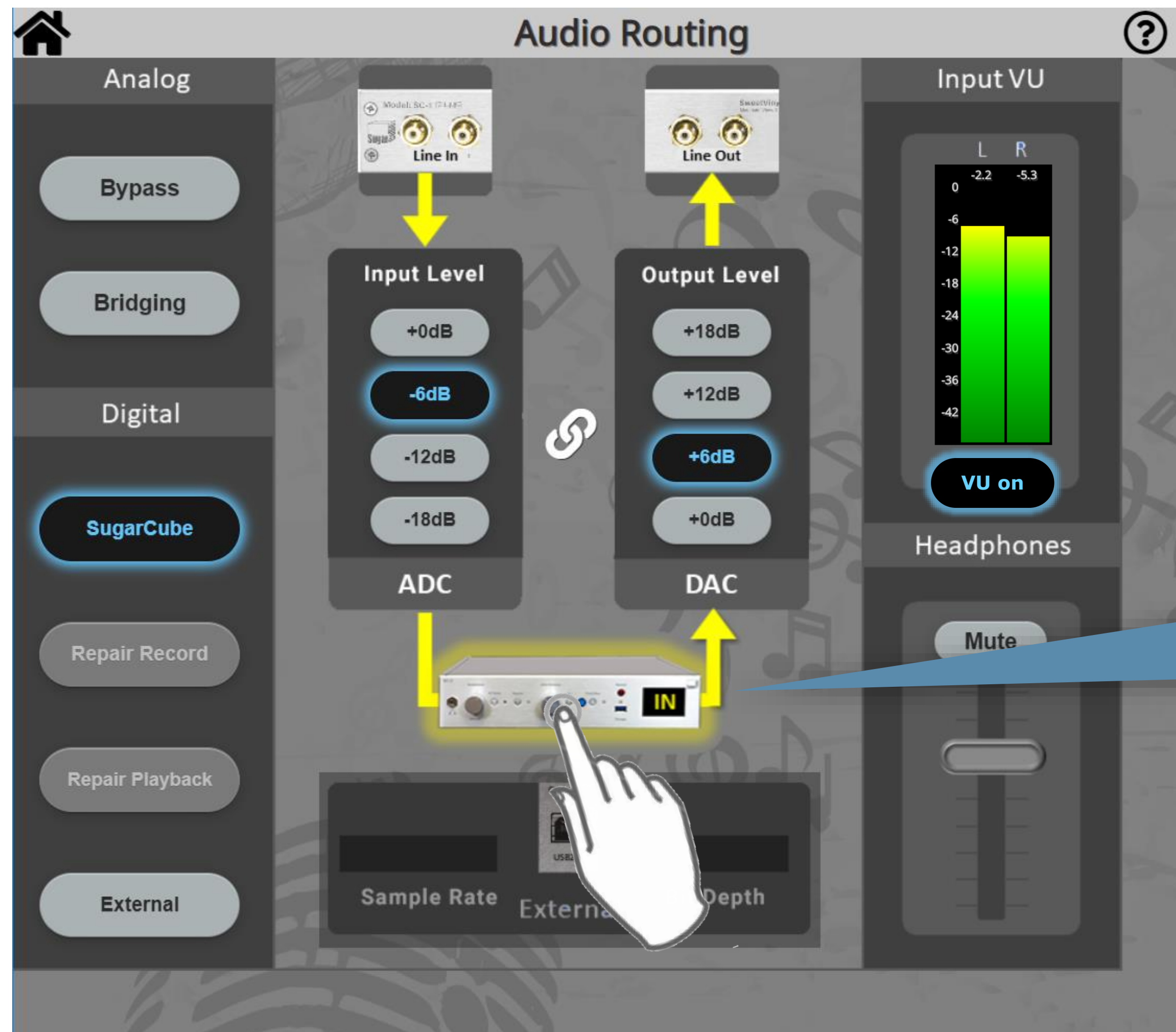
In **Bridging Mode**, **Line-In** is connected to **Line-Out** as in Bypass mode, in addition the signal is passed to the attached computer via the SugarCube's **ADC**.

5.14 Audio Routing Menu — SugarCube Mode



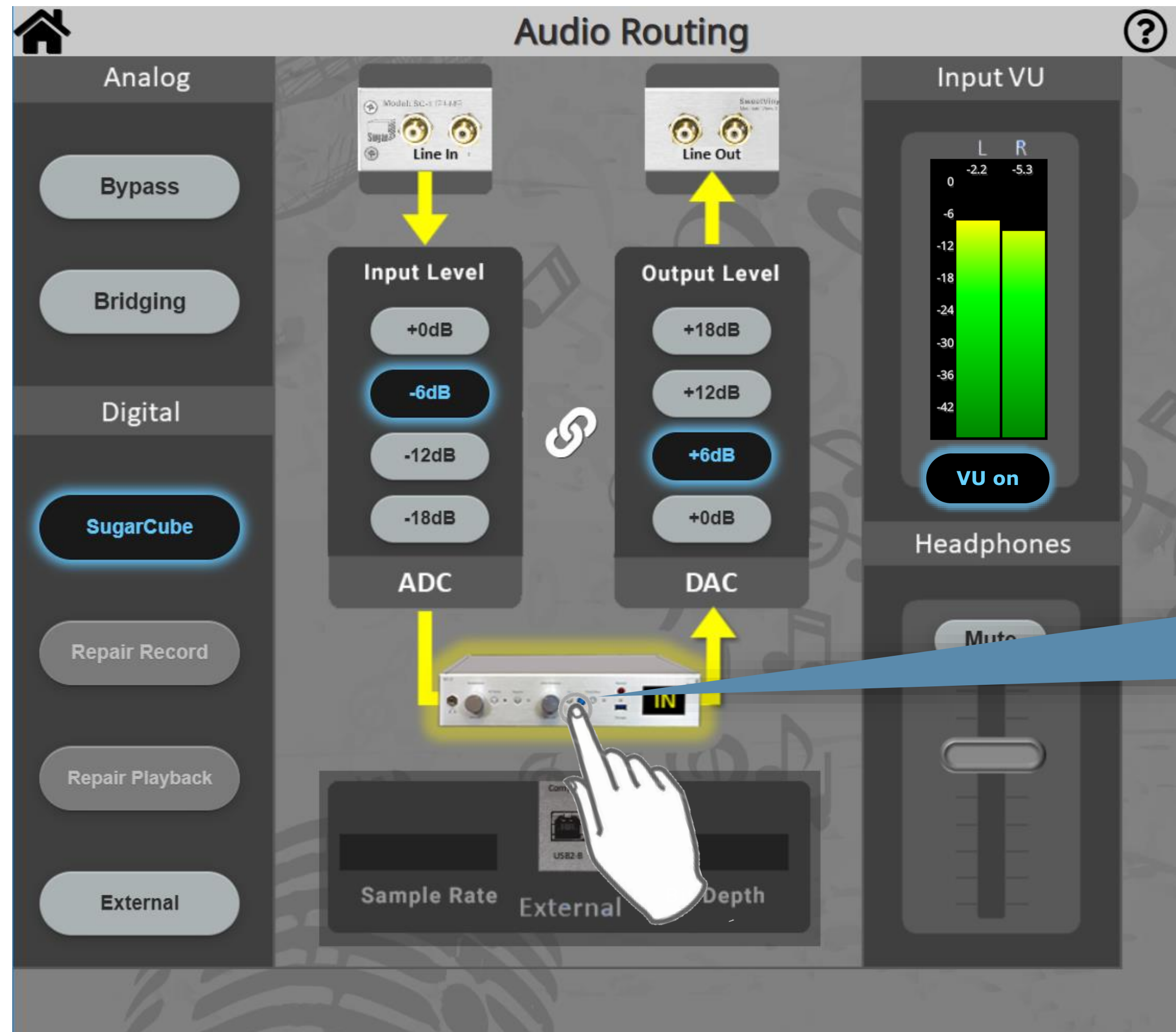
In **SugarCube** mode, audio is processed by the SugarCube as standard on all Sugar Cubes. In addition, the **Un-Repaired Digital** signal is passed to the attached computer via the **ADC**.

5.15 Audio Routing Menu — SugarCube Mode



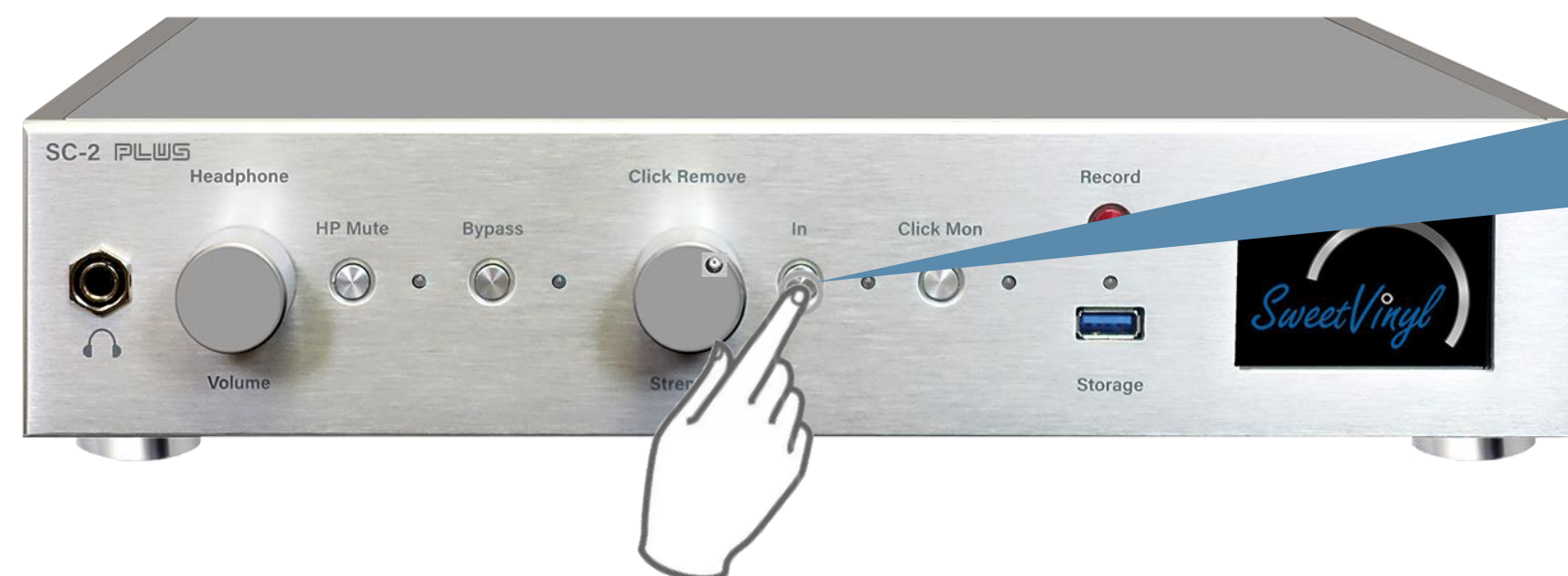
In **SugarCube** mode, audio is processed by the SugarCube as standard on all Sugar Cubes. In addition, the **Un-Repaired Digital** signal is passed to the attached computer via the **ADC**.

5.16 Audio Routing Menu — SugarCube Mode



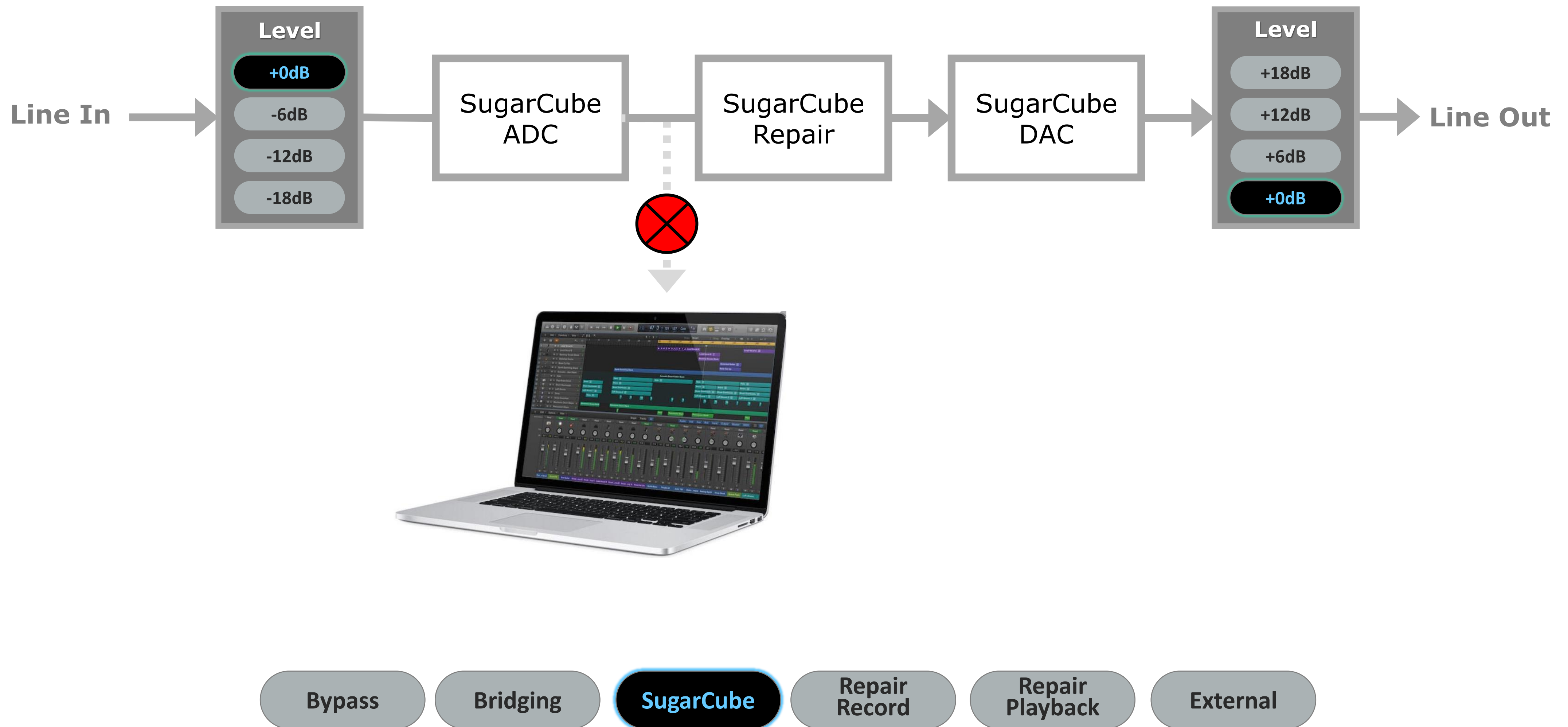
Click on the SugarCube **icon** to engage/dis-engage repair functions

Same as clicking on the front panel 'In' button



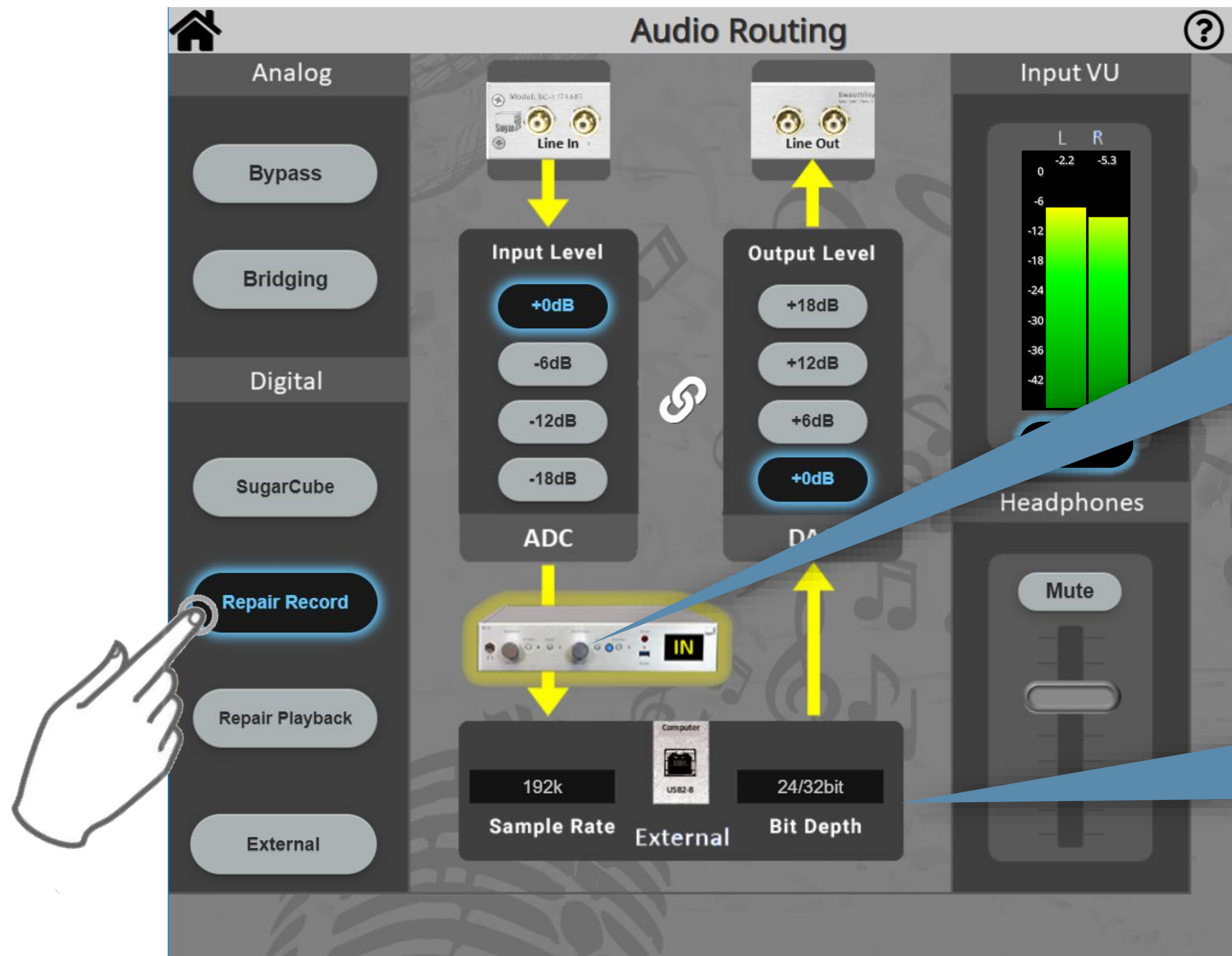
5.17 Logical Flow — SugarCube Mode

Normal SugarCube operations without a digital signal from ADC to the workstation



In **SugarCube** mode, **Line-In** is processed by the SugarCube where audio is repaired and sent to **Line Out** as standard on all SugarCubes.

5.18 Audio Routing — Repair Record Mode



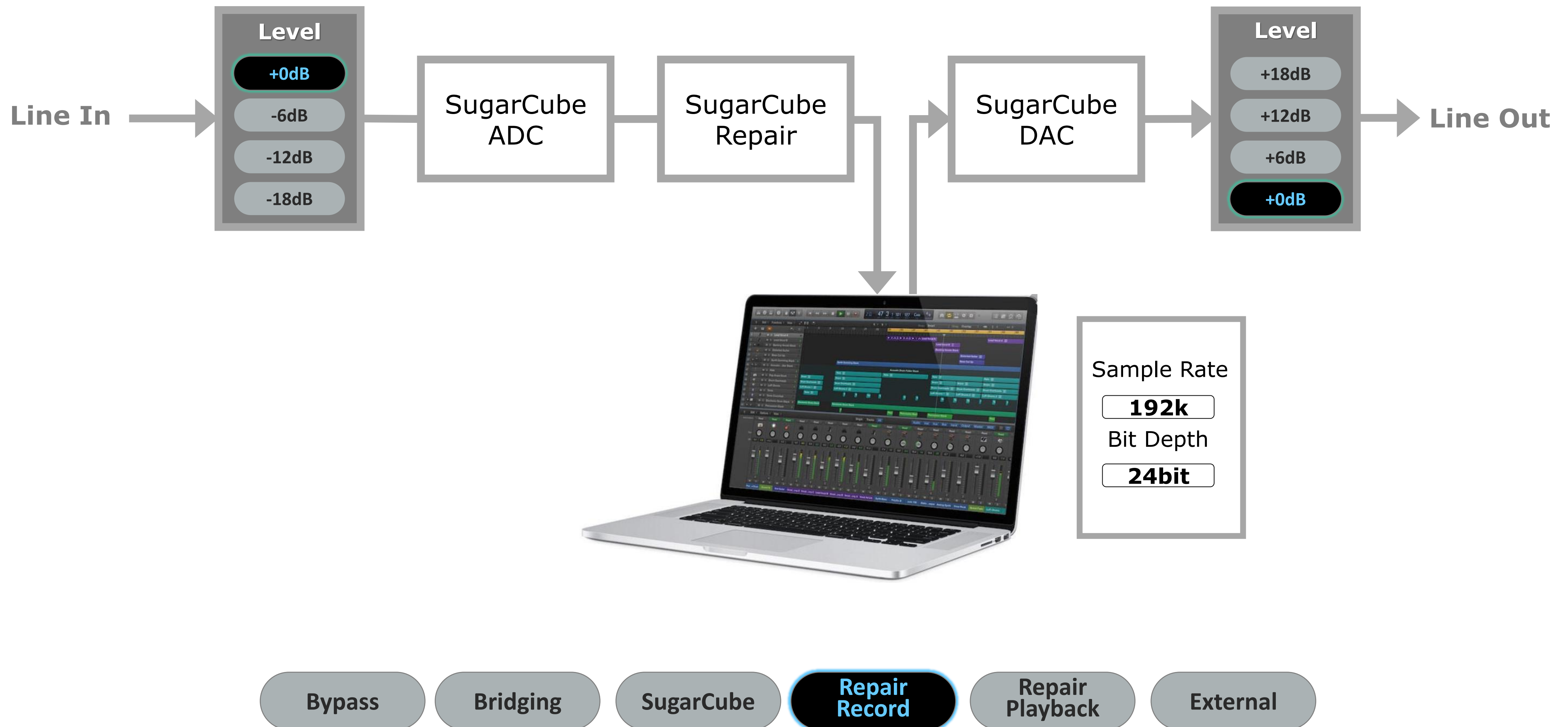
'**Repair Record**' inserts the SugarCube repair processing **after** the **ADC** to repair audio before outputting to the external device

If the connected computer operates at **192k/24bit**, the system will be able to utilize SugarCube repair functions

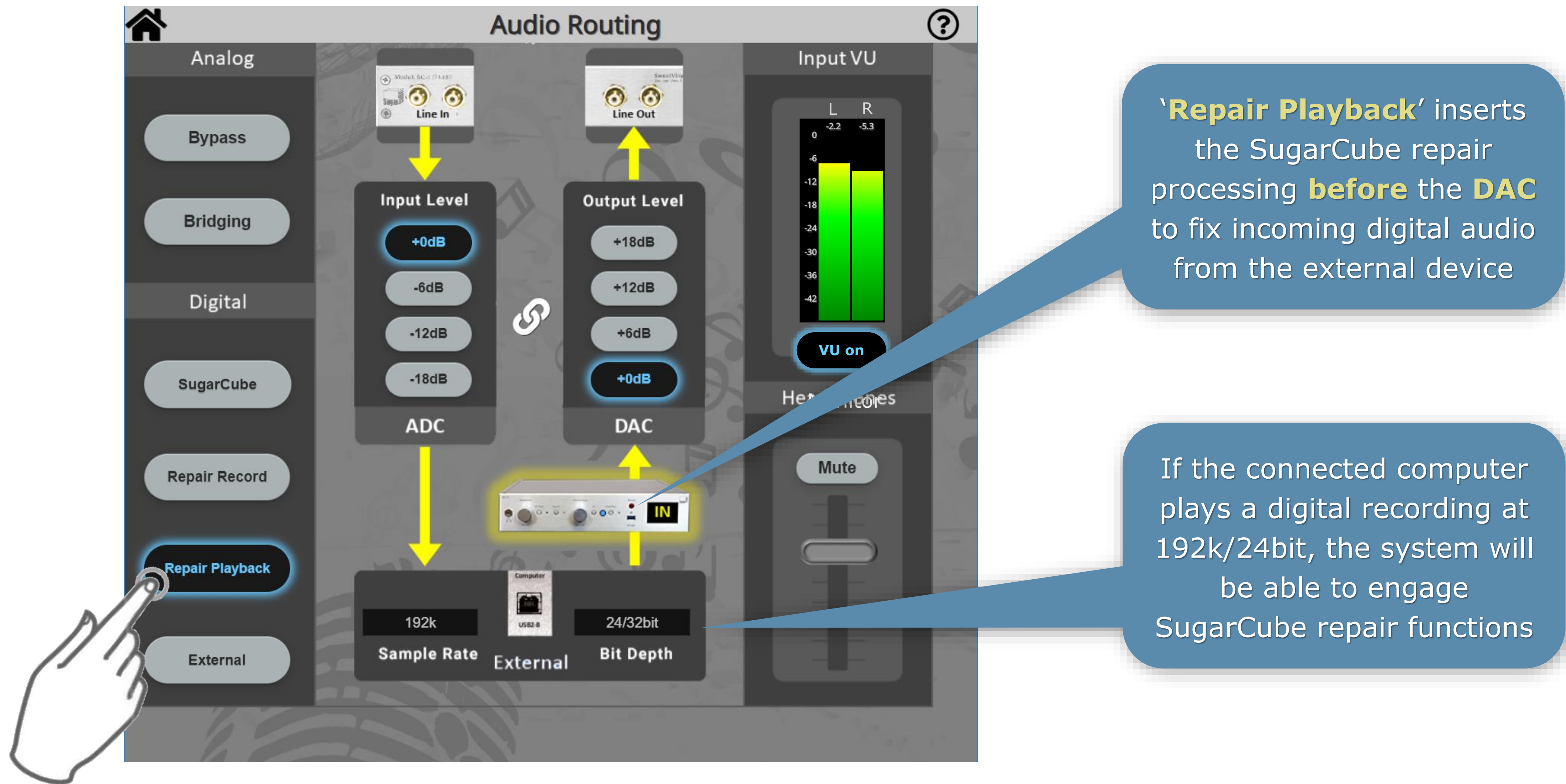
Note: Please **press again** the button of **any mode** of operation (even though it is already selected) if the external device **loses lock** with the SugarCube (you will certainly know due to obvious digital noise.) This resets the internal timing and restores the audio.

5.19 Logical Flow — Repair Record Mode

**SugarCube repairs audio before sending out to workstation.
Workstation directly drives the DAC.**



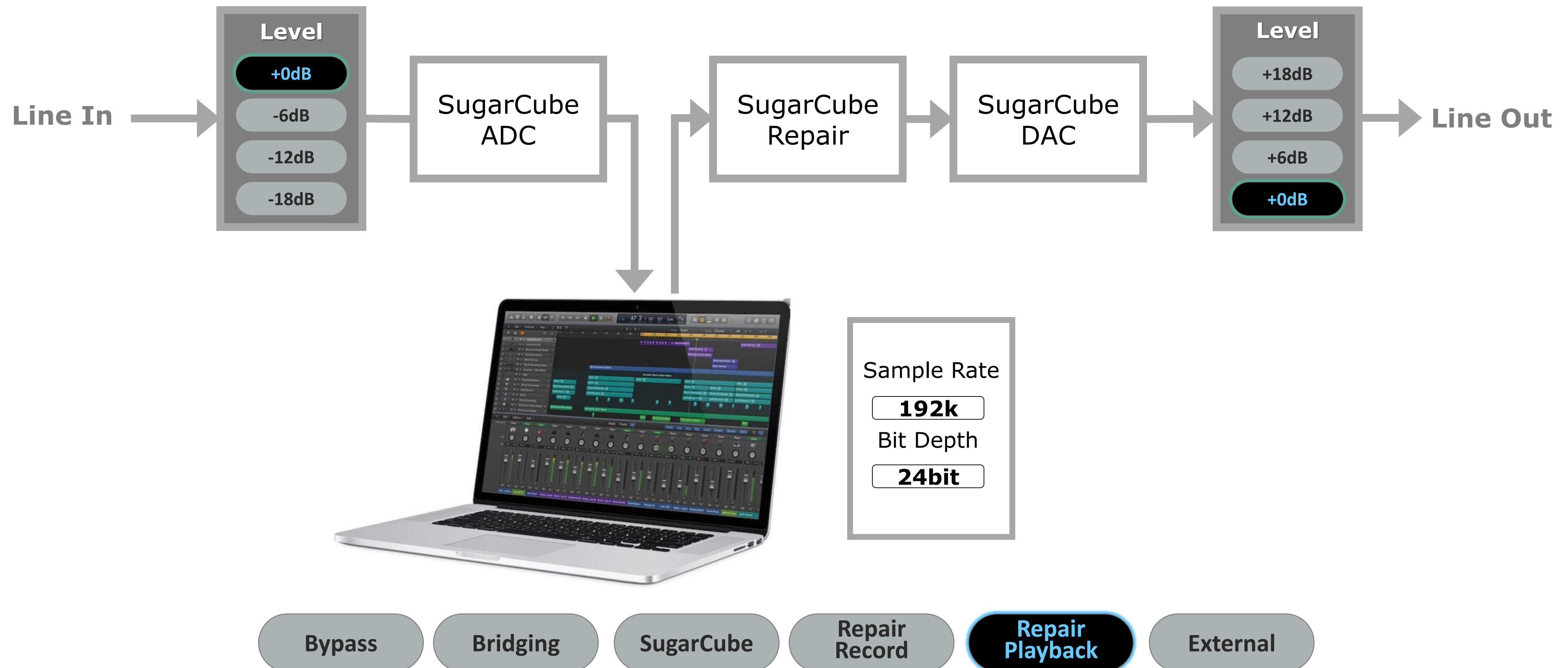
5.20 Audio Routing — Repair Playback Mode



The **Repair Playback** mode is particularly useful allowing you to play any digital file from your library to remove clicks & pops and/or remove tape hiss from previous recordings from any source from the past.

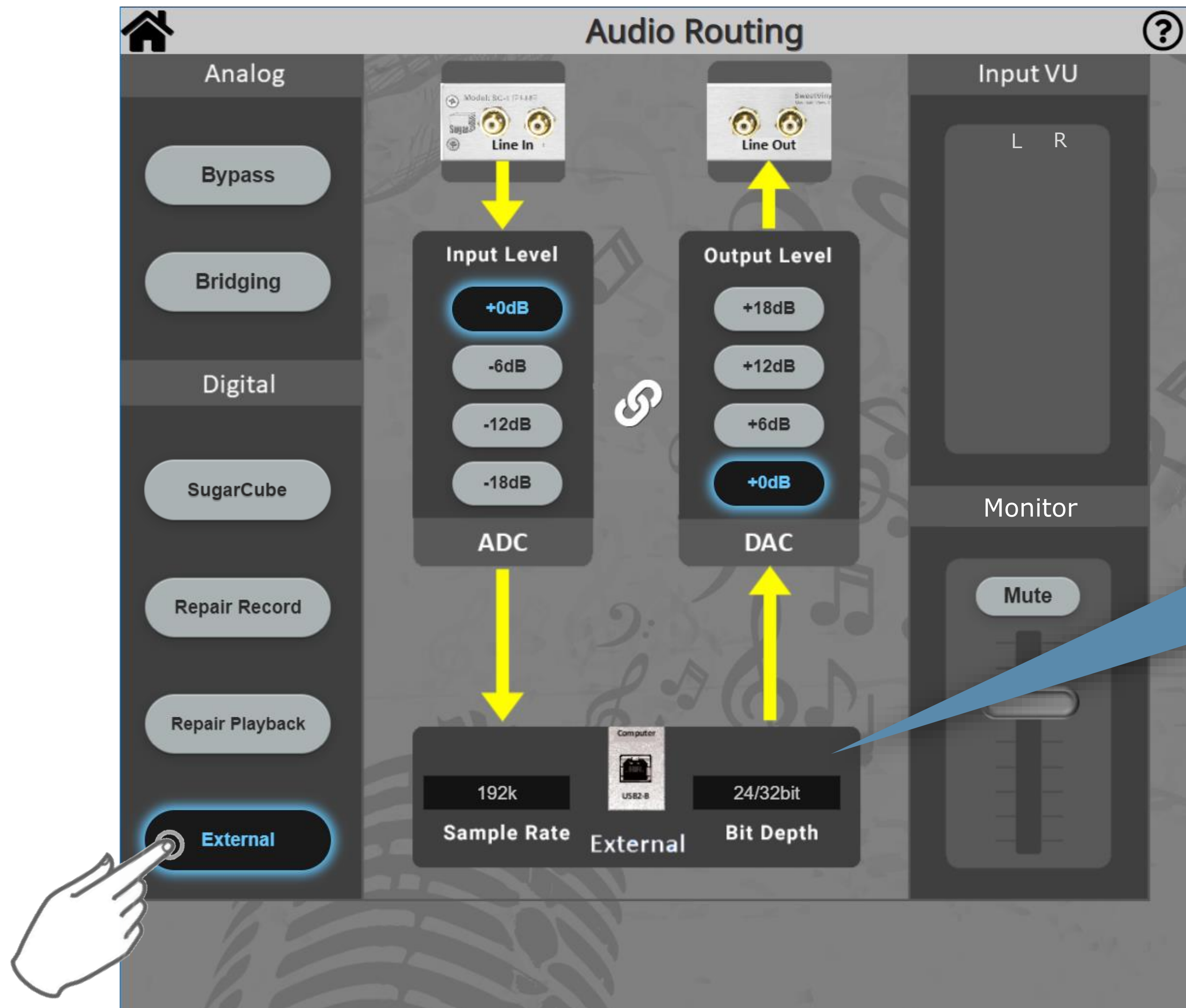
5.21 Logical Flow — Repair Playback Mode

Digitized audio goes directly to workstation. Workstation output feeds SugarCube for repair before DAC.



Repair Playback is our **recommended mode** for operation. All of your files will be **recorded** “in the raw” with no SugarCube processing applied. Then, on **playback**, you can adjust click & pop removal as well as apply noise reduction (if needed) on a recording-by-recording basis or use no processing at all if desired.

5.22 Audio Routing Menu — External Mode

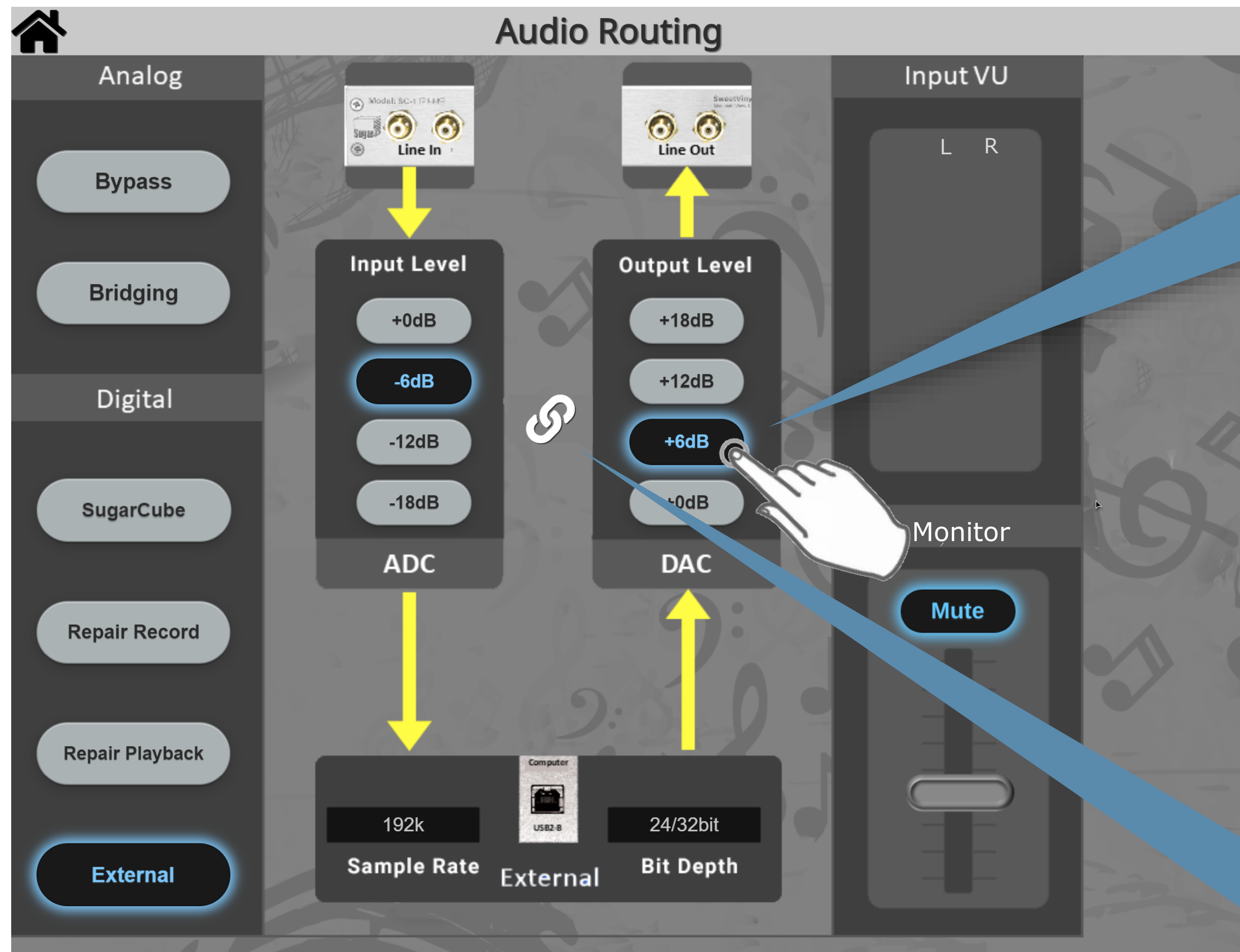


External mode allows an external device to use the **ADC** and **DAC** at **any sample rate**.

In **External** mode, the **SC1-Plus** acts like **any other** studio-quality **ADC/DAC** pair and will function at any standard sample rate and bit depth from 192k down to 44.1k and at 16/24/32 bit sample sizes.

Note: Please **press again** the button of **the current mode** of operation if the external device **loses lock** with the SugarCube (you will certainly know due to obvious digital noise.) This resets the internal timing and restores the audio.

5.23 Audio Routing Menu — External Mode



Output Level can be changed during playback to any of the four available levels.

However, when locked, input level will change in the **opposite** direction.

5.24 Audio Routing Menu — External-Mode Levels

Audio Routing

Other than in **SugarCube Only** mode, you can **unlock** input and output level controls by clicking on the chain icon.

When unlocked, **Input Level** and **Output Level** can be set **independently**.

External Modes

Input Level

- +0dB
- 6dB
- 12dB
- 18dB

ADC

Output Level

- +18dB
- +12dB
- +6dB
- +0dB

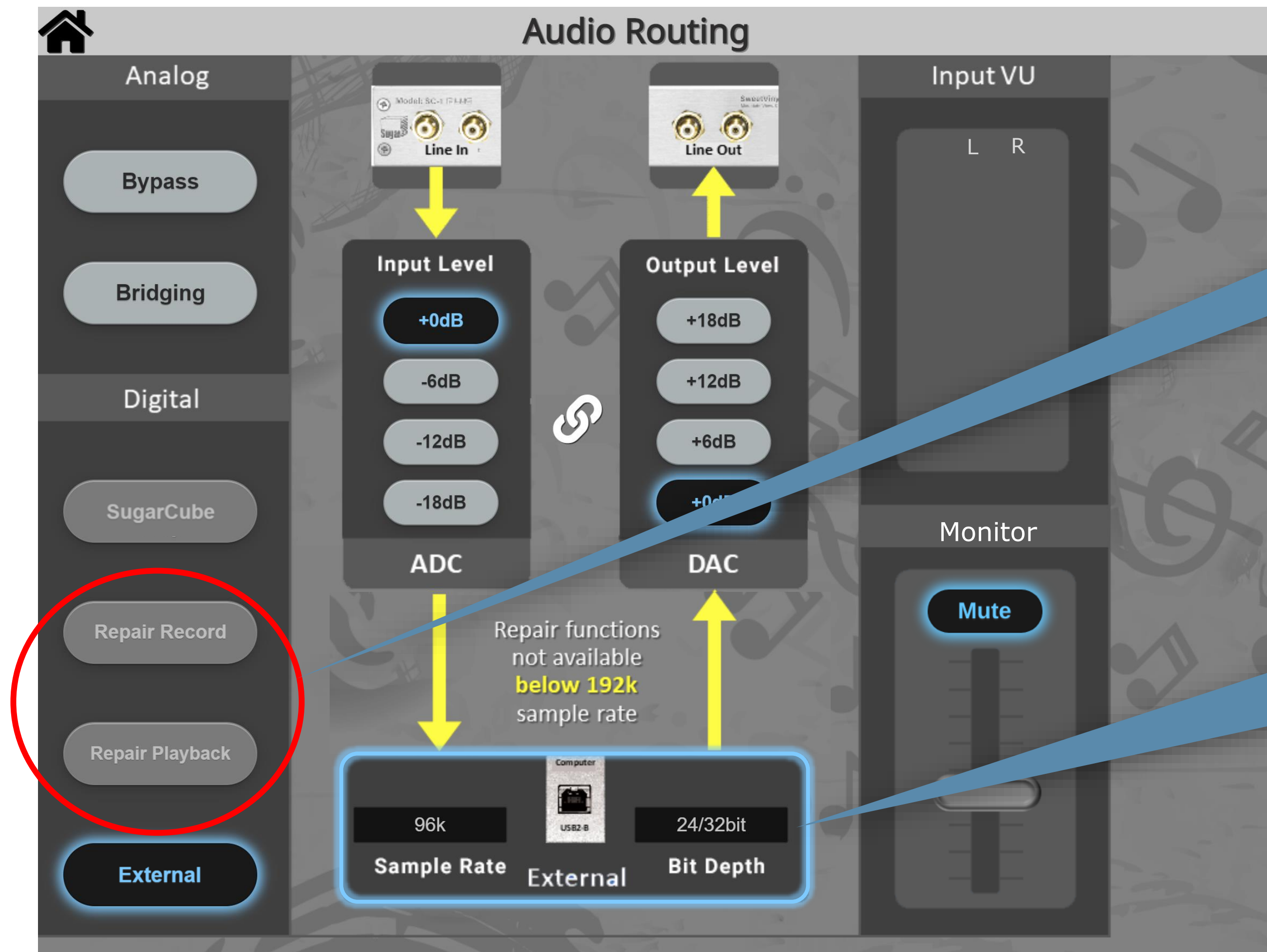
DAC

Sample Rate: 192k

Bit Depth: 24/32bit

External

5.25 Audio Routing — External Device **Not** Set to 192k/24bit



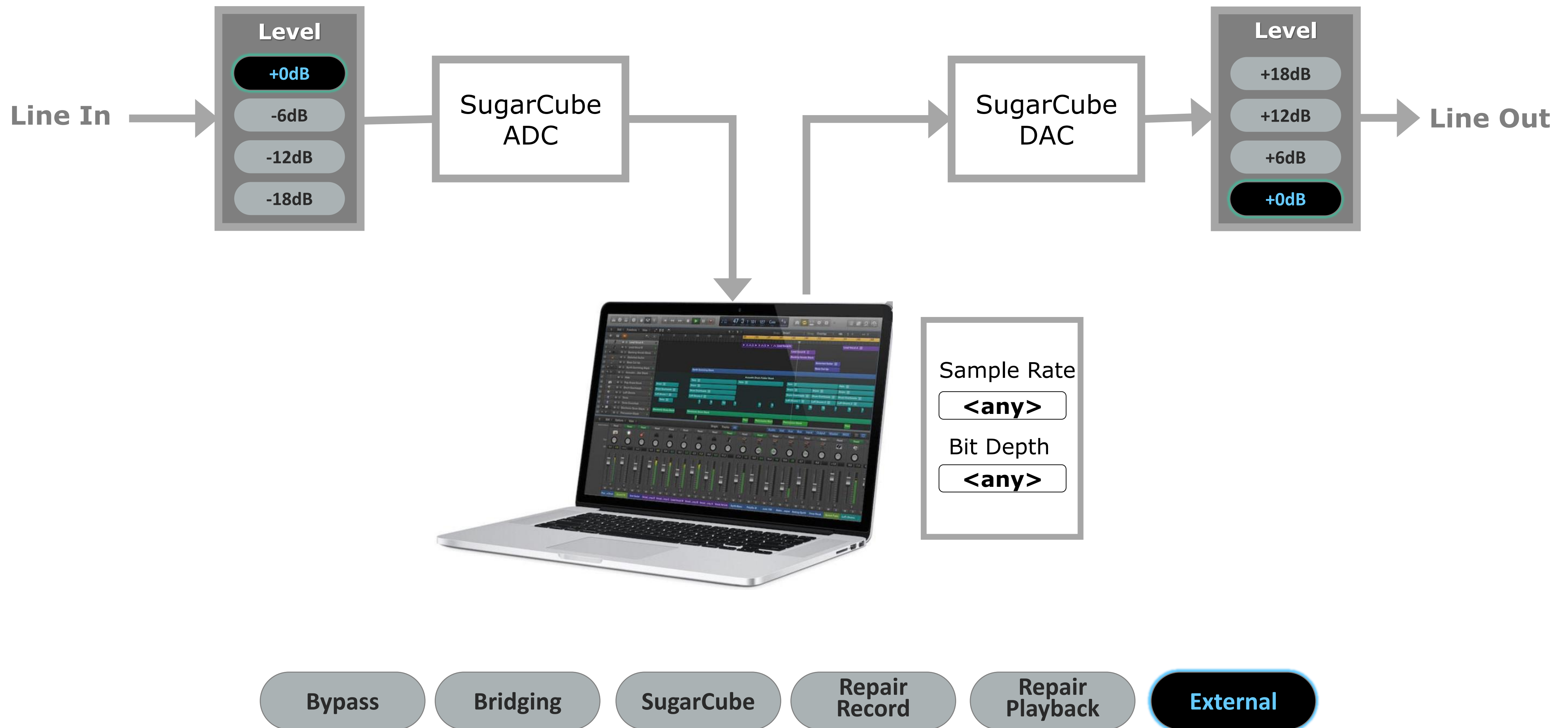
These repair modes are **not available** below **192k/24bit**

The **SugarCube** will operate as an excellent **ADC/DAC** at any sample rate or bit-depth just **no repair functions** are available **below 192k/24bit**.

In **External** mode, it is fine to operate at **any sample rate and bit-depth below 192k/24bit**, but SugarCube **repair functions** do not operate **below 192k/24bit**.

5.26 Logical Flow — External Mode

**SugarCube repair functions are out of the loop.
Workstation uses ADC & DAC directly**



In **External** mode, the SugarCube will operate at **any sample rate and bit depth** from 192k/24bit down to 44.1/16bit, just like any other studio-quality ADC/DAC pair.

FAQ

Some common questions
& answers about using
the SC-Plus series



Basic connectivity

- **Q: Line-In level setting**, why so many choices?
- **A:** You can select four steps each 6dB apart to accommodate low- to high-level output equipment. Whatever level you select for input is mirror the opposite way for output (i.e.-- choosing -6dB line in results in +6dB applied to line out so line in always equals line out.)
- **Q: What is bridging?**
- **A:** This is an analog routing mode where line in goes directly to line out through the bypass relay just like in Bypass mode but is tapped at a high impedance (about 100k ohms) and passed through the internal analog-to-digital converter and then sent out to whatever is attached to the SC1-Plus digital interface. The attached computer can operate at any sample rate or bit depth in this mode.
- **Q: How do I connect through tape loops?**
- **A:** Tape Out goes to the SugarCube Line In and the SugarCube Line Out goes to Tape In. Sometimes the tape loop is labeled Aux or Rec but is the same thing.
- **Q: How do I connecting an external computer?**
- **A:** The SC1-Plus appears as a standard ADC/DAC to a connected PC, Mac, or music server. No special drivers are needed for any device that can accommodate Class 2 USB audio. Also, we have no drivers for devices that do not support Class 2 USB.
- **Q: My computer only goes to 48k, how to I fix that?**
- **A:** You can always use the SC1-Plus in **External** mode as an ADC/DAC at 48k (or any other sample rate), but you cannot then use the other modes of the Plus. Your external computer will be internally disconnected from the SugarCube when not in **External** mode.

Repair modes

- **Q:** when I use the **noise reduction**, do I have to **start at the lead in groove** or can I play any track on the record?
 - **A:** This is a yes and no answer: for the first time you play any record using noise reduction (NR), you must **start the noise reduction process then drop the needle** on the lead in groove.
 - **A:** Then, if you do not stop the process (that is leave it running), if you are playing, for example, a box set of records from the same album, the “**fingerprint**” of the noise that the SugarCube created stays in memory and, if you don’t turn off the SugarCube, you can drop the needle anywhere or play other side without having to **stop** then **re-start NR** then drop the needle in the lead in groove. The **NR** process will continue to work perfectly.
 - **A:** Furthermore, you can extend the above concept if you have LPs from the same label and era (example: late-1950’s Blue Note), the noise “**fingerprint**” that the SugarCube picked up on the first side of the first LP should work for all other LPs of your Blue Note collection when you use **NR**.
 - **A:** However, if you play an LP from say, EMI, that label used a different vinyl plastic formula in making their LPs so their (EMI’s) noise “**fingerprint**” will be different and will require **stopping** the **NR** process and **re-starting** then dropping the needle once again in the lead in groove.
 - **A:** Finally, you could use **NR** and only one time drop the needle for a “fingerprint” in the lead in groove then never do it again, **NR** will still work but it will not be optimal and there may even be a drop in volume.
 - **A:** Bottom line: **NR** is a very powerful tool that can do an amazing job of lowering noise in recordings up to -20dB. It is well worth your time to **start** and **stop NR** when the LP is **different**.

Repair modes

- **Q:** What does **EQ** do?

- **A:** LPs are made with a process that boosts the highs and diminishes the lows during the cutting of the master disk. This process was fundamental to record making all the way back to Thomas Edison's days in the early 1900's. However, until the mid to late 1950s, each manufacturer (record label) had their own process that used differing amounts of boost and cut. So, an RCA Victor record would not sound the same played on a non-RCA turntable, a standard business practice at the time.
- **A:** More importantly, if you are playing 78s (using a 78-rpm phono cartridge) through your system, your phono pre-amp is almost surely using the RIAA equalization which became fully adopted in the early 1960s. RIAA has a much more drastic cut of high-frequencies and a correspondingly strong boost of the bass frequencies on playback than did any of the 78 recordings made prior to the emergence of the LP in the early 1950s. If you listen to a 78 even with a 78 needle, your phono pre-amp will cut much more treble and correspondingly boost much more bass than needed making the playback sound muddy and lacking in detail.
- **A:** When you apply **EQ** on the repair page with the correct equalization choice from our menu, you 78 playback will sound remarkably cleaner and balanced and with detail that you did not know was there.
- **A:** Recall that a 78 record is a direct-to-disc recording as the tape recorder had not yet been invented. When you remove the clicks and pops (which are far more pronounced than on vinyl LPs) and then you lower the loud surface noise of a lacquer disc, and then correct the **EQ** curve, the sound quality is surprisingly delightful.

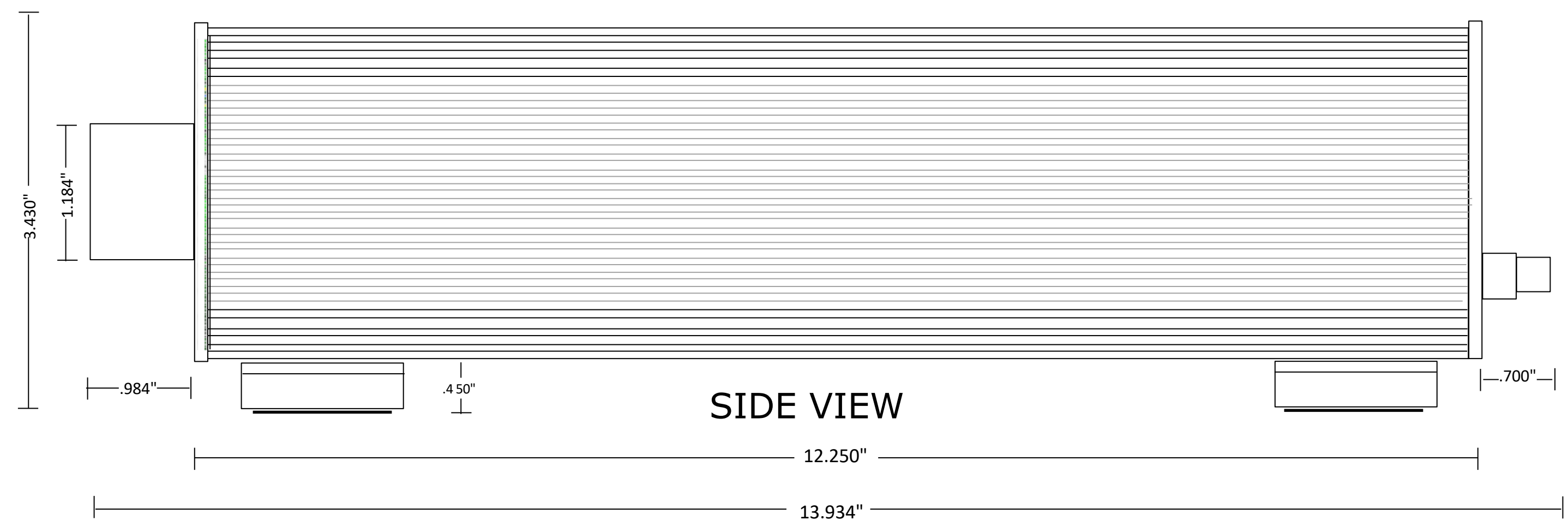
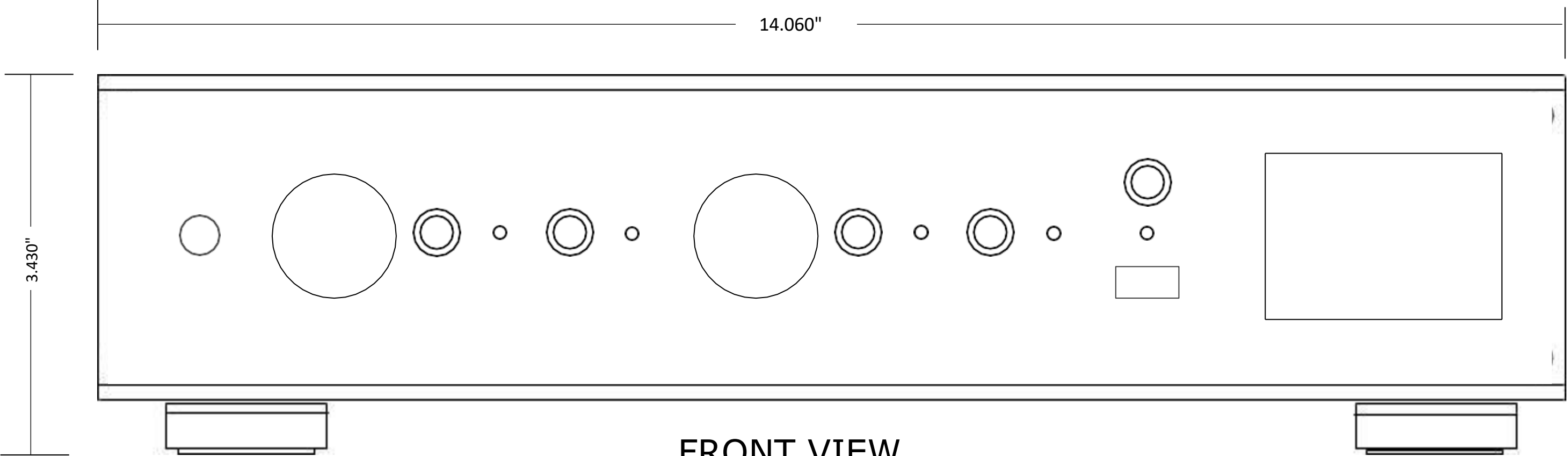
Audio routing and use with external devices

- **Q:** Why can't my computer run **Vinyl Studio** at any sample rate I want?
- **A:** You can run at any sample rate/bit depth that you wish if you are operating your SC1-Plus either in the **Bridging** mode or **External**-only mode. When you want to use the SugarCube processing with external recording, you need to operate at 192k/24bit to match our internal digital processing
- **Q:** Why would you want to remove clicks & pops during playback (**Repair Playback** mode) instead of during recording (**Repair Record** mode?)
- **A:** There are a reasons to operate your SC1-Plus with **Repair Record** as well as with **Repair Playback**. (1) for the **Repair Playback** mode, you can record batches of LPs without any processing and then apply click and pop removal and/or noise reduction on playback on a case-by-case basis just like might you do when you listen to an LP. If you were to use **Repair Record**, you would want to be sure you had all your settings optimal because once recorded, you cannot change the settings. However, if you were making a lot of recordings to save, say, MP3's for your car or mobile phone (i.e., non-critical listening) then it would be best to use **Repair Record** mode.
- **Q:** Why do I only hear noise or nothing at all from my **external DAC** when connected to the Plus except in **SugarCube** mode?
- **A:** The **External DAC** mode is a carryover from our other SugarCube models. In this mode, the **External DAC** is receiving audio from the SugarCube processor inside the Plus. So, in **External** mode, the SugarCube is **disconnected** from the audio pathway and the ADC and DAC are **independent audio units** now connected to just your computer as the graphic shows.
- **A:** Also, the **External DAC** will work in the **Repair Playback** mode because the SugarCube has control of the **DAC** but the not the **Repair Record** mode where the **external computer** has control.

Technical Specs



SugarCube SC2-Plus Mechanical:



Technical Specifications

- Line-In maximum level = +22.7 dBu
(approx. 9.9V rms or 28V p-p)
- Line-Out level always equals Line-In level when locked
- THD+N = 0.0031% (A-weighted)
- Frequency Response = 35Hz to 45kHz +/-0.1dB
- Frequency Response = 15Hz to 75kHz -3dB
- DC in = 5 volts, center-pin positive
- Average Power Dissipation = 10 watts
- Internal cache capacity for recordings = 10
- Internal SugarCube processing operations at 192k/32bits
- External drive formats: NTFS, exFAT, FAT32

General Info

Support

Thank you for purchasing this SugarCube product.

Please contact customer support at: support@sweetvinyl.com

Trademarks

SweetVinyl™ and SugarCube™ are trademarks of SweetVinyl Inc.
Any non-SweetVinyl trademarks are used for reference purposes only.

Compliance

For the current EU Declaration of conformity, visit:
<http://support.sweetvinyl.com/regulatory>

For regulatory compliance information in North America, visit:
<http://support.sweetvinyl.com/regulatory>

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